04 Logic Synthesis with Design Compiler

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Course Outline

- **Logic Synthesis Flow (Design-Compiler)**
  - Reading Target Library and Source HDL code
  - Setup Design Constraints
  - Design Optimization and Report Analysis
  - Gate-level Simulation and Timing Violations

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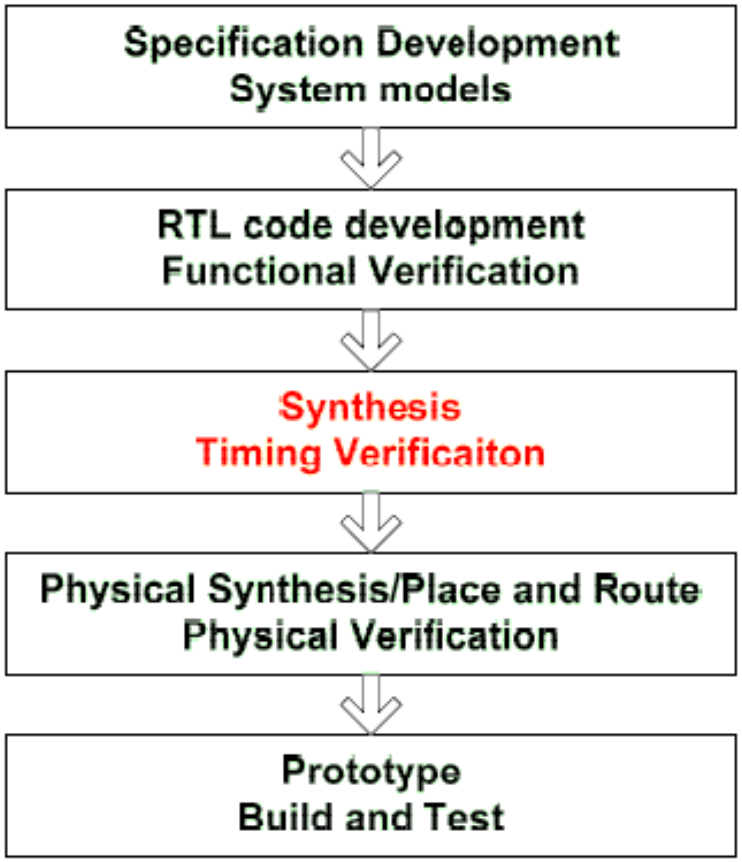
Cell-based Design Flow



System Architecture

RTL

Synthesis

Physical Design

System Integration and Software Test

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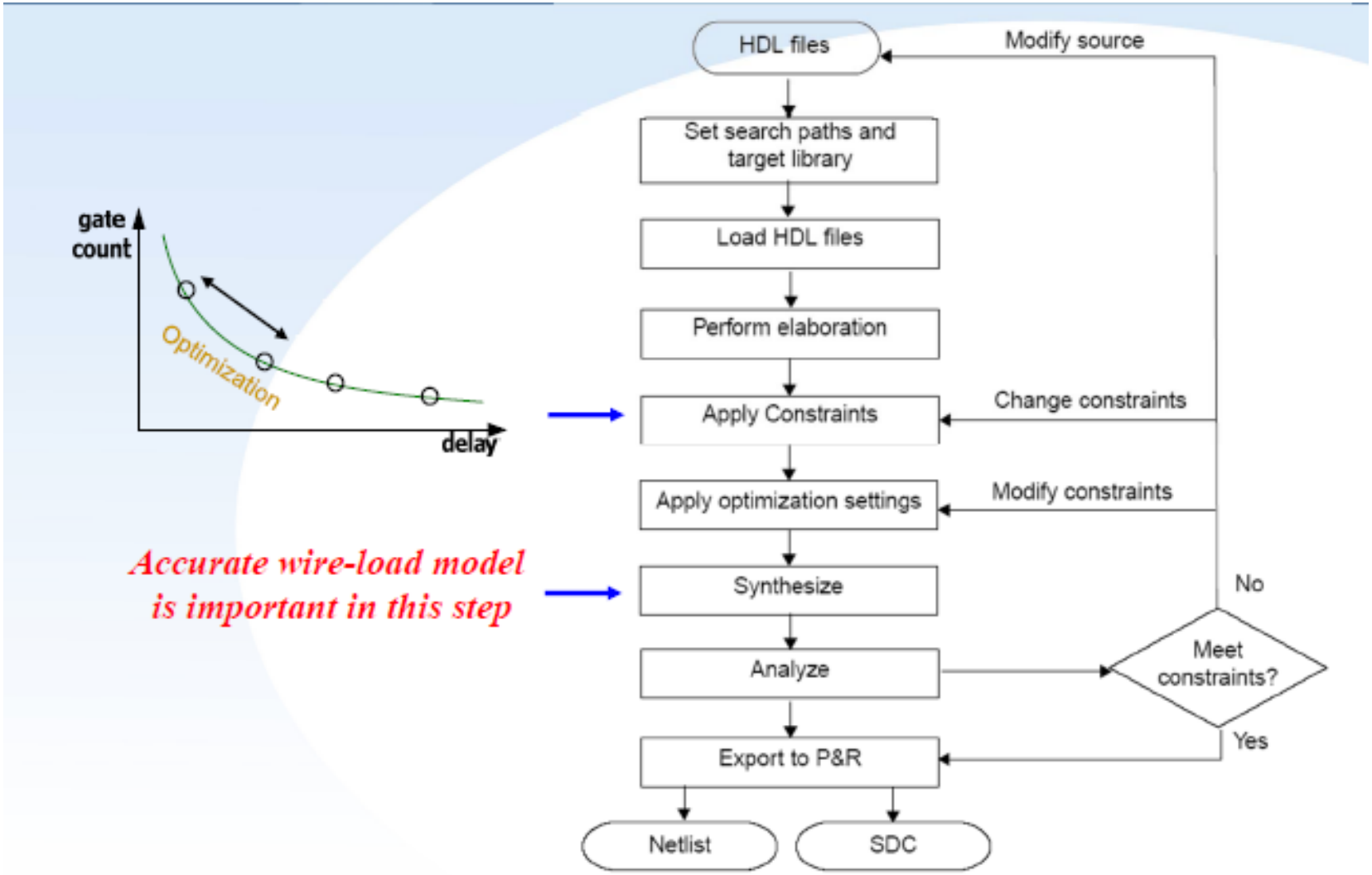
Design Synthesis Flow



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Specify Target Libraries

- **Target Library**
  - During mapping, logic synthesizer selects functionally correct gates from the library. It also calculates the timing of the circuit, using the vendor-supplied timing data for these gates.
  - Example :

```
## Specified UMC 0.18um library ##
set search_path [format "%s %s" $search_path "." ]
set synthetic_library [list dw_foundation.sldb  ]
set link_library [list slow.db dw_foundation.sldb  ]
set target_library [list slow.db ]
set symbol_library [list umc18.sdb ]
set alib_library_analysis_path "./umc18"
```


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Specify Macro Libraries

- **Hard MACRO timing Library**
  - Hard macro library is only used for timing analysis,and those macro cells will not be automatic selected during logic optimization. (such as SRAM)
  - Example: If we have a RAM timing model: SRAM.db

```
## Specified UMC 0.18um library ##
set search_path [format "%s %s" $search_path "." ]
set synthetic_library [list dw_foundation.sldb  ]
set link_library [list slow.db SRAM16x24.db dw_foundation.sldb ]
set target_library [list slow.db ]
set symbol_library [list umc18.sdb ]
set alib_library_analysis_path "./umc18"
```


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Loading HDL Files

- **Load design**
  - Reads a design HDL file into internal database. The logic synthesizer optimization process works only on the designs loaded in memory.
  - If design is a pure gate-level netlist, -netlist option can speed up the load design time.
  - Syntax : **read_file –format verilog file1.v \ file2.v \....\ filen.v**
- **Specify the top design name**
  - The current_design command sets the working design for many dc_shell commands.
  - Syntax : **current_design <module_name>**

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Specify Interconnect Mode

- **Specify Interconnect Mode**
  - Specify the mode to handle hierarchical wire models.
  - Syntax : **set_wire_load_mode** {top | enclosed |segmented}
  - Example : set_wire_load mode “enclosed”
- **Specify Interconnect Model**
  - Select the wire load model for calculating wire capacitance.
  - Syntax: **set_wire_load_model –name** model_name
  - Example: set_wire_load_model –name “umc18_wl20”

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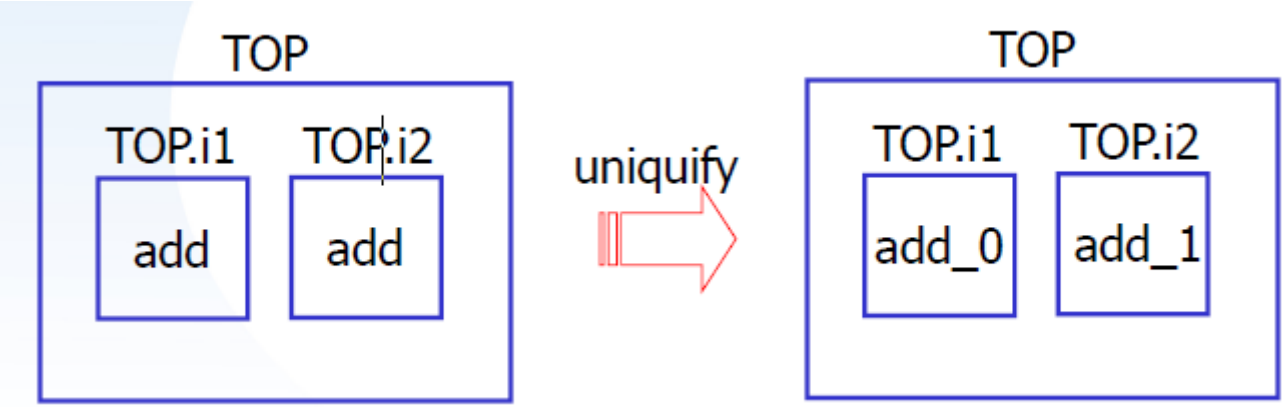
Uniquify Design

- **Remove multiply-instantiated hierarchy in the current design.**
  - ➢Creating a unique design for each cell instance.
  - ➢Syntax : **uniquify**



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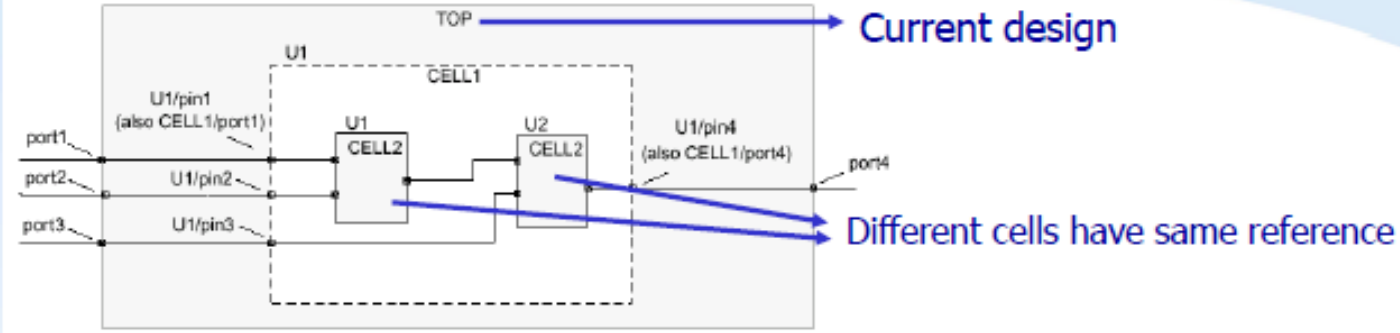
Design Object



Object Name	Meaning
Design	Circuit description that performs functions
Current design	It is the active design (the design being worked on)
Current instance	It is the active cell within the current design.
Cell	An instantiation of a design instance
Reference	The original design that a cell points to
Port	The input or output of the current design
Pin	The input or output of a cell
Net	The wire connects between ports to pins and pins to each other
Clock	Clock source

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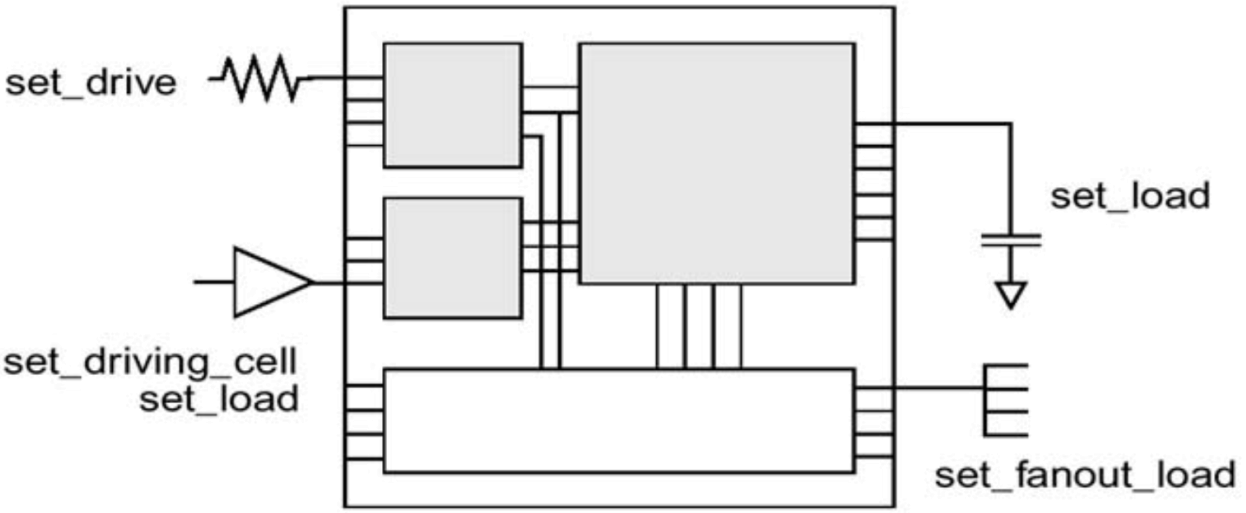
Define Design Environment

- **Design Environment**
  - ➢ You must define the environment in which the design is expected to operate. You define the environment by specifying system interface characteristics.



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Input Drive Characteristic

- **The transition time delay at an input port**
  - The transition time delay function is **Time = arrival_time + [drive_resistance *load(net)]**
- **Set drive Command**
  - to set the driving resistance on the top-level ports of the design when the input port drive capability cannot be characterized with a cell in the technology library.
    - Syntax : **set_drive** resistance port_list
- **Set driving cell Command**
  - Specify driving characteristics on ports that are driven by cells in the technology library. The default of drive resistance is zero.
    - Syntax : **set_driving_cell –library <lib_name> -lib_cell <cell_name> -pin <pin_name> port_list**

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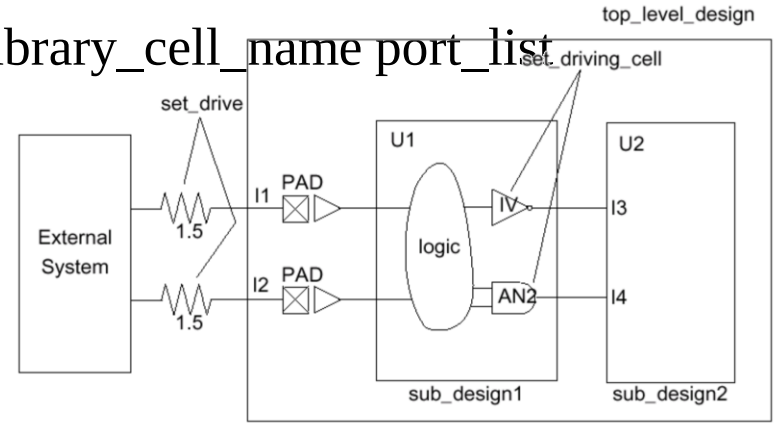
Input Drive Characteristic (cont'd)

- **Example of drive characteristics**
  - Because ports I1 and I2 are not driven by library cells, use the set_drive command to define the drive resistance.
    - current_design top_level_design
    - set_drive resistance port_list
  - Change the current design to sub_design2. An IV cell drives port I3 and because IV has only one output and one input,define the drive capability as follows.
    - set_driving_cell –lib_cell library_cell_name port_list

Logic Synthesizer will look up the drive resistance of cell IV in the .lib



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Output Load Characteristic

- **Defining loads on input and output ports**
  - By default, it assumes zero capacitive load on input and output ports. Use the set_load command to set a capacitive load value on input and output ports of the design.
    - Syntax : **set_load** capacitance_value objects
    - Example : To set a load of 30pF on output port set_load 30 {out1}
- **Defining fanout loads on output ports**
  - Fanout load is a unitless value that represents a numerical contribution to the total fanout. Load is a capacitance value.
  - To specify the expected fanout load values on output ports with the set_fanout_load command.
    - Syntax: **set_fanout_load** value port_list
    - Example: To set a fanout load 4 on the output port set_fanout_load 4 {out2}

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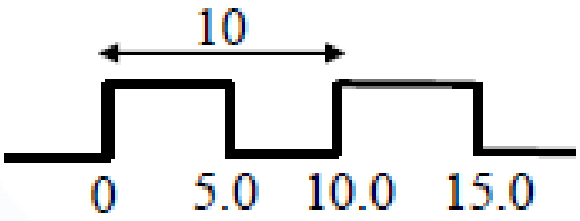
Define Design Constraints

- **Design Optimization Constraints**
  - ➢ User define timing and area optimization goals for logic synthesizer.
  - ➢ Timing constraints
    - ■ Defining a clock
    - ■ Specifying I/O timing requirements
    - ■ Specifying combinational path delay requirements
    - ■ Specifying timing exceptions
  - ➢ Area constraints (set_max_area)
  - ➢ Power constraints

Clock Constraints

- **Define the clocks**
  - For synchronous designs, the clock period constrains all register-to-register paths in the design.
  - Defines the period and waveform for the clock
    - Syntax: **create_clock** -name clock_name source_objects –period period_value -waveform edge_list
    - source_objects : a list of pins or ports on which to apply this clock.
    - -period period_value : the period of the clock waveform
    - -waveform edge_list : Specifies the rise and fall edge times of the clock waveforms
    - Ex: The clock applied on port CLK with a period of 10.0, rise at 0.0 and fall at 5.0.
      create_clock -name CLK1 CLK -period 10 -waveform { 0.0 5.0 }



Duty-cycle of CLK is
(5.0 - 0.0)/(10.0 - 0.0)=50%

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Setup Time and Hold Time

- **Setup Time**
  - **Data must arrive and stabilize at the capture flip-flop before the next active clock edge.**
- **Maximum-delay check**

$$T_{clk} + T_{skew} > T_{cq} + T_{comb(max)} + T_{setup}$$

- **Hold Time**
  - **New data must not reach the capture flip-flop too early after the same active clock edge.**
- **Minimum-delay check**

$$T_{cq} + T_{comb(min)} > T_{hold} + T_{skew}$$

T_{cq}: *Flop clock to Flop q delay*
T_{setup}: *Required setup time of a Flop*
T_{hold}: *Required hold time of a Flop*
T_{skew}: *Delay between clock edges of two adjacent flops*

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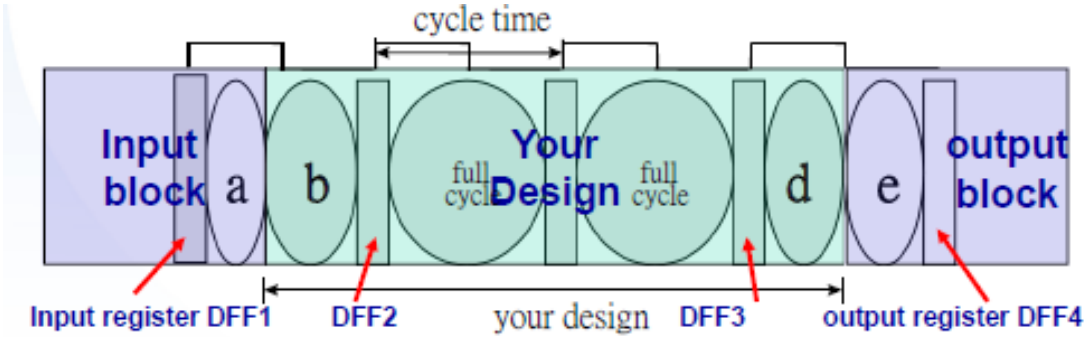
I/O Timing Constraints

- **Input delay**
  - clock cycle >= (DFF1clk-Qdelay + a) + b + DFF2setup
  - Input delay = (DFF1clk-Qdelay + a)
  - The default timing is to arrive at the input port at time 0 relative to the clock rise edge time
- **Output delay**
  - clock cycle >= DFF3clk-Qdelay + d + (e +DFF4setup)
  - Output delay = (e +DFF4setup)
  - The default is no constraint of any paths which end at an output port



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Setup Input/Output Delay

- **Input Delay**
  - Defines the arrival time relative to a clock edge on input ports.
    - Syntax: **set_input_delay** delay_value -clock clock_name port_pin_list
    - delay_value : Specifies the path delay.
    - -clock clock_name : Specifies the clock to which the specified delay is related.
    - Ex: set_input_delay 2.0 –clock CLK [all_inputs]
- **Output Delay**
  - Defines the data required time at the output port.
    - Syntax: **set_output_delay** delay_value -clock clock_name port_pin_list
    - delay_value : Specifies the path delay.
    - -clock clock_name : Specifies the clock to which the specified delay is related.
    - Ex: set_output_delay 2.0 –clock CLK [all_outputs]

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Path Delay Constraints

- **For combinational circuit design, the max/min delay of the design can be constrained.**
- **Setup max delay**
  - To define the maximum delays for the specified paths
    - Syntax : **set_max_delay** delay_value -from from_list -to to_list
    - from_list : specifies a list of path start points
    - to_list : specify a list of path end points
    - Ex: This example specifies that all paths from in1 to out1 must have delays less than 10 units set_max_delay 10 -from in1 -to out1
- **Setup min delay**
  - To define the minimum delays for the specified paths
    - Syntax : **set_min_delay** delay_value -from from_list -to to_list
    - from_list : specifies a list of path start points
    - to_list : specify a list of path end points

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Set False Path

- **A false path is a timing path that cannot propagate a signal. Timing constraints are removed from these paths so that the logic synthesizer does not optimize them.**
- **Identifies false paths in a design, and breaks or disables specific instance timing arcs in a design.**
  - This command selectively disables the arrival time information depending on the precedence rules.
  - Syntax: **set_false_path** –from pin_list –to to_list
  - Ex: To set a false path for any path beginning at p1 or p2 and terminating at t1 or t2, group the pins as follows: set_false_path –from {p1 p2} –to {t1 t2}

Power Constraints

- **Set the total power constraints during design optimization.**
  - total_power = dynamic_power + leakage_power
- **Syntax:**
  - **set_max_dynamic_power power [unit]**
  - **set_max_leakage_power power [unit]**
  - Ex: set_max_dynamic_power 0.0

Try to minimize the power consumption

Design Optimization

- **Optimization**
  - The logic synthesizer maps the design to an optimal combination of specific target library cells, based on the design's functional, speed, area, and power requirements.
- **Optimization process**
  - Logic Synthesizer performs the following three levels of optimization
    - Architectural optimization : depend on how your coding style affects architectural optimization.
    - Logic-level optimization
    - Gate-level optimization

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Design Optimization (cont'd)

- **Structuring**
  - Applies Boolean and algebraic algorithms and transformations to achieve logic optimization and logic structuring. These transformations are technology independent.
- **Mapping**
  - This process uses gates (combinational and sequential) from the target technology libraries to generate a gate-level implementation of the design whose goal is to meet timing, area and power goals.
  - Syntax: **compile_ultra**

Change Naming Rules

- **After synthesis, we must change the names of ports, cells, and nets in a design.**
  - ➢ We must follows the strict naming rule required for automatic placement routing (APR) tools and layout tools.

```
## Change netlist naming rule ##
set bus_inference_style "%s\[%d\]"
set bus_naming_style "%s\[%d\]"
set hdlout_internal_busses true

change_names -hierarchy -rule verilog

define_name_rules name_rule -allowed "a-z A-Z 0-9 _" -max_length 255 -type cell
define_name_rules name_rule -allowed "a-z A-Z 0-9 _[]" -max_length 255 -type net
define_name_rules name_rule -map {{"\\*cell\\*" "cell"}}
change_names -hierarchy -rules name_rule
```


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Report Analysis: Timing

- **Report Design Timing:**
  - Generates a timing report that provides information about the various paths in the design.
  - Syntax:
  - **report_timing** > report.timing

```
Startpoint: pointer_reg_0_
          (rising edge-triggered flip-flop clocked by CLK)
Endpoint: pointer_reg_2_
          (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
STACK              umc18_wl20            slow

Point                                    Incr       Path
-----------------------------------------------------------
clock CLK (rise edge)                    0.00       0.00
clock network delay (ideal)              0.00       0.00
pointer_reg_0_/CK (DFFTRX2)              0.00       0.00 r
pointer_reg_0_/QN (DFFTRX2)              0.89       0.89 r
U293/Y (NAND2X1)                         0.46       1.35 f
U318/Y (NOR2X1)                          0.55       1.90 r
U317/Y (BUFX12)                          0.36       2.26 r
U282/Y (NOR2X1)                          0.31       2.57 f
U279/Y (NAND2X1)                         0.50       3.07 r
U278/Y (NOR2BX1)                         0.36       3.43 f
U295/Y (OAI2BB1X1)                       0.46       3.89 f
U296/Y (NAND2X1)                         0.29       4.18 r
U297/Y (OAI31X1)                         0.29       4.47 f
pointer_reg_2_/RN (DFFTRX2)              0.00       4.47 f
data arrival time                                   4.47

clock CLK (rise edge)                    5.00       5.00
clock network delay (ideal)              0.00       5.00
pointer_reg_2_/CK (DFFTRX2)              0.00       5.00 r
library setup time                      -0.52       4.48
data required time                                  4.48
-----------------------------------------------------------
data required time                                  4.48
data arrival time                                  -4.47
-----------------------------------------------------------
slack (MET)                                        0.01
```

Timing slack must > 0 (constr. met)

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Report Analysis: Area

- **Report Design Area:**
  - Generates an area report that provides information about area of each modules in the design.
  - Syntax: **report_area** > report.area

```
Combinational area:       6606.230435
Noncombinational area:    9350.510147
Net Interconnect area:    142334.058716

Total cell area:          15956.740582
Total area:               158290.799298
```

Estimated wire area

  - In UMC 0.18um cell-library, area (um**2**)/9.97 is gate count.
    - 9.97 is the area of min. 2-input NAND gate in the cell-library.
    - Ex: 15956/9.97 = 1600 = 1.6k (gate count)
  - The same RTL design synthesized in different process always get almost the same gate count.

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Save Gate-Level Design

- **Writes out a netlist stored in the database into Verilog format.**
  - Syntax: **write –format format_name –output file_name -** hierarchy

```
module STACK ( DATA_OUT, FULL, EMPTY, CLK, RESET, PUSH, POP, DATA_IN );
  output [15:0] DATA_OUT;
  input [15:0] DATA_IN;
  input CLK, RESET, PUSH, POP;
  output FULL, EMPTY;

  BUFX8 U349 ( .A(n184), .Y(n604) );
  BUFX8 U350 ( .A(n277), .Y(n611) );
  AOI21X4 U351 ( .A0(EMPTY), .A1(POP), .B0(n146), .Y(n288) );
  BUFX8 U352 ( .A(n187), .Y(n603) );
  BUFX8 U353 ( .A(n266), .Y(n617) );
  AOI22X1 U354 ( .A0(n616), .A1(n440), .B0(n601), .B1(n441), .Y(n266) );
  AOI22X1 U355 ( .A0(n601), .A1(n440), .B0(n612), .B1(n441), .Y(n271) );
  ..........
endmodule
```


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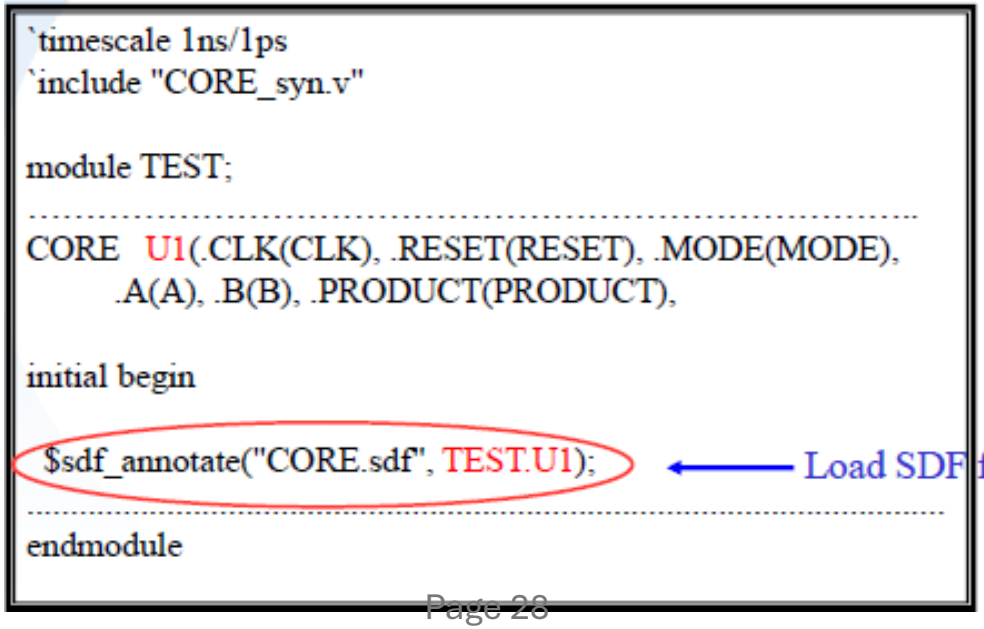
Save Design Timing File (SDF)

- **A SDF file can be generated from Design-Compiler for gate-level simulation. It contains the interconnect delay and gate delay.**
  - Syntax: **write_sdf** DESIGN.sdf



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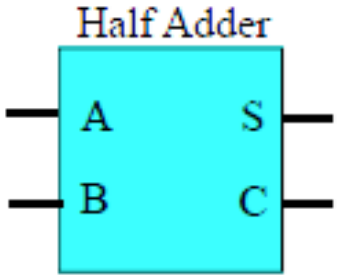
Example SDF File

- **Example of Cell Delay**

```
(CELL
  (CELLTYPE "HA1CHD")
  (INSTANCE i_3/i_63)
   (DELAY
    (ABSOLUTE
    (IOPATH A C  (0.20:0.20:0.20) (0.19:0.19:0.19))
    (IOPATH A S  (0.30:0.30:0.30) (0.26:0.26:0.26))
    (COND B == 0 (IOPATH A S  (0.24:0.24:0.24) (0.31:0.31:0.31)))
    (COND B == 1 (IOPATH A S  (0.30:0.30:0.30) (0.26:0.26:0.26)))
    (IOPATH B C  (0.21:0.21:0.21) (0.18:0.18:0.18))
    (IOPATH B S  (0.28:0.28:0.28) (0.24:0.24:0.24))
    (COND A == 0 (IOPATH B S  (0.21:0.21:0.21) (0.27:0.27:0.27)))
    (COND A == 1 (IOPATH B S  (0.28:0.28:0.28) (0.24:0.24:0.24)))
    )
   )
 )
```



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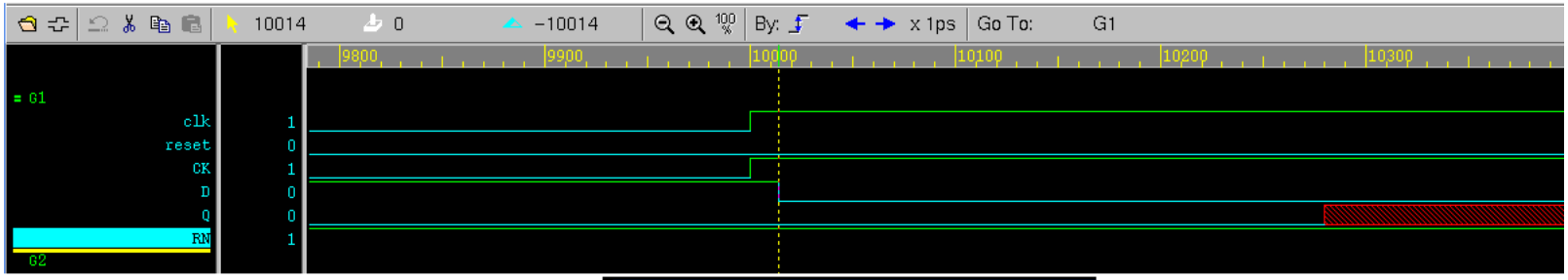
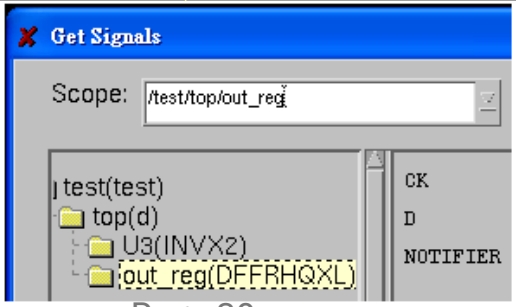
Timing Violation

- **Timing violation happens in gate-level simulation**
  - **$hold → hold time violation ; $setup → setup time violation**

```
Warning!  Timing violation
          $setuphold<hold>( posedge CK &&& (flag == 1):10 NS, negedge D:10014 P
S,  0.15 : 150 PS,  0.02 : 20 PS );
           File: ./umc18.v, line = 8087
          Scope: test.top.out_reg
           Time: 10045 PS
```





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Timing Violation (cont'd)

CHIP.sdf : DFFRHQL(standard cell)

```
(INSTANCE)
(DELAY
  (ABSOLUTE
  (INTERCONNECT B[6] U8/I (0.000:0.000:0.000))
  (INTERCONNECT A[6] U7/I (0.000:0.000:0.000))
  (INTERCONNECT U8/Z U6/I (0.000:0.000:0.000))
  (INTERCONNECT U7/Z U5/I (0.000:0.000:0.000))
  (INTERCONNECT U6/Z U4/I (0.000:0.000:0.000))
  (INTERCONNECT U5/Z U3/I (0.000:0.000:0.000))
  (INTERCONNECT s1/U1/Z sum reg\[0\]/D (0.000:0.000:0.000))
```

Timing check specification of DFFRHQL in file ADFPN16.v

```
BUFFD2BWP16P90LVT U1 ( .I(n14), .Z(n13) );
BUFFD2BWP16P90 U2 ( .I(n6), .Z(n14) );
XOR2D2BWP16P90 U3 ( .A1(Ci), .A2(n8), .Z(n6) );
CKBD1BWP16P90LVT U4 ( .I(n10), .Z(S) );
CKBD1BWP16P90LVT U5 ( .I(n9), .Z(n8) );
CKBD1BWP16P90LVT U6 ( .I(n1), .Z(n9) );
CKBD1BWP16P90LVT U7 ( .I(n11), .Z(n10) );
CKBD1BWP16P90LVT U8 ( .I(n12), .Z(n11) );
```


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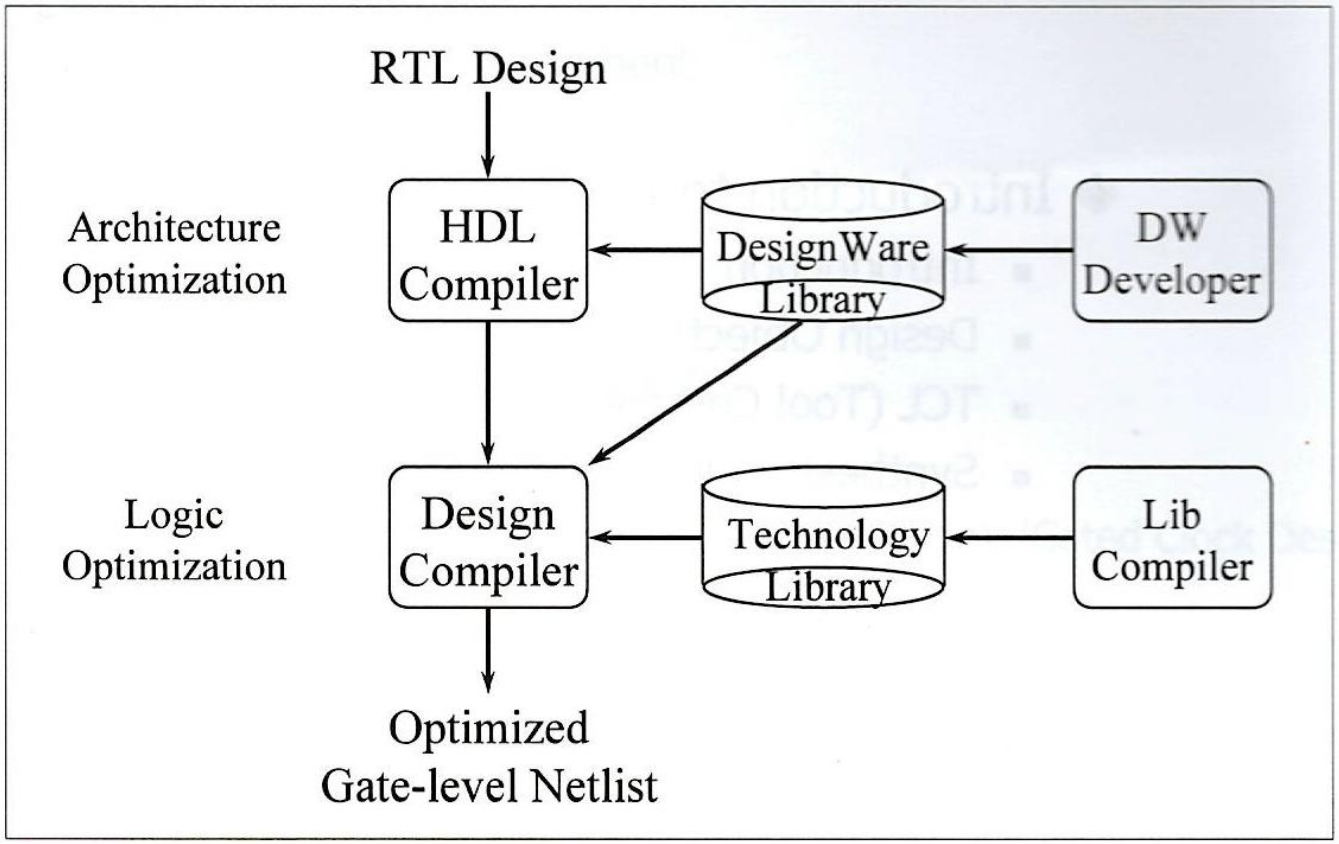
False Timing Violations

- **Following mistakes may cause timing violations**
  - In test pattern, input data varies at wrong clock edge.
  - In test pattern, reset and clock varies at the same time.
  - In test pattern, forgetting to add standard delay format (SDF) file.
- **Those three mistakes are man-made.**

Logic Synthesis Overview



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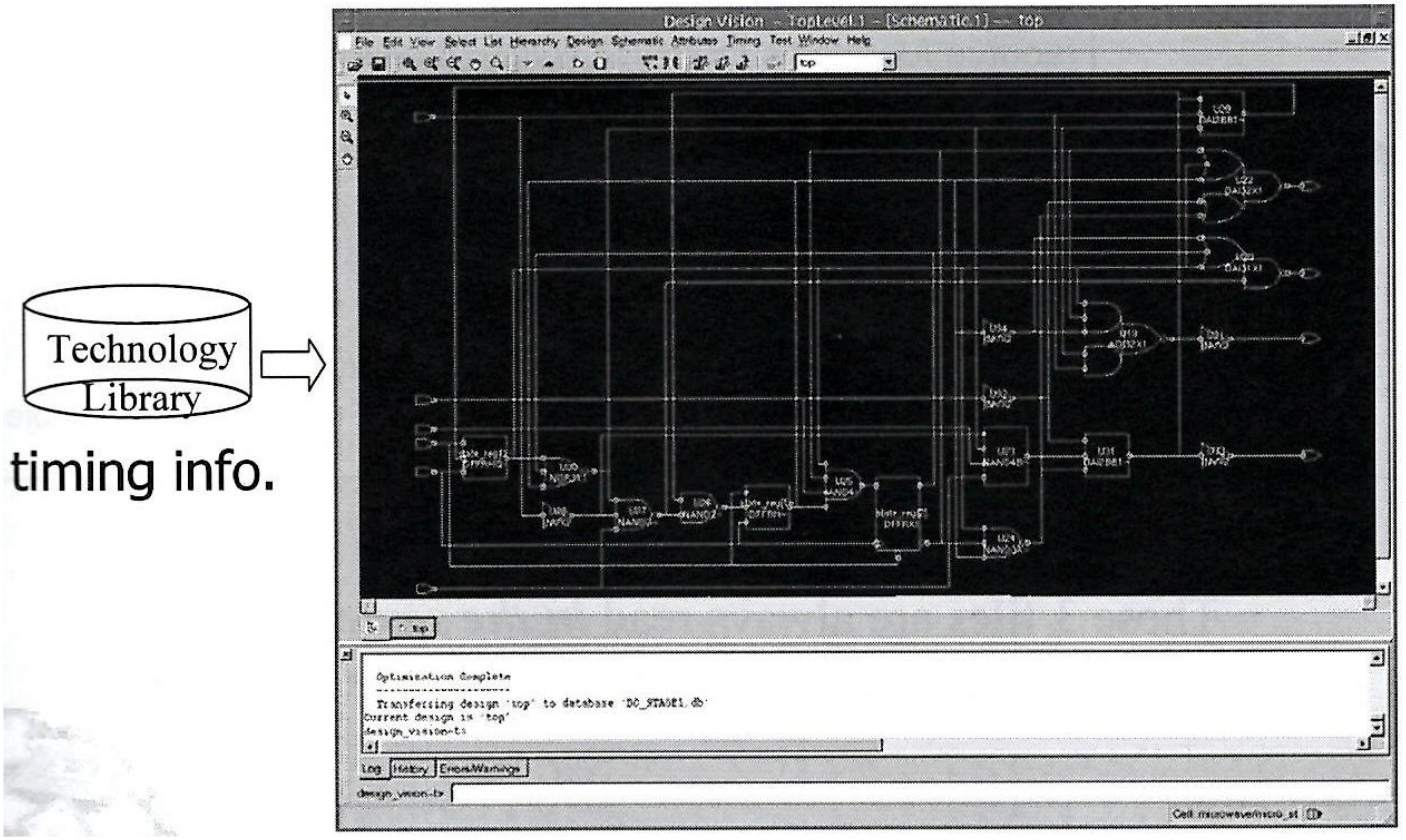
Design Compiler

- Design Compiler maps Synopsys design block to gate level design with a user specified library



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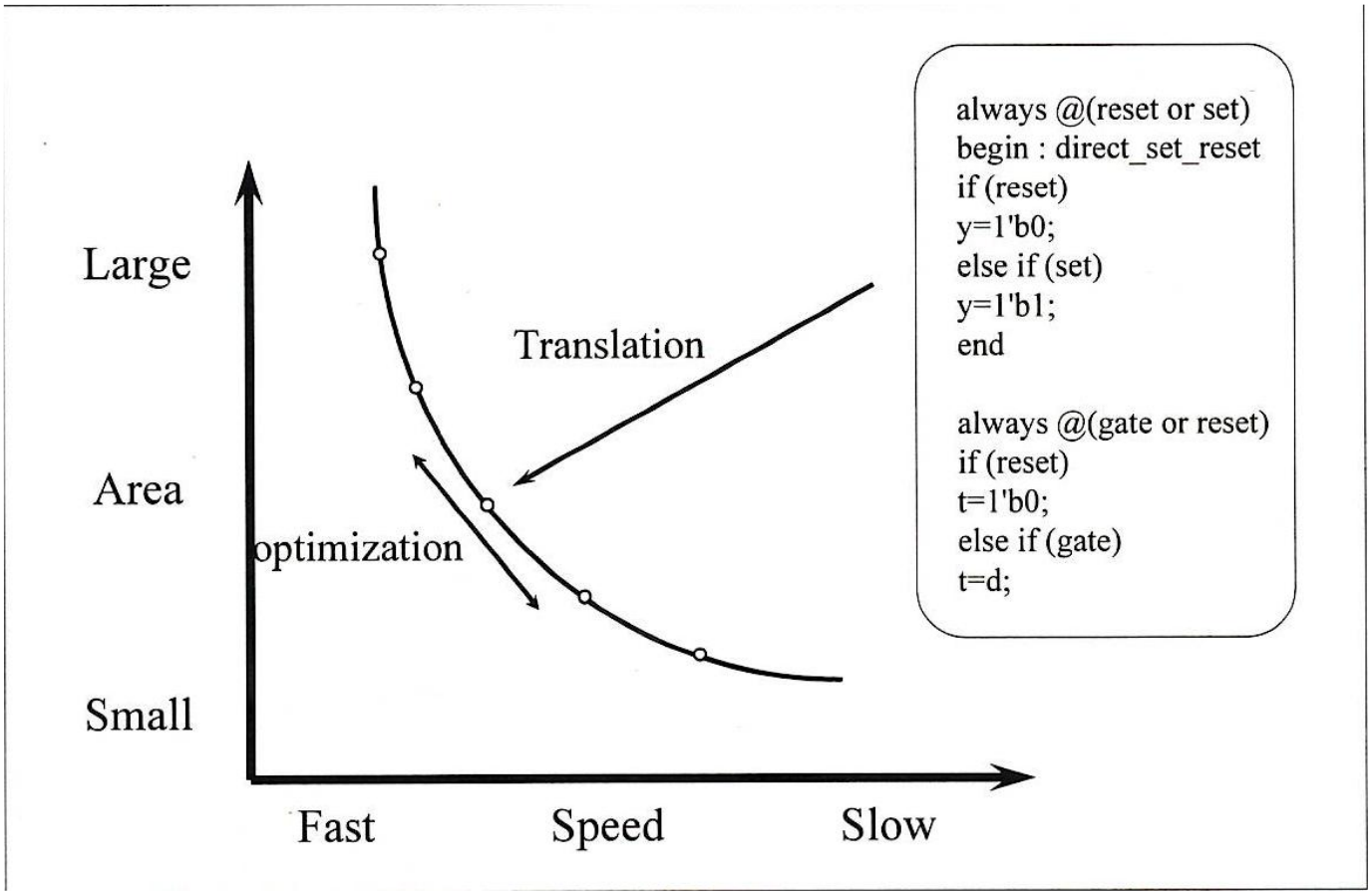
Synthesis is Constraint Driven



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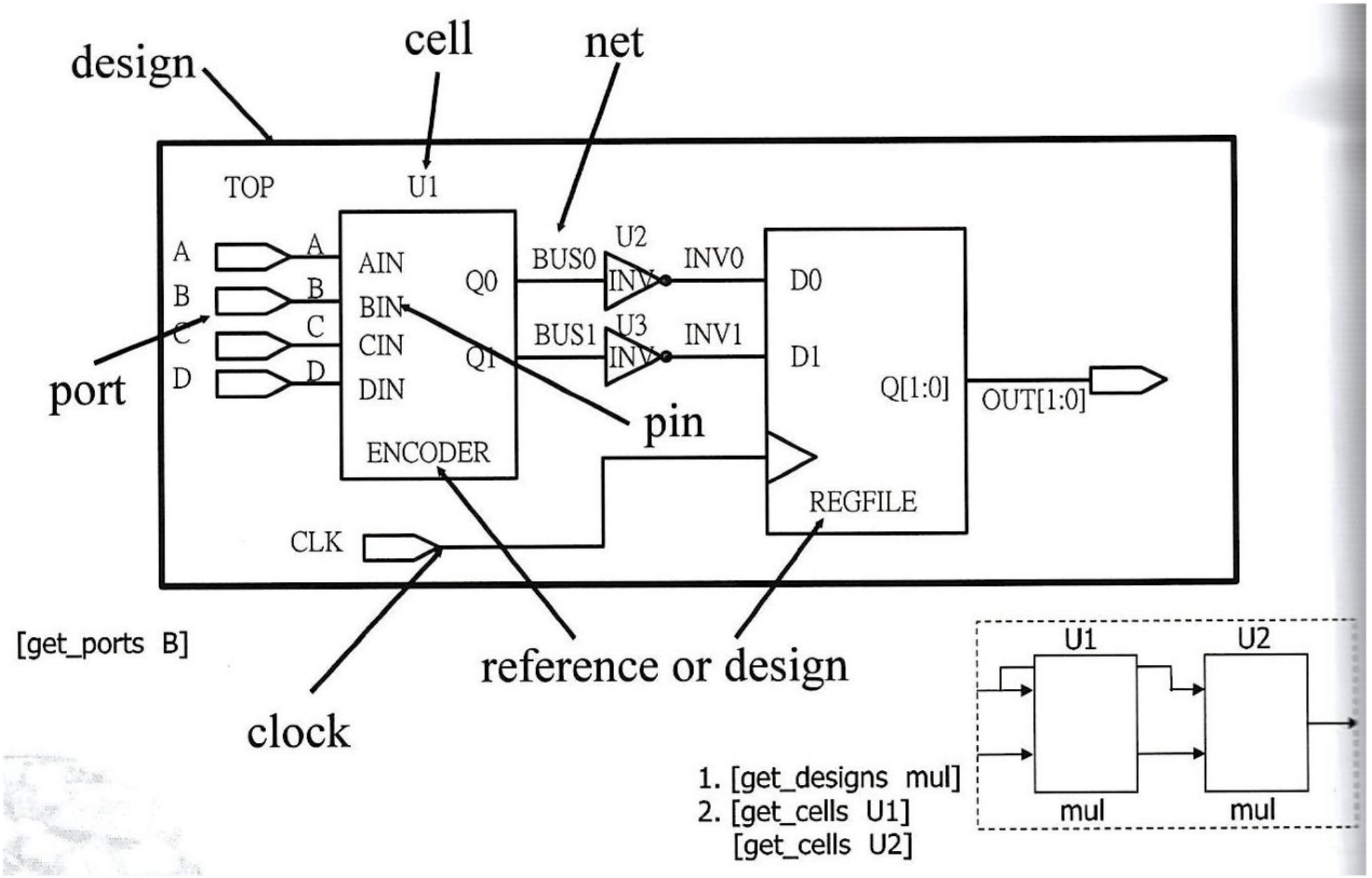
Design Object(Schematic Perspective)



Design Object

- Design: A circuit that performs one or more logical functions
- Cell: An instantiation of a design within another design
- Reference: The original design that a cell "points to"
- Port: The input or output of a design
- Pin: The input or output of a cell
- Net: The wire that connects ports to pins and/or pins to other each other
- Clock: Waveform applied to a port or pin identified as a clock source

Synthesizable Code for Verilog-1995

- Verilog Basis
  - parameter declarations
  - (1) wire, wand, wor declarations
  - reg declarations
  - input, ouput, inout declarations
  - (2) continuous assignments
  - module instantiations
  - gate instantiations
  - (3) always blocks
  - task statements
  - function definitions
  - (4) for, while loop
  - (5) disable
  - (6) if ... else if ... else
  - case, casex, casez

- Synthesizable Verilog primitives cells
  - and, or, not, nand, nor, xor, xnor
  - bufif0, bufif1, notif0, notif1

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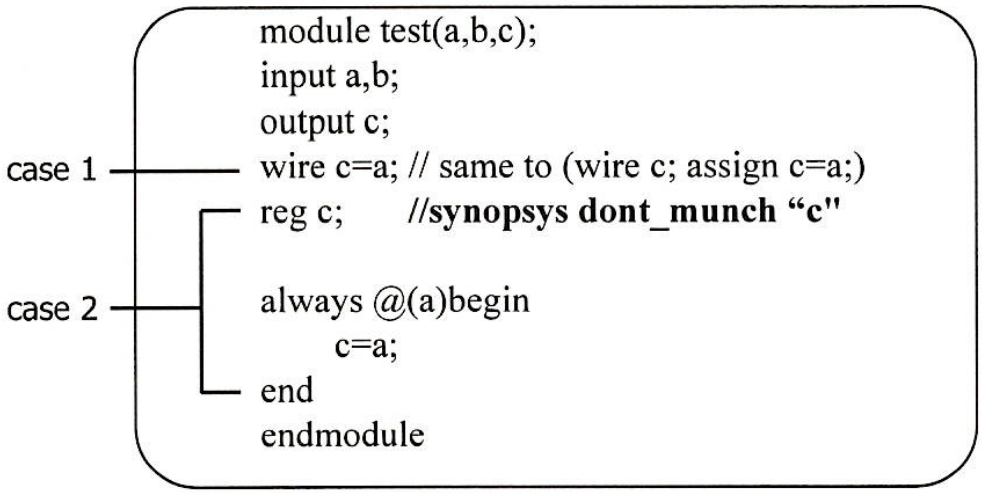
Wire & Register(1/2)

- wire(wand, wor, tri)
  - Physical wires in a circuit
  - Cannot assign a value to a wire within a function or a begin.....end block (ex: always block)
  - A wire does not store its value
  - An undriven wire defaults to a value of Z (high impedance).
  - Input, output, inout port declaration -- wire data type (*default*)

Wire & Register(2/2)

- reg: a variable in Verilog
- Use of "reg" data type is not exactly synthesized to a really register.
- Compare to use of wire & reg
  - wire ➔ usually use "assign" and "assign" *does not* appear in "always" block
  - reg ➔ only use "a=b" , always appear in "always" block



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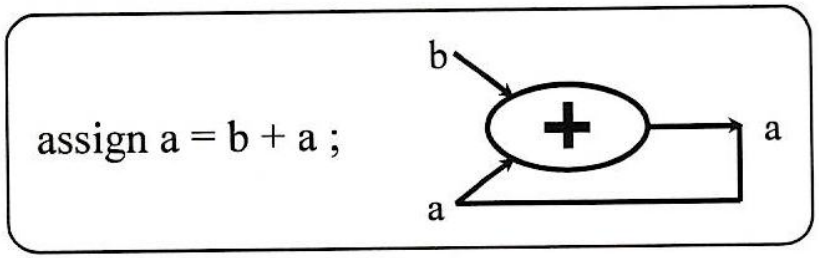
Continuous Assignment

- Avoid logic loop
  - HDL Compiler and Design Compiler will automatically open up asynchronous logic loops
  - Without disabling the combinational feedback loop, the static timing analyzer can't resolve
  - An error example:



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Always Block

- Example

```
always @ (event-expression ) begin
        statements;
end
```

- *Flip-Flop:* if event-expression contains *posedge* or *negedge*
- *Combination Logic:* if event-expression isn't contains *posedge* or *negedge*
- *Latches:* a variable assigned within an always @ block that is not fully specified

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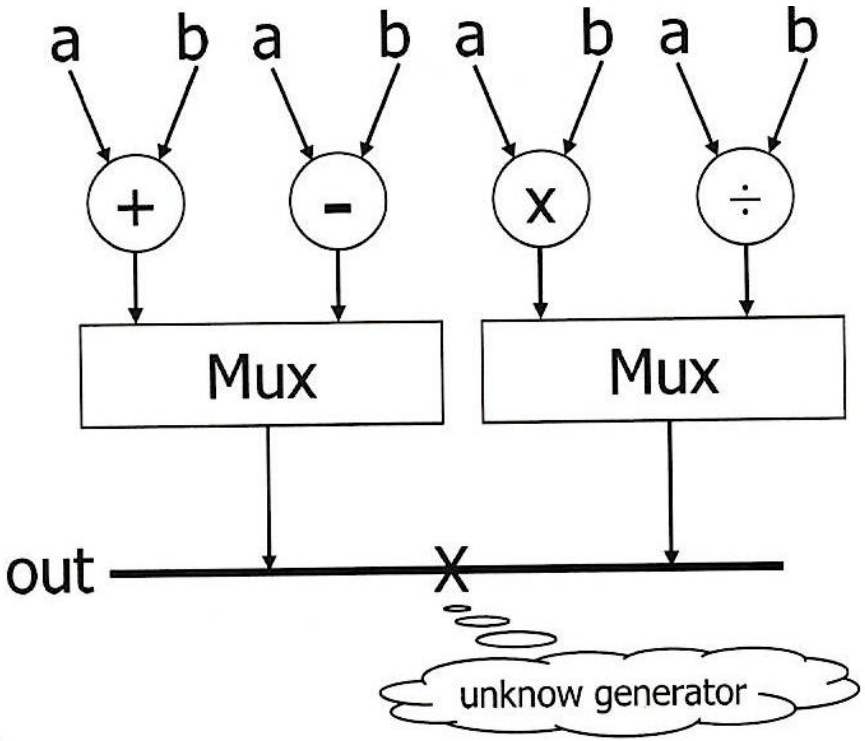
Example for error always syntax

```
module top(clk, a, b, out, …);
output  out;
input   a, b, clk;
reg     out;
          :
          :
always  @(posedge clk)begin
   if(en1)    out=a+b;
   else       out=a-b;
end
          :
          :
always  @(posedge clk)begin
   if(en2)    out=a*b;
   else       out=a/b;
end
          :
endmodule
```



Synthesis Unsupported

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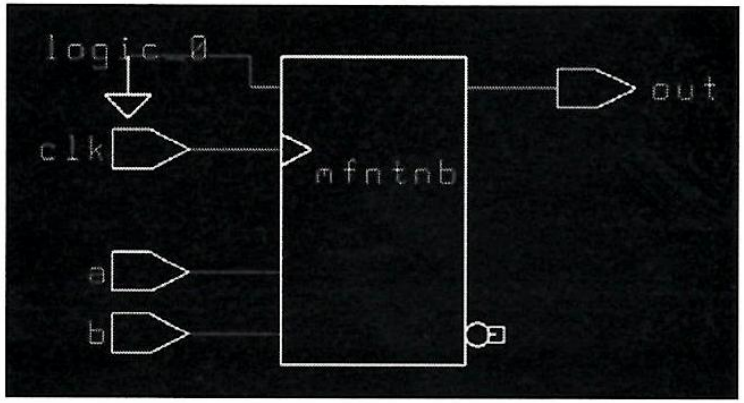
Register Inference(1/2)

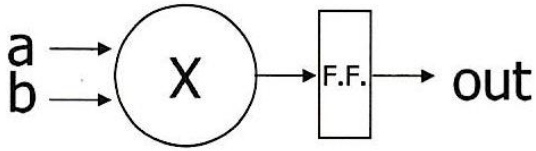
- A register (flip-flop) is implied when you use the *@(posedge clk)* or *@(negedge clk)* in an always block
- Any variable that is assigned a value in this always block is synthesized as a D-type edge-triggered flip-flop

```
always @(posedeg clk)
  out <= a & b ;
```



ex:

```
always @(posedeg clk)
  out <= a * b ;
```



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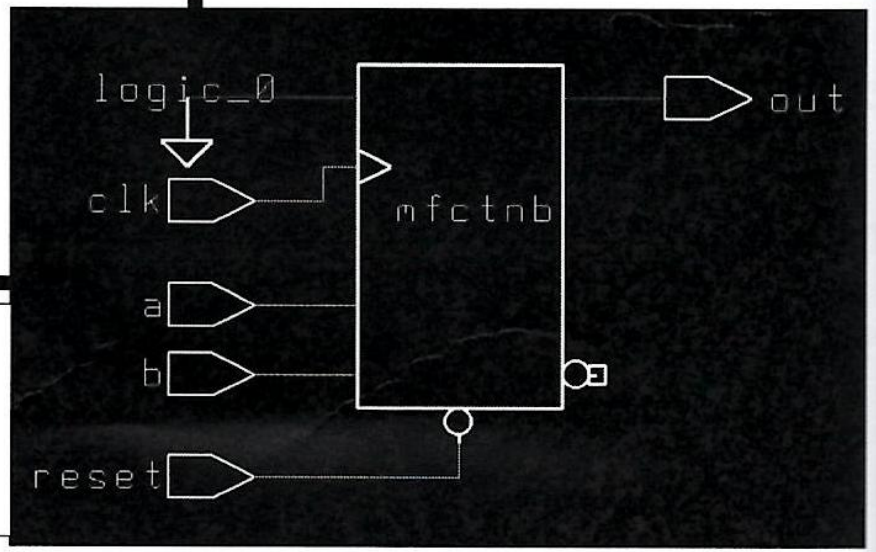
Register Inference(2/2)

- Inference of positive edge clock with active low asynchronous reset flip-flops

```
always @(posedeg clk or negedge reset)
begin
if (!reset)
  out <= 0;
else
  out <= a & b ;
end
```

clocking specified in else



note: if in always block's sensitive list, you use edge trigger signal, then, all signal in this sensitive list must be edge trigger form

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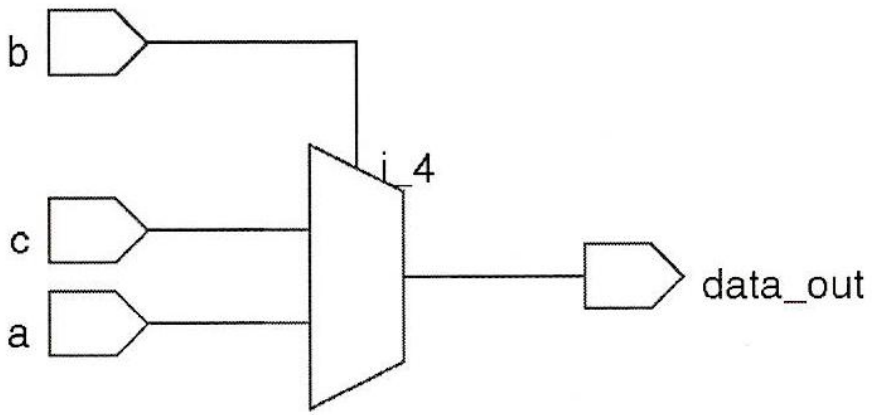
Combinational Logic Inferencr

- Not all variables declared as ***reg*** data type need a flip-flop or a latch for logic implementation

```
reg data_out;
always @(a or b or c)
if(b)
   data_out = a;
else
   data_out = c;
```



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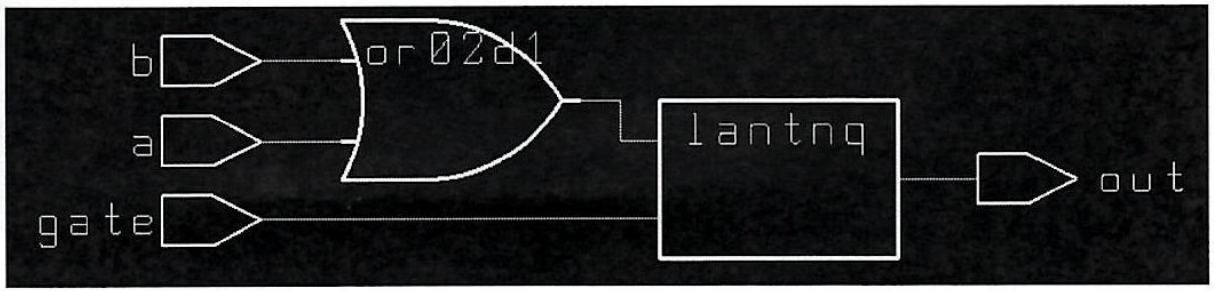
Latch Inference

- A variable assigned within an always block that is *not fully specified*

可以用一個 * 取代全部, **ex: always @(*)**

```
always @(a or b or gate) begin
  if (gate)
     out = a | b ;
end
```

The conditional expression becomes the latch enable



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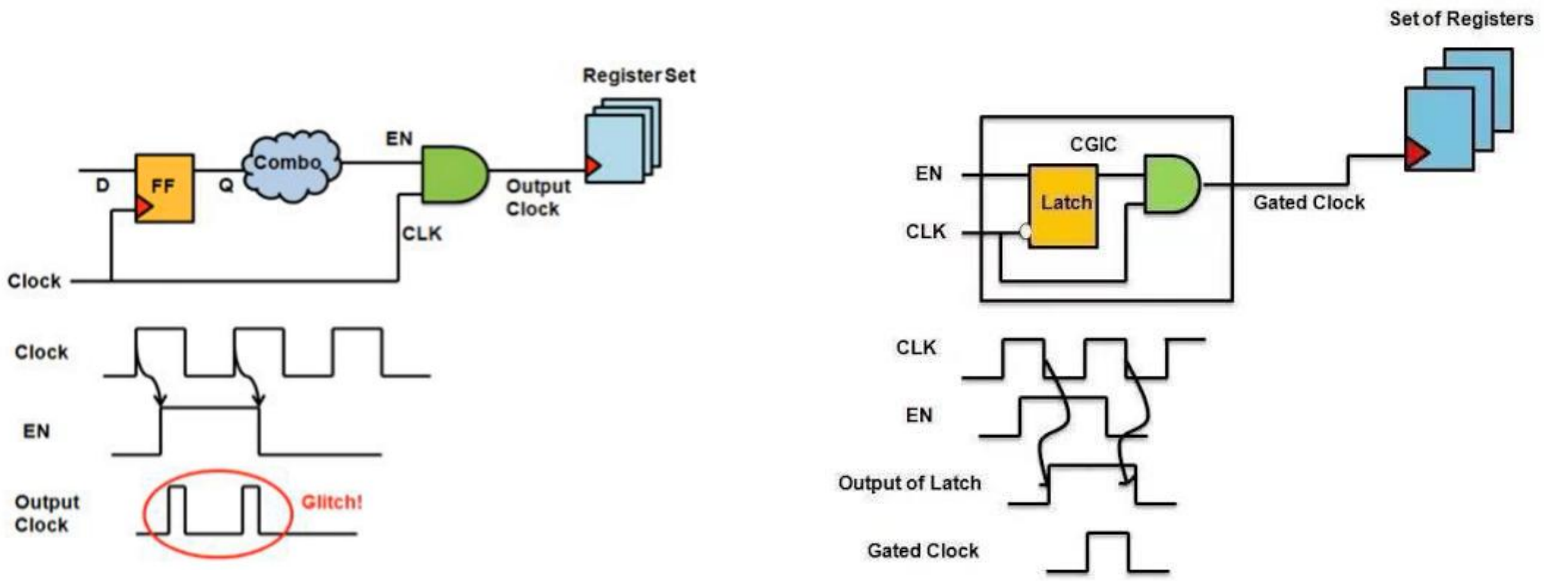
Clock Gating Coding Style



- Dynamic Power Reduction
  - Disables the clock when data does not need updating, significantly reducing switching power consumption in registers and the clock distribution network.
- Glitch Elimination
  - Directly gating the clock with a simple AND gate can easily produce glitches when the enable signal (EN) toggles during a clock high phase, causing functional failure in downstream registers.

Clock Gating Coding Style

If statement

```
always@ (posedge clk)
    if (enable) Q <= D_in;
```

If + Loop statement

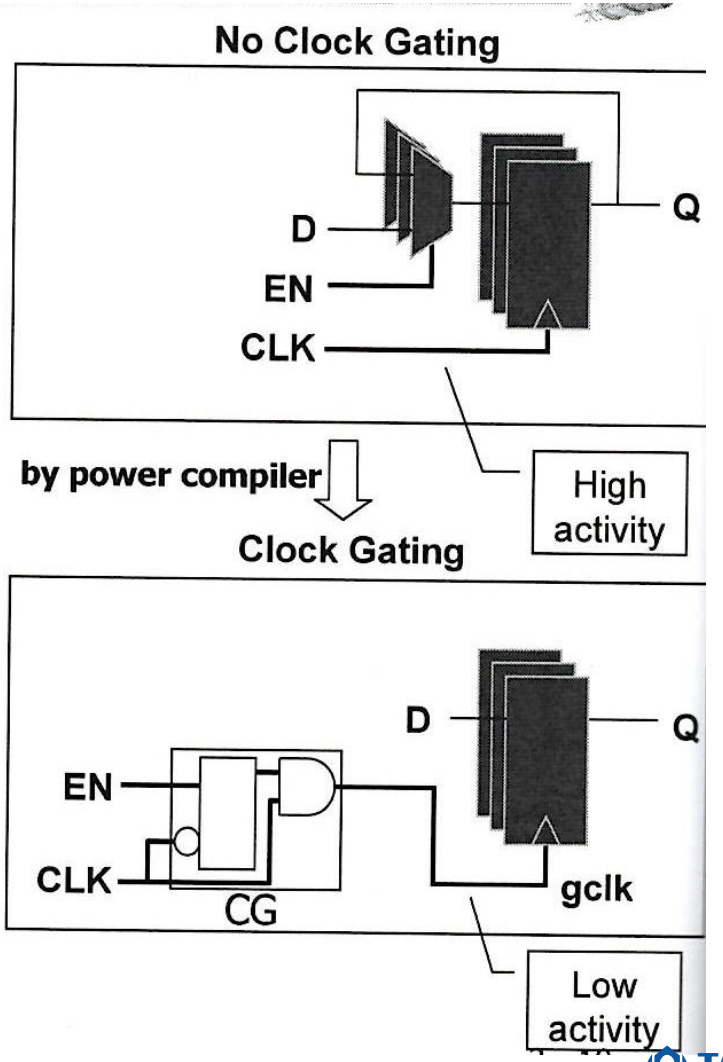
```
always @ (posedge clk)
  if (enable)
    for (i=0; i<8; i=i+1)
      s[i] = a[i] ^ b[i];
```

Conditional Assignment

```
always@ (posedge clk)
    Q <=(enable)? D_in : Q;
```

Case statement

```
always@ (posedge clk)
    case (enable)
        1'b1: Q  <=  D_in;
        1'b0: Q  <=  Q;
    endcase
```



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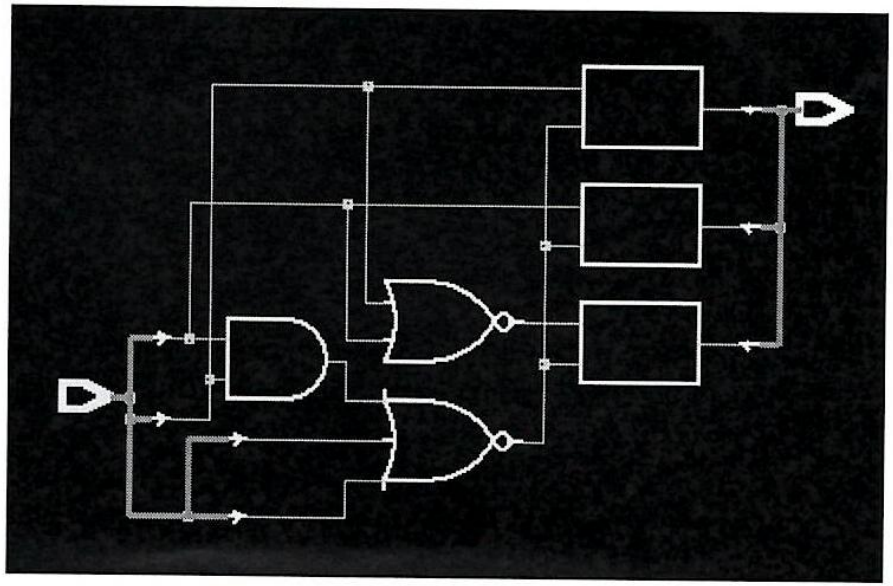
Latch Inference with Case

- If a case statement is not a full case, it will *infer a latch*

```
always @(bcd) begin
  case (bcd)
       4'd0:out=3'b001;
       4'd1:out=3'b010;
       4'd2:out=3'b100;
  endcase
end
```



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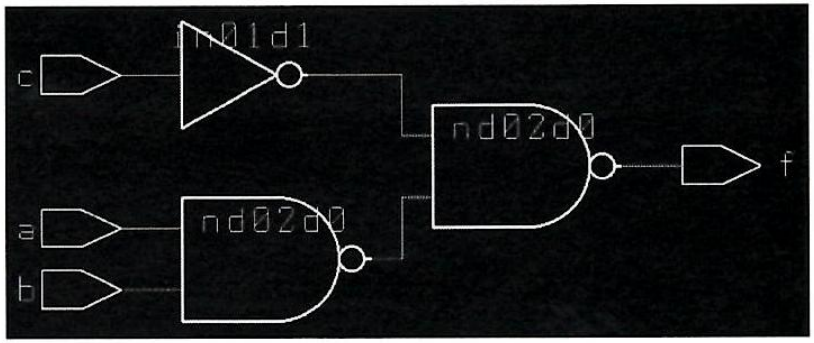
Combinational Always Block

- Sensitivity list must be specified completely, otherwise synthesis may mismatch with simulation

```
always @(a or b or c)
f=a&b|c;
```

```
always @(a or b)
f=a&b|c;
```




```
Warning: Variable 'c' is being read
    in routine train line 6 in file '/raid/raid09/hsieh/synopsys/vhdl_test/train1.v',
    but does not occur in the timing control of the block which begins
    there.  (HDL-180)
```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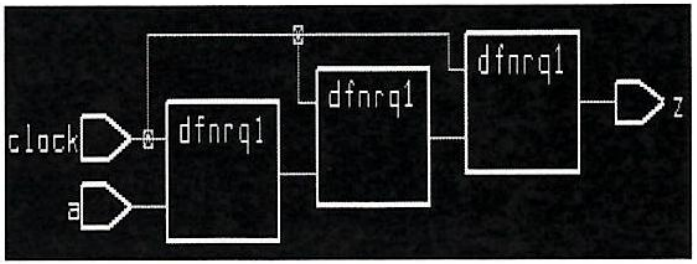
Non-Blocking and Blocking(1/2)

- For the most straightforward code: (you get what you expect)
  - Use *non-blocking* assignments within sequential always block.
  - Example:

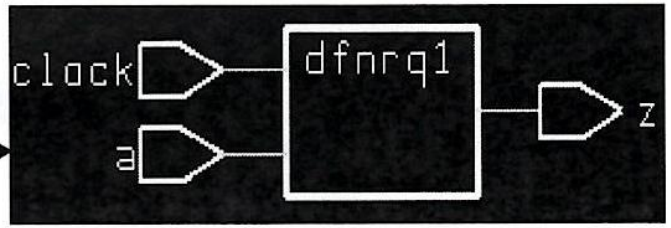
```
always @(posedge clock) begin
    x <= a;
    y <= x;
    z <= y;
end
```

Usually you expect




```
always @(posedge clock) begin
    x = a;
    y = x;
    z = y;
end
```

May not you expect



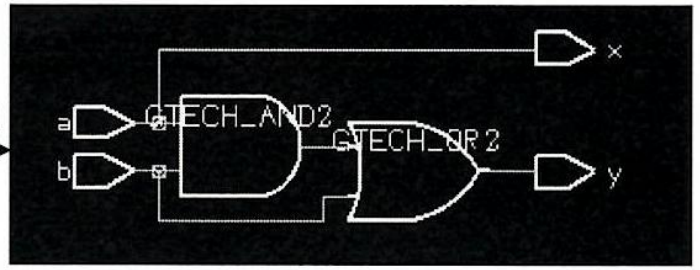
TAIWAN TECH

Non-Blocking and Blocking(2/2)

- For the most straightforward code: (you get what you expect)
  - Use *blocking* assignments within combinational always block.
  - Example:

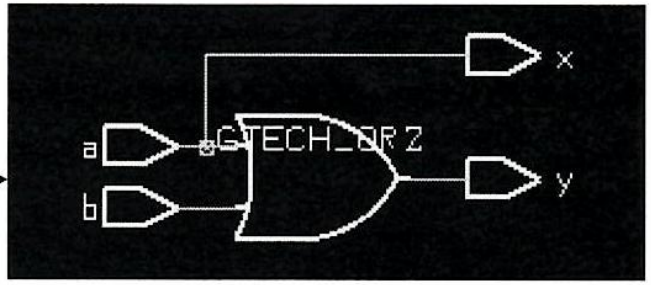
```
always @(a or b or x) begin
   x = a & b;
   y = x | b;
   x = a;
end
```

Usually you expect




```
always @(a or b or x) begin
   x <= a & b;
   y <= x | b;
   x <= a;
end
```

May not you expect



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For Loop

- Provide a shorthand way of writing a series of statements
- Loop index variables must be integer type
- Step, start & end value must be constant
- In synthesis, for loops loops are "*unrolled*", and then synthesized
- Example

```
always @(a or b) begin
  for( k=0; k<=3; k=k+1 ) begin
            out[k]=a[k]^b[k];
            c=(a[k]|b[k])&c;
  end
end
```

→

```
out[0] = a[0]^b[0];
out[1] = a[1]^b[1];
out[2] = a[2]^b[2];
out[3] = a[3]^b[3];
c = (a[0] | b[0]) & (a[1] | b[1]) &
     (a[2] | b[2]) & (a[3] | b[3]) & c
```


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disable

- DC supports the disable statement when you use it in named blocks and when it is used to disable an enclosing block.
- When a disable statement is excuted, it causes the named block to terminate.

```
always @(posedge clock) begin: compare
   for (i = 7; i >= 0; i = i - 1) begin
      if (a[i] != b[i]) begin
           greater_than = a[i];
           less_than = ~a[i];
           equal_to = 0;
           //comparison is done so stop looping
           disable compare;
      end
   end
// If you get here a == b
// If the disable statement is executed, the next three
// lines will not be executed
   greater_than = 0;
   less_than = 0;
   equal_to = 1;
end
```


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Normal case & if… else statement

- Provide for more complex conditional actions, each condition expression controls a multiplexer
- legal only in function & always construct Syntax

```
always @(*) begin
  if ( expr ) begin
     ... statements ...
  end
  else  begin
     ... statements ...
   end
end
```

```
always @(*) begin
  if ( expr ) begin
    case
      ... statements ...
    endcase
  end
  else  begin
   case
      ... statements ...
    endcase
  end
end
```

```
always @(*) begin
  case (CS)
   init: begin
      if ( event1) begin   ... end
      else          begin   … end
      temp =(event2)? 1'b1:1'b0;
    end
      :
  endcase
end
```


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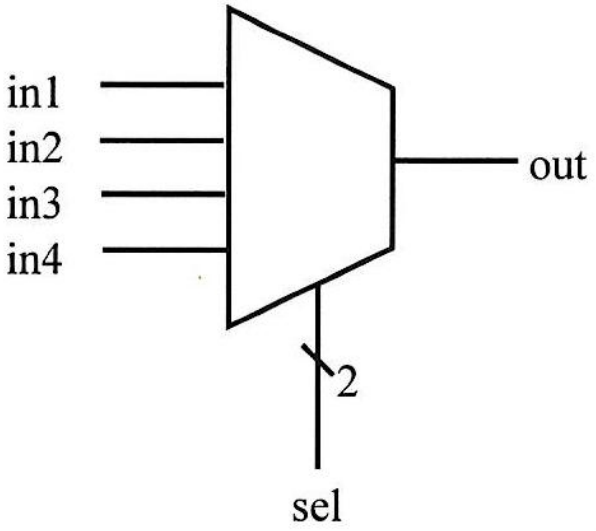
Conditional Operator

- The value assigned to LHS is the one that results TRUE from the expression evaluation
- This can be used to model multiplexer
- Can be nested

```
assign out = ( sel == 2'b00) ? in1 :
             (sel == 2'b01 ) ? in2 :
             (sel == 2'b10 ) ? in3 :
             (sel == 2'b11 ) ? in4 :
                  1'bx;
```



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casex v.s. casez statement

- casex: allows ? and x and z bits in casez items.
- casez: allows ? and z bits in casez items but not in expressions.

Ex:

```
always @ (opcode or a or b or c)
casex(opcode)
  4'b1zzx : begin  // Don't care  2:0 bits
              out = a;
              $display("@%0dns 4'b1zzx is selected, opcode %b",$time,opcode);
            end
  4'b01?? : begin  // bit 1:0 is don't care
              out = b;
              $display("@%0dns 4'b01?? is selected, opcode %b",$time,opcode);
            end
  4'b001? : begin  // bit 0 is don't care
              out = c;
              $display("@%0dns 4'b001? is selected, opcode %b",$time,opcode);
            end
  default : begin
              $display("@%0dns default is selected, opcode %b",$time,opcode);
            end
endcase
```


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Finite State Machine (1/2)

- Finite State Machine Structure:

```
module FSM(…..);
input clk, reset;
                :
/* Define states and encodings */
parameter [2:0]
        Grant_A=3'b001, Wait_A=3'b011, Timeout_A1=3'b111,
        Grant_B=3'b010, Wait_B=3'b110, Timeout_B1=3'b101;
reg [2:0]  current_state, next_state;
                :
always @ (current_state or data1 or data2)  begin
                :
end
always @(posedge clk or negedge reset)begin
                :
end

endmodule
```


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Finite State Machine (2/2)

- Use explicit FSM to describe asynchronous reset FSM
- Use if or case statement to allow compact description of state machine logic

```
always @(current_state or data1 or data2) begin
        case (current_state)
         S0: begin
              result = data1;
                 :
              next_state = S1;
              end
         S1: begin
              :
        endcase
end
```

use 1st always block to describe combinational logic

```
always @(posedge clk or negedge reset) begin
        if (!reset)
          current_state <= S0;
        else
          current_state <= next_state;
end
```

use 2nd always block to synthesize state vector, to describe state transition

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Example for merging two always block to one

```
`define DEL    1
module mem_test(clk,reset,BistMode,Q,BistFail);
input          clk, reset, BistMode;
output [7:0]   Q;
output         BistFail;
reg    [1:0]   CS,NS;
  :
parameter [1:0]  C_WRITE=2'd1,  C_READ=2'd2, OVER=2'd3;
  :
always @(posedge clk or negedge reset) begin
   if(!reset) begin   #`DEL NS=C_WRITE;   A=0;  D=0; end
   else begin         #`DEL CS=NS;            D=D+1;
     case(CS)
      C_WRITE:begin
              if(A<511)  begin A=A+1;  NS=C_WRITE; end
              else       begin A=1;    NS=C_READ;  end
              WEB=1'b0;  CSB=1'b1;    OE=1'b0;
            end
      C_READ:begin
              if(A<511)  begin A=A+1;  NS=C_READ;  end
              else       begin A=1;    NS=OVER;    end
              WEB=1'b1;  CSB=1'b1;    OE=1'b1;
            end
      OVER:begin
              A='d511;   WEB=1'b0;   CSB=1'b0;   OE=1'b0;
              NS=OVER;
            end
     endcase
   end
endmodule
```


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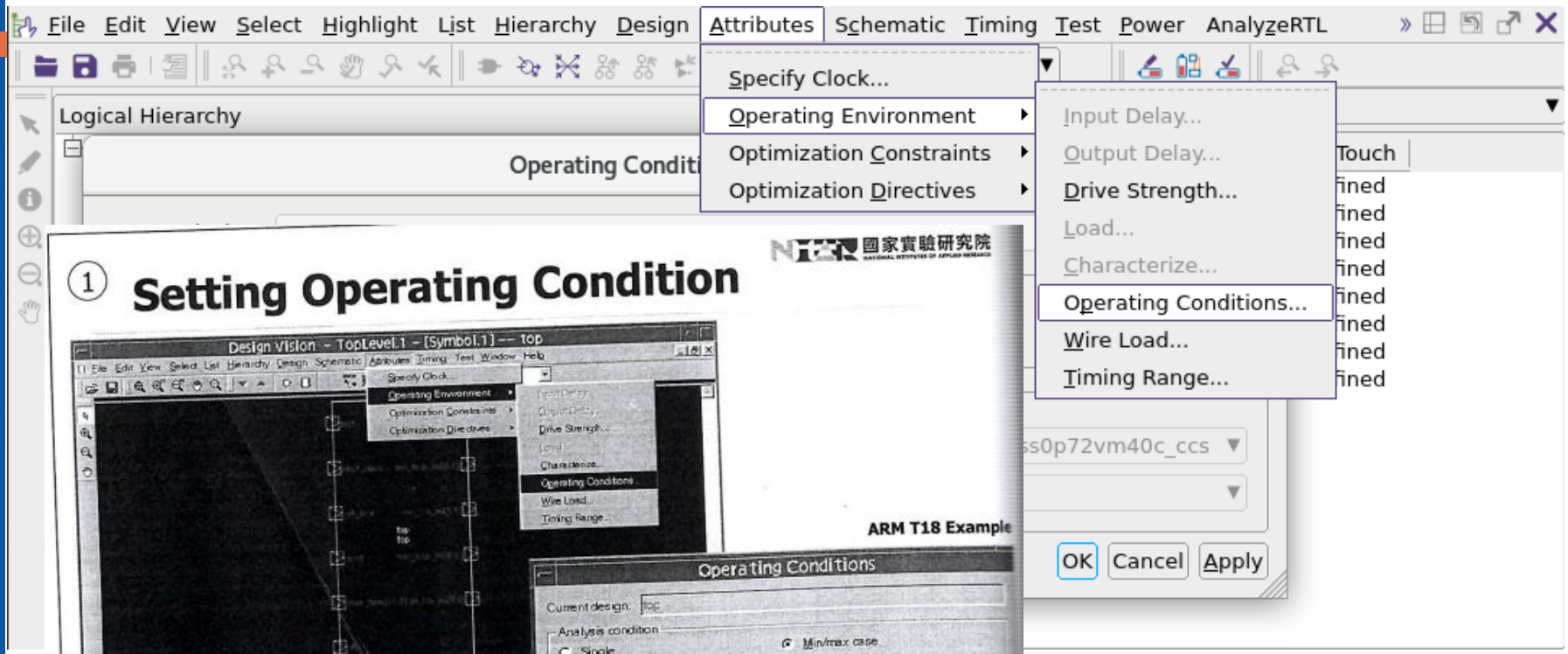
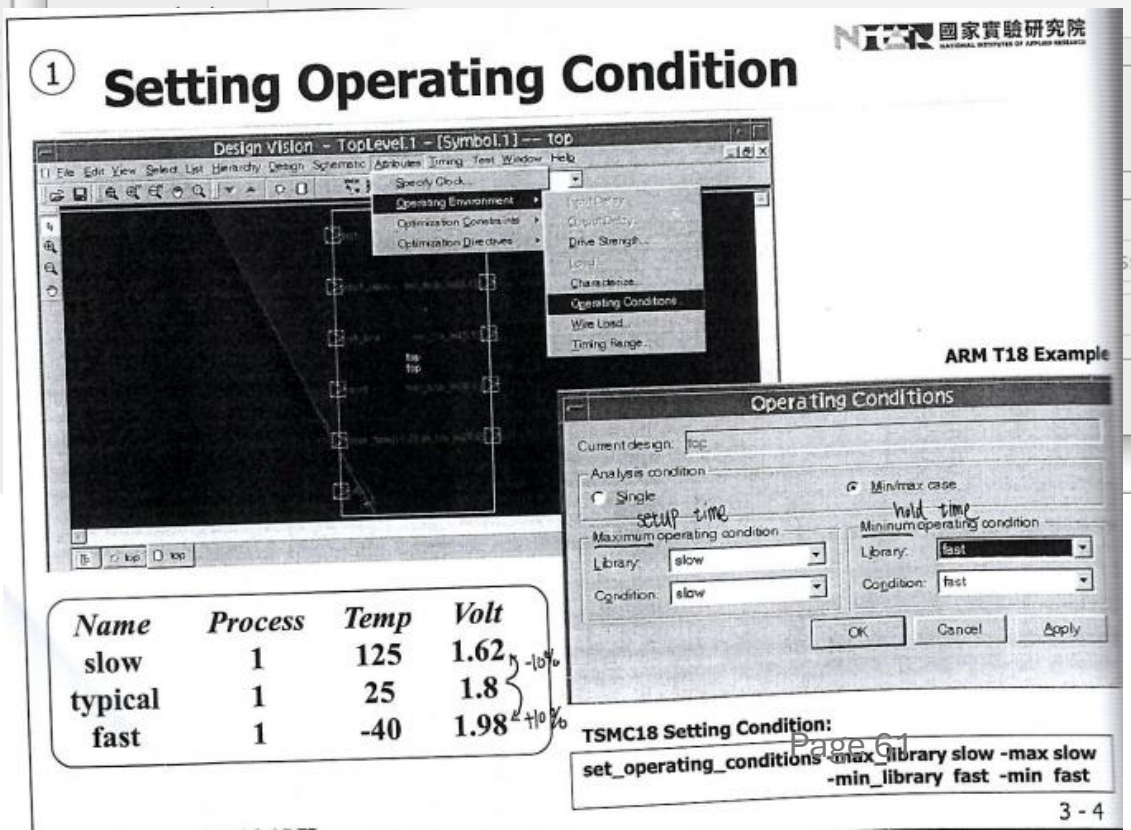
Setting Operating Condition





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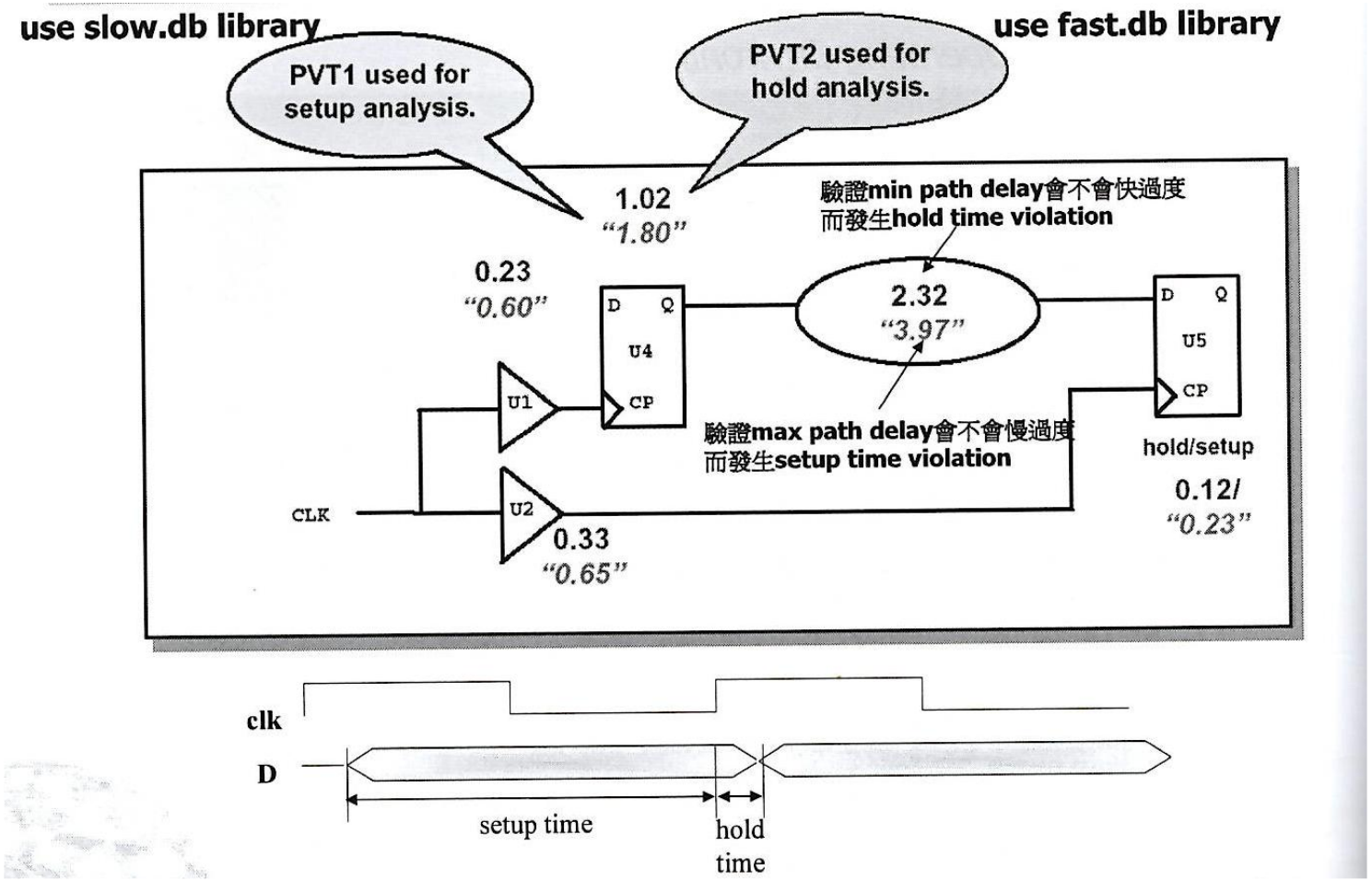
Static Timing Analysis(STA) with Min/Max Analysis



Operating Condition(ex: ADFP 16)

Example for T16 (FinFET process)

T16 NAND: 0.72V @125C

Output Capacitance	Input Transition		
	0.0018	0.0049	0.0112
0.004	0.0032	0.0042	0.0058
0.011	0.0040	0.0050	0.0068

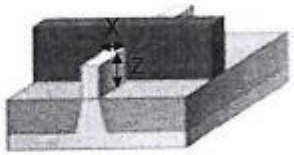
Planar

Width = W

-40C: slower

T16 NAND: 0.72V @ -40C

Output Capacitance	Input Transition		
	0.0018	0.0049	0.0112
0.004	0.0034	0.0044	0.0061
0.011	0.0044	0.0055	0.0074



FinFET (1 Fin)

Width = x + 2z

S.S. WANG / 2026 ADFP 3 - 6

Setting Design Environment

- Setting Operating Condition
- Setting Input Driving Strength
- Setting Output Loading
- Setting Input/Output Delay
- Setting Wire Load Model
- No WLM: Topographical Technology

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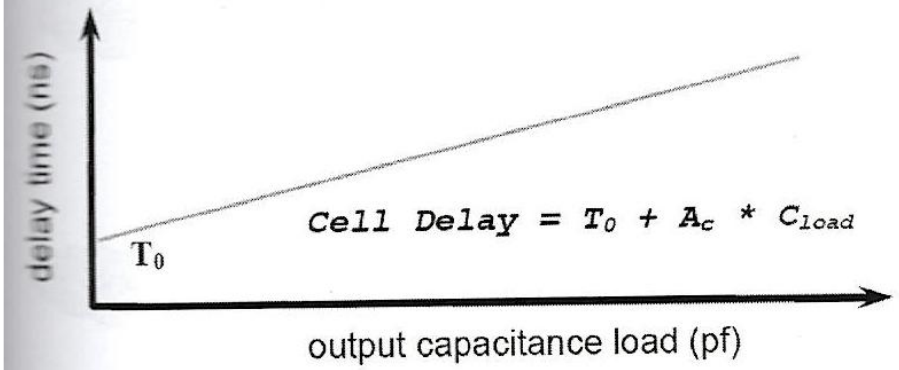
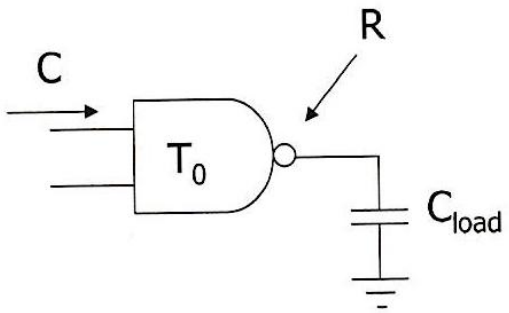
Input Drive Impedance (Linear Cell Delay Model)

- $T_{delay} = T_0 + A_c * C_{load}$
  - T_0 : cell pin to pin intrinsic delay
  - A_c : drive impedance *(unit: ns/pf)*



T_0 : cell pin to pin intrinsic delay
(delay without any loading)
A_c : drive impedance



TSMC18/90: slow.lib

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Non-linear Cell Delay Model

- Cell delays are calculated from tables in the technology library:
  - The tables are commonly indexed by input transition versus output load
  - Cell output transition is also calculated from a table

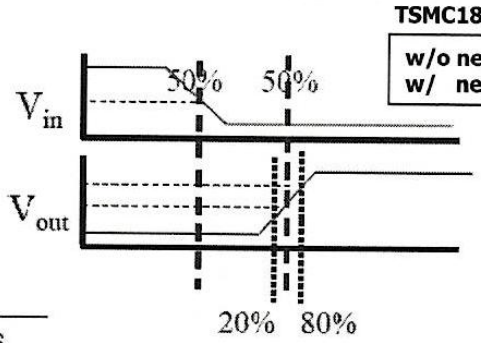
$$D_{cell(I2)} = f(D_{transition(I1)}, C_{eq})$$

$$D_{transistion(I2)} = g(D_{transition(I1)}, C_{eq})$$

slow.lib

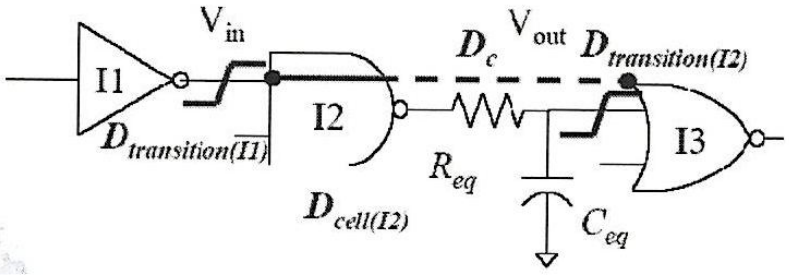
Output Capacitance	Input Transition 0	0.5	1
0.1	0.123	0.234	0.456
0.2	0.222	0.432	0.801

Delay Table



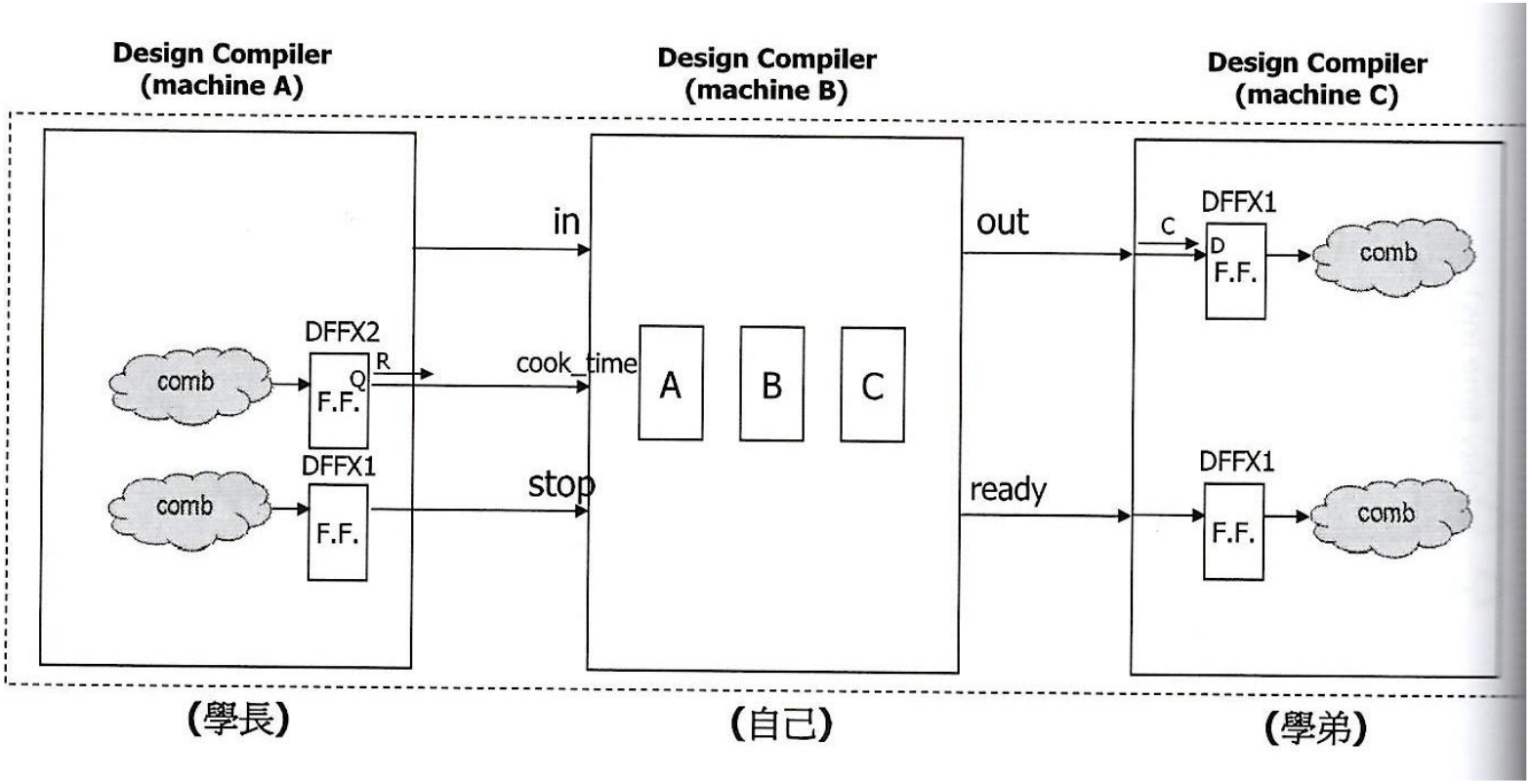
TSMC18

w/o negative delay time: tsmc18.v
w/ negative delay time: tsmc18_neg.v



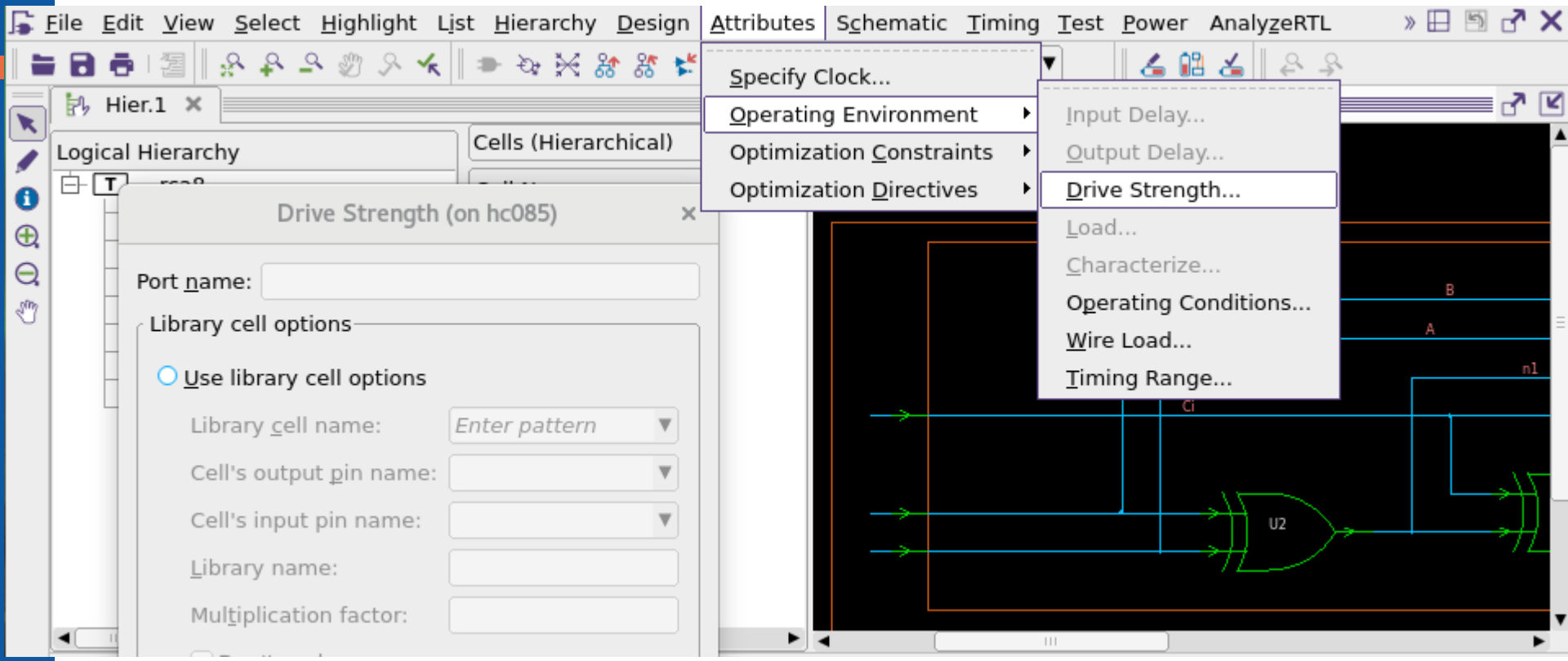
TAIWAN TECH

Example:Driving Strength/loading



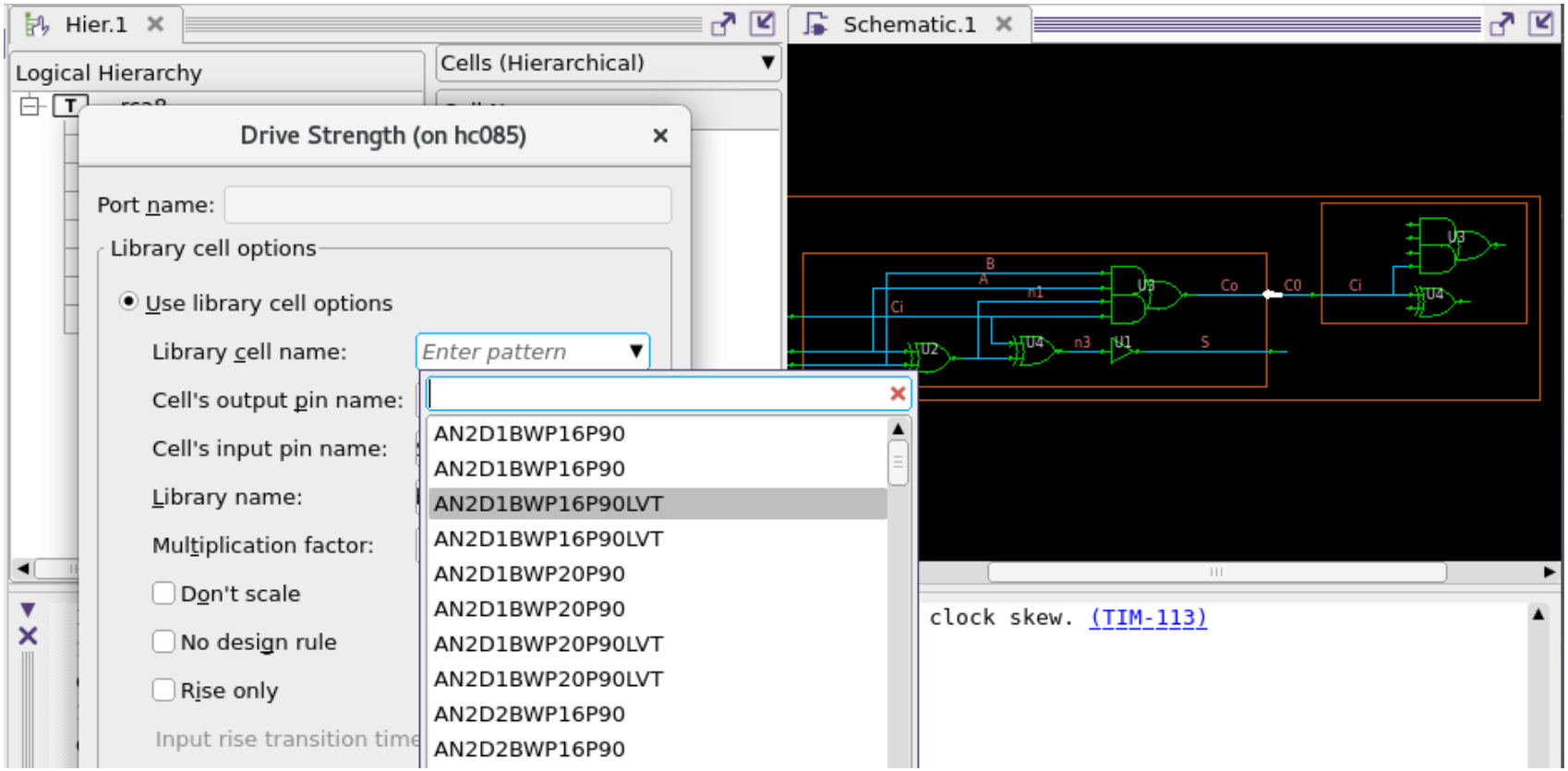
TAIWAN TECH

Setting Input Drive Impedance



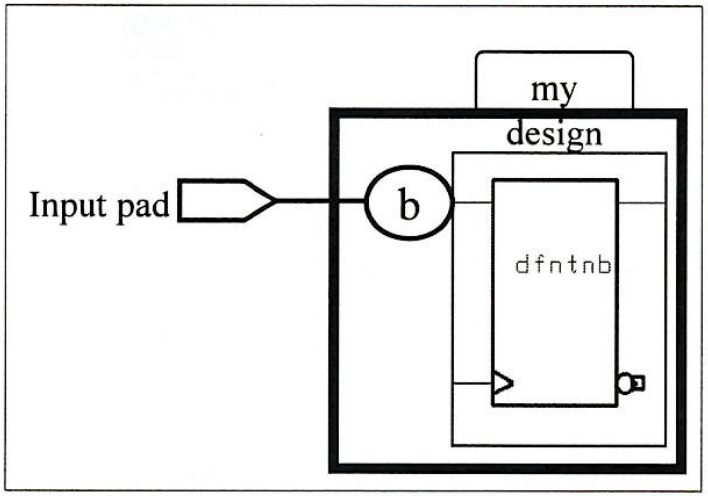
TAIWAN TECH

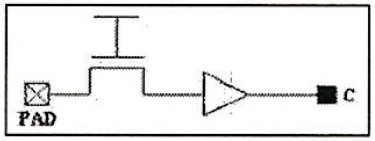
Setting Input Drive Impedance

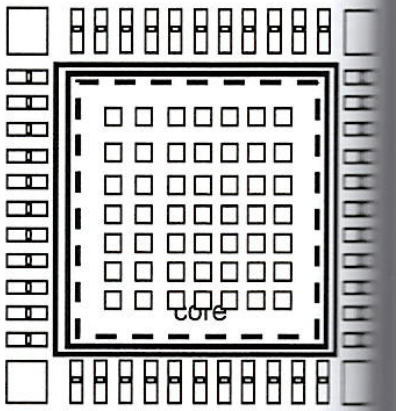


Input Drive Strength for Pads

- If your design is as follow
- Assumed that we use input pad PDIDGZ (PDIDGZ_33), by the list as follow, we can set the input drive strength as:








```
set_driving_cell -library tpz973gvwc -lib_cell PDIDGZ -pin {C} [get_ports cook_time]
```

補充: Appoint all primary input ports: [all_inputs]

```
set_driving_cell -library tpz973gvwc -lib_cell PDIDGZ -pin {C} [all_inputs]
```

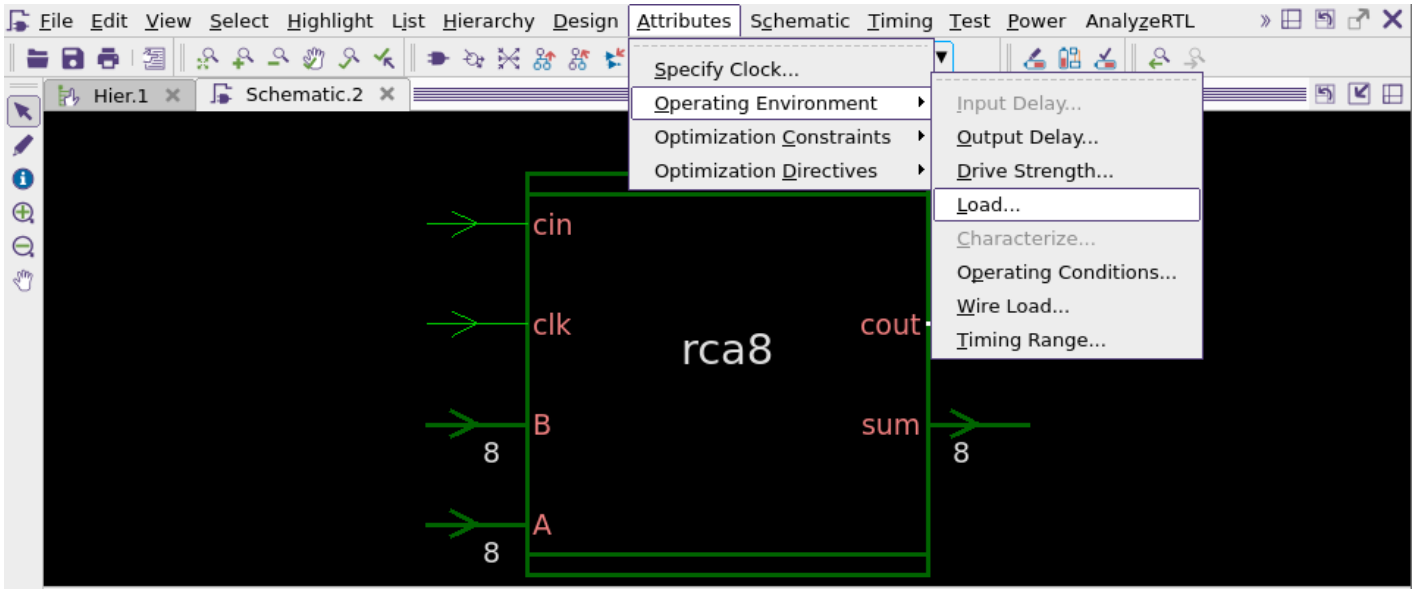

TAIWAN TECH

Setting Design Environment

- Setting Operating Condition
- Setting Input Driving Strength
- Setting Output Loading
- Setting Input/Output Delay
- Setting Wire Load Model
- No WLM: Topographical Technology

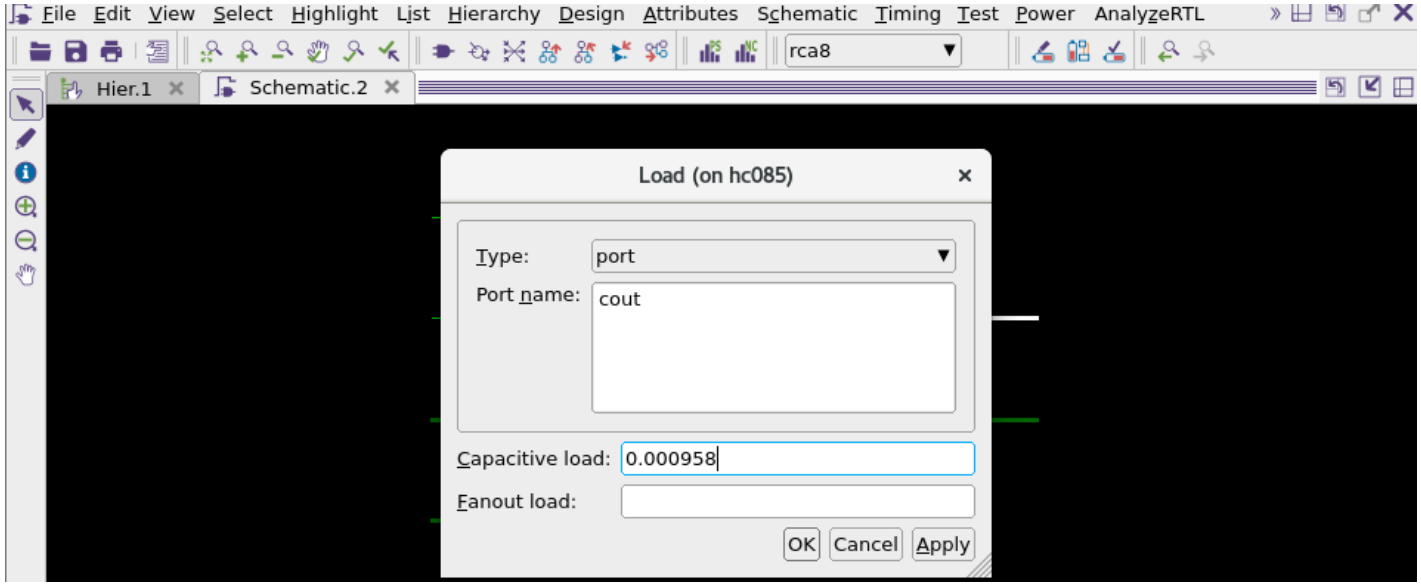
TAIWAN TECH

Setting Output Loading

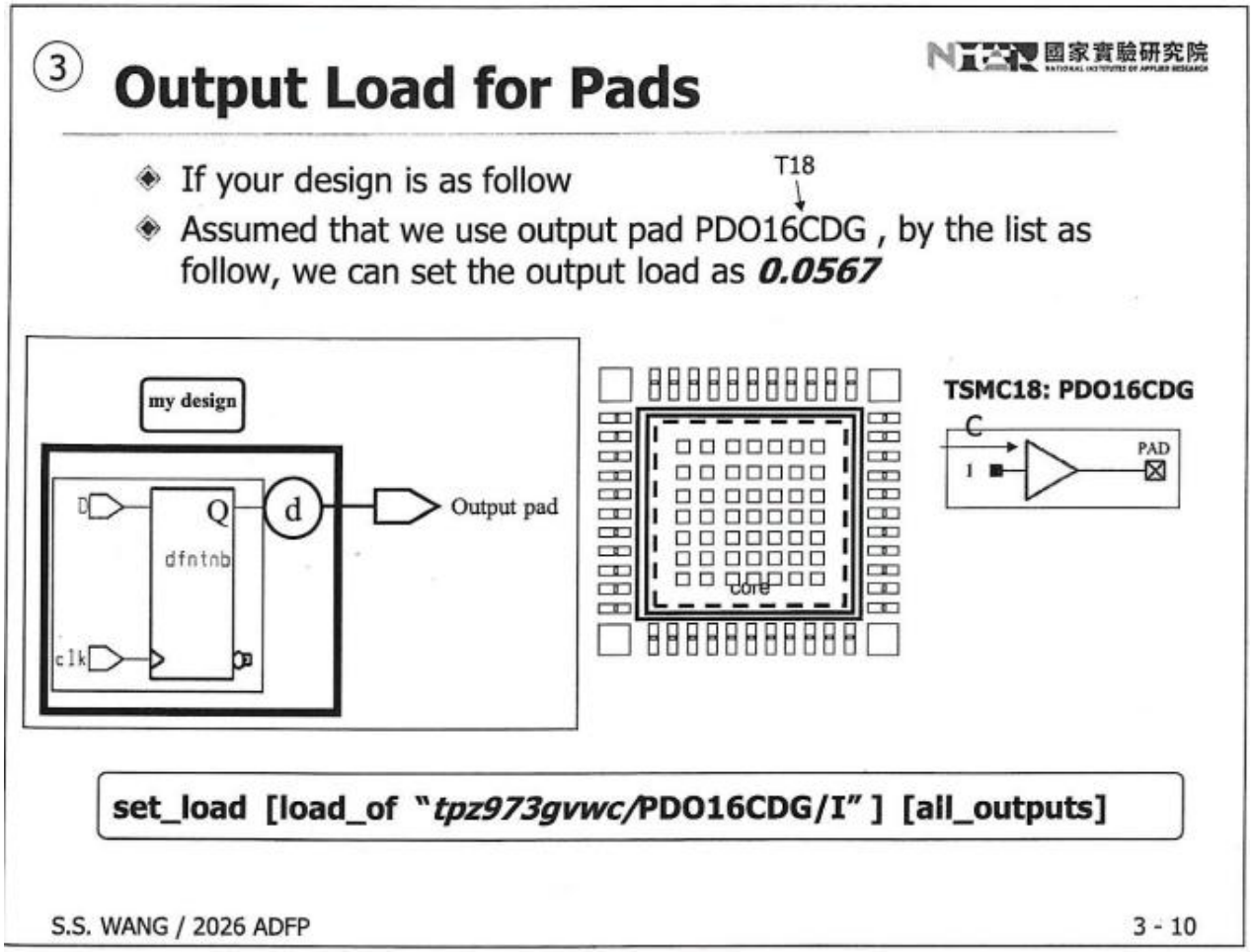


TAIWAN TECH

Setting Output Loading



Output Load for Pads



TAIWAN TECH

Setting Design Environment

- Setting Operating Condition
- Setting Input Driving Strength
- Setting Output Loading
- Setting Input/Output Delay
- Setting Wire Load Model
- No WLM: Topographical Technology

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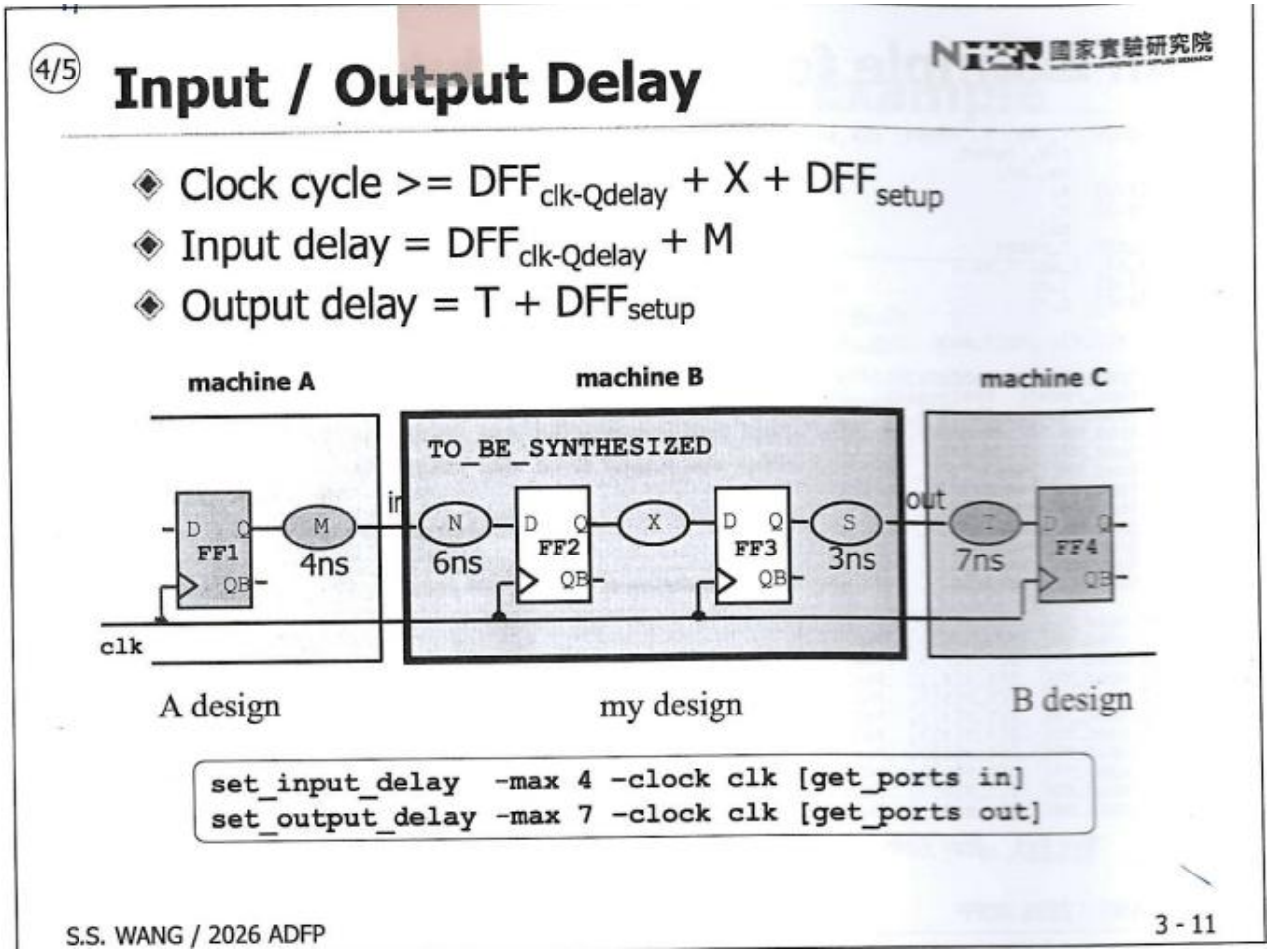
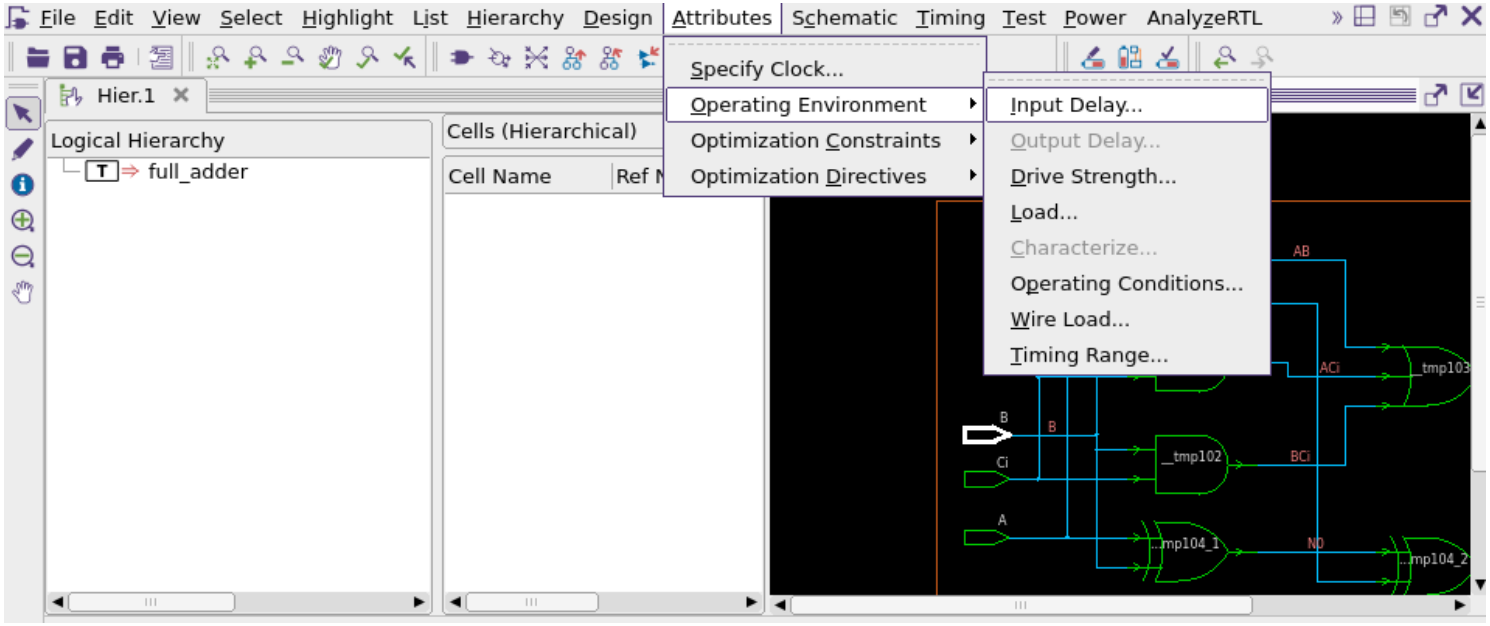
Input / Output Delay



Setting Input Delay(1/2)



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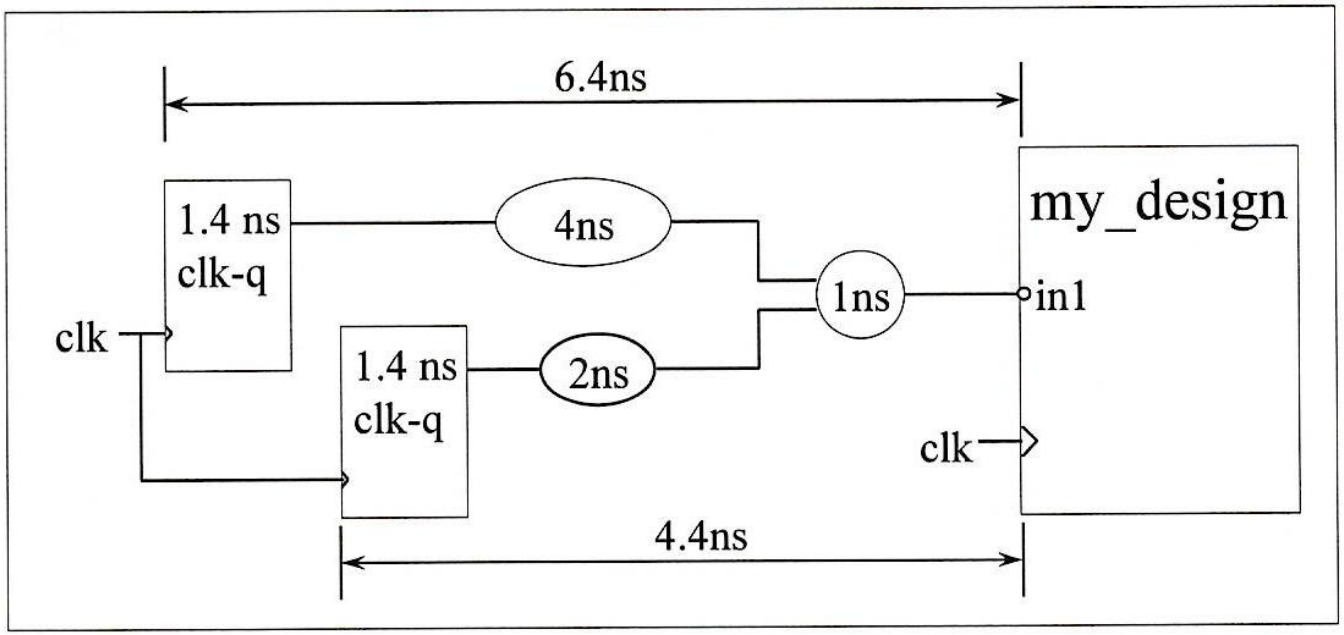
Setting Input Delay(2/2)

- Example




```
dc_shell-xg-t>set_input_delay -clock clk -max 6.4 [get_ports in1]
dc_shell-xg-t>set_input_delay -clock clk -min 4.4  [get_ports in1]
```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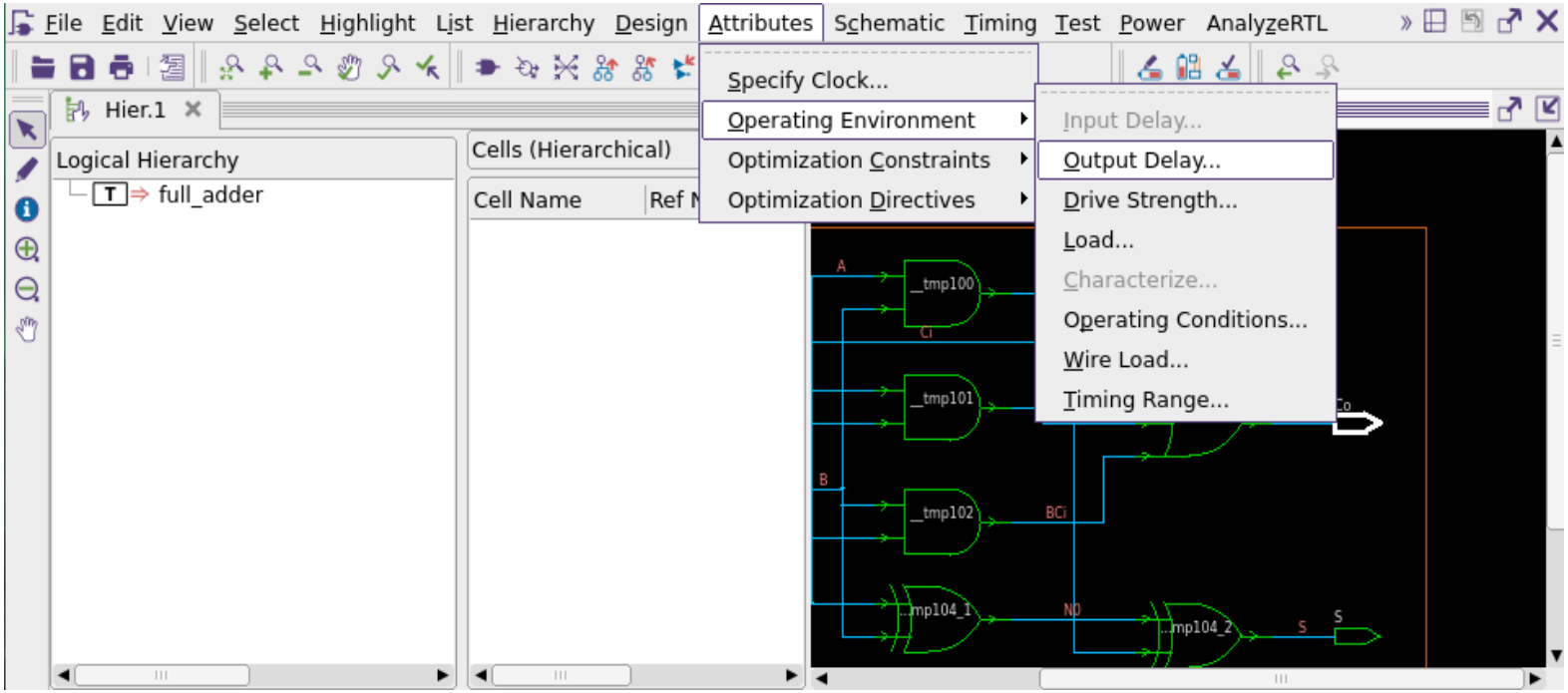
Setting Output Delay(1/2)



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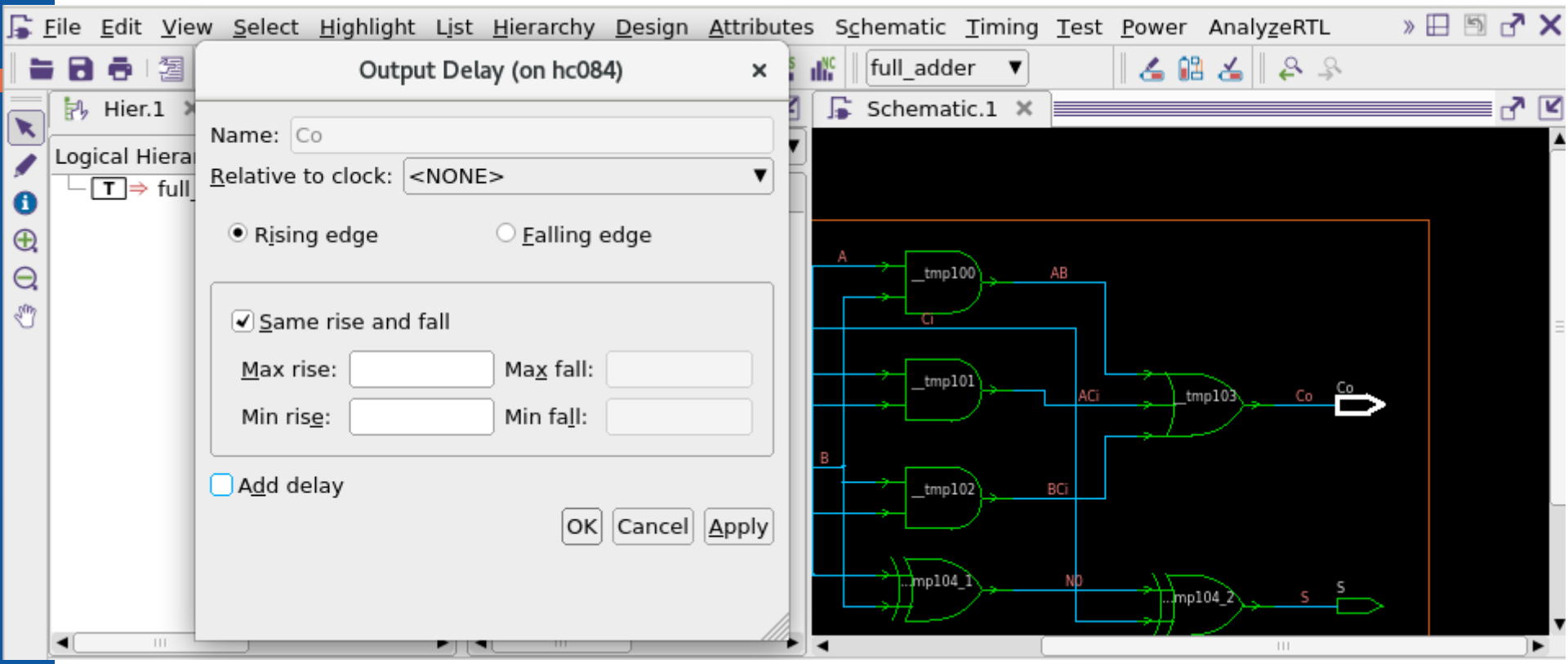
Setting Output Delay(2/2)



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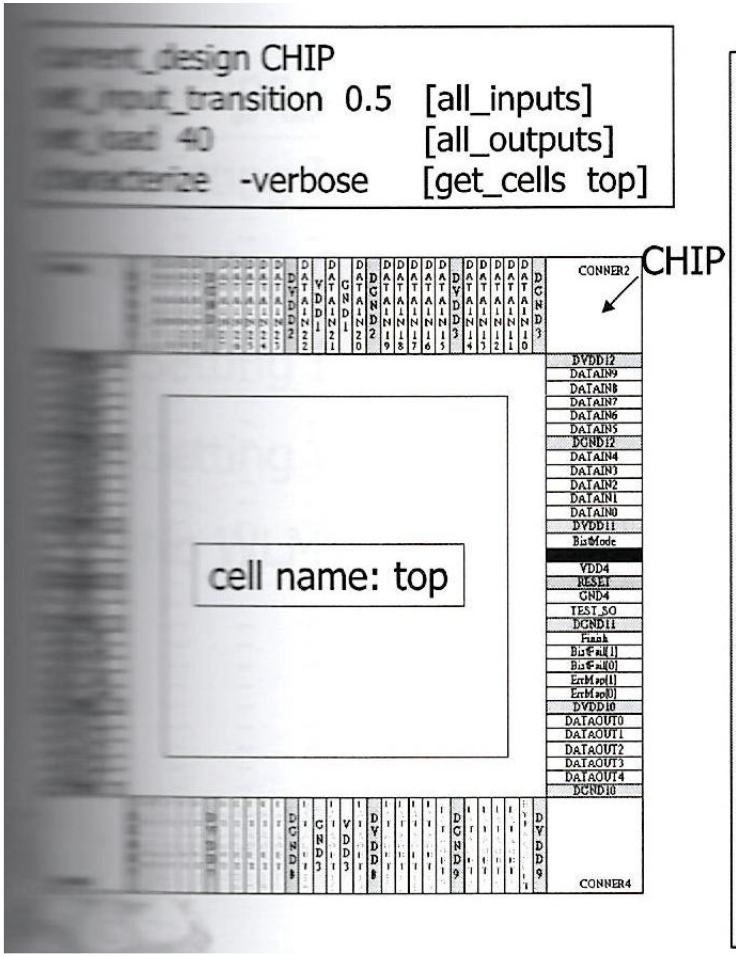
Setting Input/Output Delay for CHIO Level



characterize log message:

```
set_load -pin_load 0.06 [get_ports Y[0]]
set_load -min -pin_load 0.08 [get_ports Y[0]]
set_load -pin_load 0.06 [get_ports Y[1]]
set_load -min -pin_load 0.08 [get_ports Y[1]]
        :
        :
set_load -pin_load 0.06 [get_ports Y[9]]
set_load -min -pin_load 0.08 [get_ports Y[9]]

set_driving_cell -lib_cell PDIDGZ -library tpz973gvwc -pin C -from_pin PAD \
-input_transition_rise 0.5 -input_transition_fall 0.5 -no_design_rule \
[get_ports clk]
set_driving_cell -lib_cell PDIDGZ -library tpz973gvwc -pin C -from_pin PAD \
-input_transition_rise 0.5 -input_transition_fall 0.5 -no_design_rule \
[get_ports {X[0]}]
        :
set_driving_cell -lib_cell PDIDGZ -library tpz973gvwc -pin C -from_pin PAD \
-input_transition_rise 0.5 -input_transition_fall 0.5 -no_design_rule \
[get_ports {X[7]}]

set_input_delay  -max -rise 1.44658 [get_ports clk]
set_input_delay  -max -fall 0.780128 [get_ports clk]
set_input_delay  -max -rise 1.44658  [get_ports reset]
set_input_delay  -max -fall 0.780128 [get_ports reset]
       :
       :
set_input_delay  -max -rise 1.44658  [get_ports {X[0]}]
set_input_delay  -max -fall 0.780127 [get_ports {X[0]}]

set_output_delay -max -rise 3.46819  [get_ports {Y[0]}]
set_output_delay -max -fall 3.86219  [get_ports {Y[0]}]
set_output_delay -max -rise 3.46819  [get_ports {Y[1]}]
set_output_delay -max -fall 3.86219  [get_ports {Y[1]}]
set_output_delay -max -rise 3.46819  [get_ports {Y[2]}]
set_output_delay -max -fall 3.86219  [get_ports {Y[2]}]
set_output_delay -max -rise 3.46819  [get_ports {Y[3]}]
set_output_delay -max -fall 3.86219  [get_ports {Y[3]}]
       :
       :
set_output_delay -max -rise 3.46819  [get_ports {Y[9]}]
set_output_delay -max -fall 3.86219  [get_ports {Y[9]}]
```


TAIWAN TECH

補充:CHIP-Level Synthesis

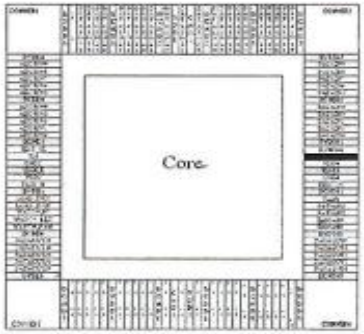
補充

An Example for CHIP Module

國家實驗研究院

```
module  CHIP (Y, so, X, reset, clk,si , se);
input           clk, reset;
input           se, si;
input  [7:0]    X;
output [9:0]    Y;
output          so;
wire    i_clk, i_reset;
wire    i_si, i_se, i_so;
wire    [7:0]   i_X;
wire    [9:0]   i_Y;

 CS top(.Y(i_Y), .so(i_so), .X(i_X), .reset(i_reset), .clk(i_clk), .si(i_si), .se(i_se));

PDIDGZ  ipad_clk  ( .PAD(clk)   , .C(i_clk));
PDIDGZ  ipad_reset( .PAD(reset), .C(i_reset));
PDIDGZ  ipad_si   ( .PAD(si),    .C(i_si));
PDIDGZ  ipad_se   ( .PAD(se),    .C(i_se));
PDIDGZ  ipad_X7( .PAD(X[7]), .C(i_X[7]) );
PDIDGZ  ipad_X6( .PAD(X[6]), .C(i_X[6]) );
PDIDGZ  ipad_X5( .PAD(X[5]), .C(i_X[5]) );
PDIDGZ  ipad_X4( .PAD(X[4]), .C(i_X[4]) );
PDIDGZ  ipad_X3( .PAD(X[3]), .C(i_X[3]) );
PDIDGZ  ipad_X2( .PAD(X[2]), .C(i_X[2]) );
PDIDGZ  ipad_X1( .PAD(X[1]), .C(i_X[1]) );
PDIDGZ  ipad_X0( .PAD(X[0]), .C(i_X[0]) );

PDO16CDG opad_Y9( .I(i_Y[9]), .PAD(Y[9]) );
PDO16CDG opad_Y8( .I(i_Y[8]), .PAD(Y[8]) );
PDO16CDG opad_Y7( .I(i_Y[7]), .PAD(Y[7]) );
PDO16CDG opad_Y6( .I(i_Y[6]), .PAD(Y[6]) );
PDO16CDG opad_Y5( .I(i_Y[5]), .PAD(Y[5]) );
PDO16CDG opad_Y4( .I(i_Y[4]), .PAD(Y[4]) );
PDO16CDG opad_Y3( .I(i_Y[3]), .PAD(Y[3]) );
PDO16CDG opad_Y2( .I(i_Y[2]), .PAD(Y[2]) );
PDO16CDG opad_Y1( .I(i_Y[1]), .PAD(Y[1]) );
PDO16CDG opad_Y0( .I(i_Y[0]), .PAD(Y[0]) );
PDO16CDG opad_so( .I(i_so),   .PAD(so) );

endmodule
```

散字越大，delay 越小



DFP

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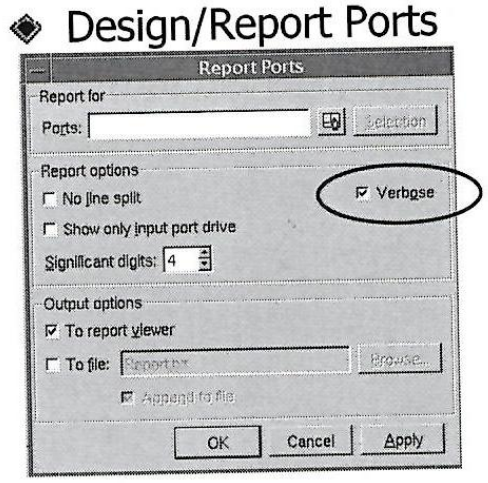
Report-Port(1/2)



Input Port	Driving Cell Rise	Driving Cell Fall	Mult	Attrs
clk	slow/BUFX4/Y	slow/BUFX4/Y	--	
cook_time[0]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[1]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[2]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[3]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[4]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[5]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[6]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[7]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[8]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[9]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[10]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[11]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[12]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[13]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[14]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
cook_time[15]	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
reset	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
set_time	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
start_cook	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	
test	slow/DFFX2/Q (D)	slow/DFFX2/Q (D)	--	

Port	Dir	Pin Load	Wire Load	Max Trans	Max Cap	Connection Class
clk	in	0.0000	0.0000	--	--	--
cook_time[0]	in	0.0000	0.0000	--	--	--
cook_time[1]	in	0.0000	0.0000	--	--	--
cook_time[2]	in	0.0000	0.0000	--	--	--
cook_time[3]	in	0.0000	0.0000	--	--	--
cook_time[4]	in	0.0000	0.0000	--	--	--
cook_time[5]	in	0.0000	0.0000	--	--	--
cook_time[6]	in	0.0000	0.0000	--	--	--
cook_time[7]	in	0.0000	0.0000	--	--	--
cook_time[8]	in	0.0000	0.0000	--	--	--
cook_time[9]	in	0.0000	0.0000	--	--	--
cook_time[10]	in	0.0000	0.0000	--	--	--
cook_time[11]	in	0.0000	0.0000	--	--	--
cook_time[12]	in	0.0000	0.0000	--	--	--
cook_time[13]	in	0.0000	0.0000	--	--	--
cook_time[14]	in	0.0000	0.0000	--	--	--
cook_time[15]	in	0.0000	0.0000	--	--	--
reset	in	0.0000	0.0000	--	--	--
set_time	in	0.0000	0.0000	--	--	--
start_cook	in	0.0000	0.0000	--	--	--
test	in	0.0000	0.0000	--	--	--
min_lsb_led[0]	out	0.0022	0.0000	--	--	--
min_lsb_led[1]	out	0.0022	0.0000	--	--	--
min_lsb_led[2]	out	0.0022	0.0000	--	--	--
min_lsb_led[3]	out	0.0022	0.0000	--	--	--
min_lsb_led[4]	out	0.0022	0.0000	--	--	--
min_lsb_led[5]	out	0.0022	0.0000	--	--	--
min_lsb_led[6]	out	0.0022	0.0000	--	--	--
min_msb_led[0]	out	0.0022	0.0000	--	--	--
min_msb_led[1]	out	0.0022	0.0000	--	--	--
min_msb_led[2]	out	0.0022	0.0000	--	--	--
min_msb_led[3]	out	0.0022	0.0000	--	--	--
min_msb_led[4]	out	0.0022	0.0000	--	--	--
min_msb_led[5]	out	0.0022	0.0000	--	--	--
min_msb_led[6]	out	0.0022	0.0000	--	--	--

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Report-Port(2/2)

	Input Delay					
	Min		Max		Related	Max
	Rise	Fall	Rise	Fall	Clock	Fanout
	--	--	--	--	--	--
	--	--	--	--	--	--
	--	--	--	--	--	--

Setting Design Environment

- Setting Operating Condition
- Setting Input Driving Strength
- Setting Output Loading
- Setting Input/Output Delay
- Setting Wire Load Model
- No WLM: Topographical Technology

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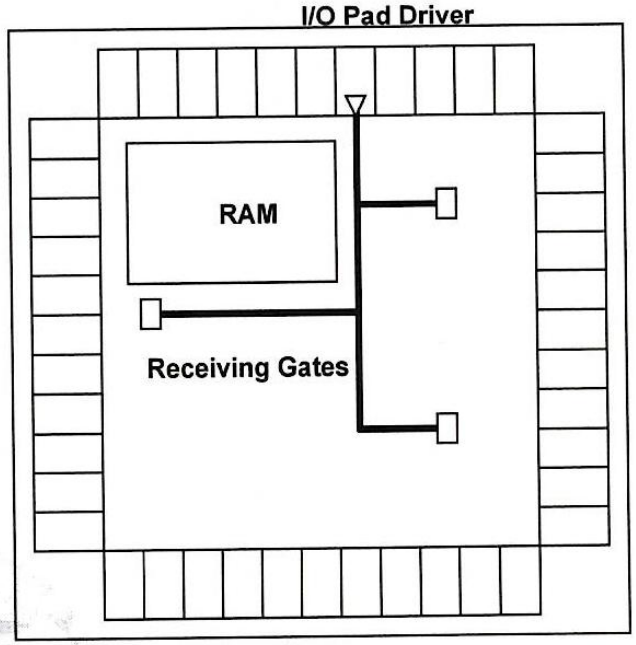
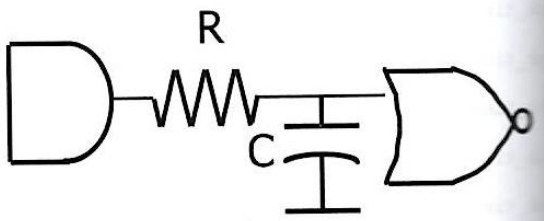
Wire Load Model

- A wire load model is an estimate of a net's RC parasitics based on the net's fanout:
  - Model is created by your vendor
  - Estimates are based on statistics from other designs the vendor has fabricated using this process



Path delay ratio		
Process	Cell delay	Net delay
0.35um	7	3
0.25um	5	5
0.18um	3	7



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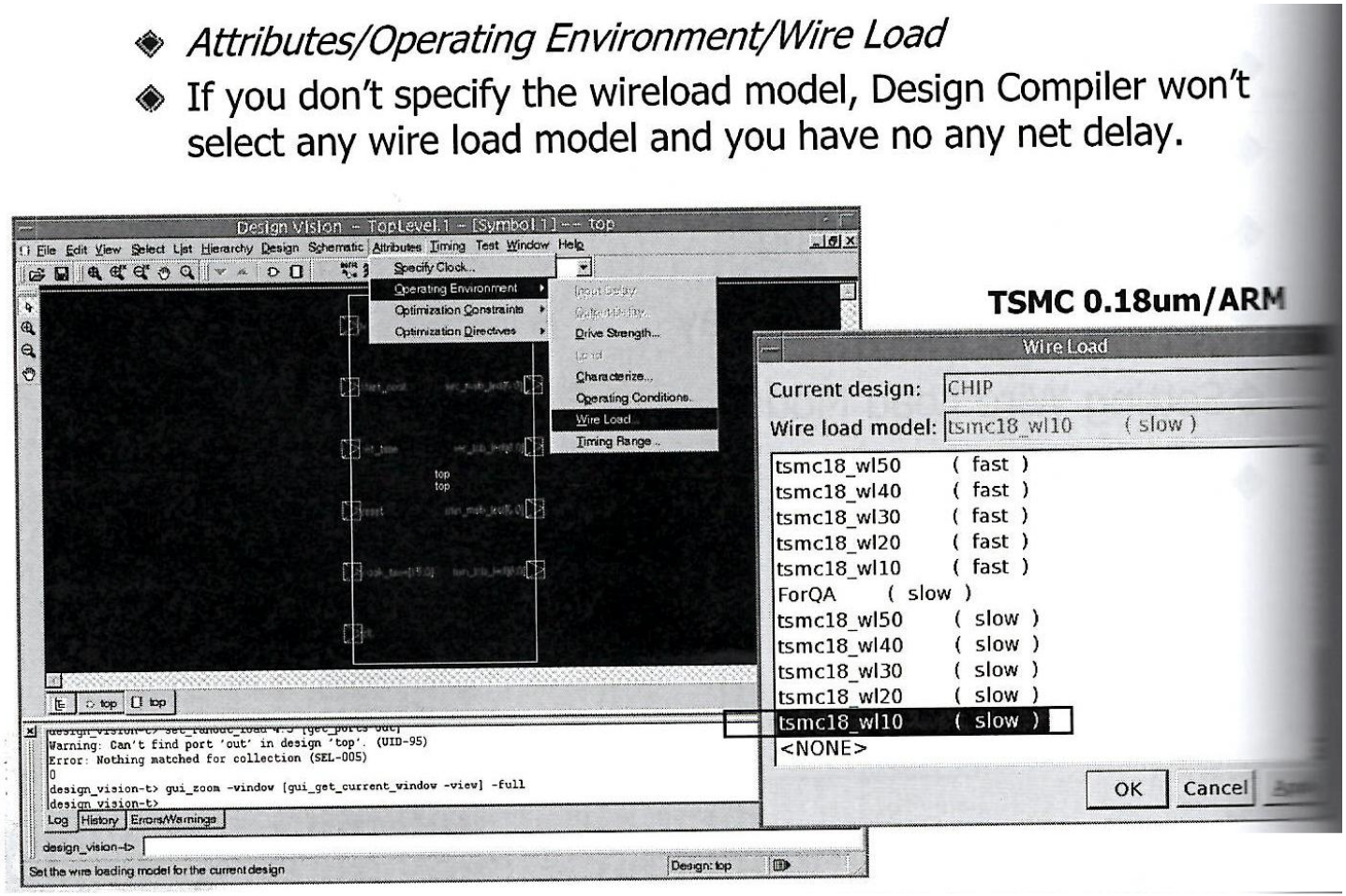
Setting Wire Load Model

- *Attributes/Operating Environment/Wire Load*
- If you don't specify the wireload model, Design Compiler won't select any wire load model and you have no any net delay.



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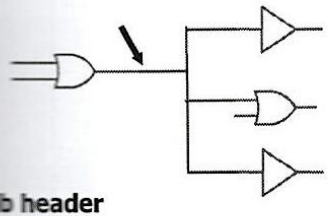
WLM Example for tsmc18_wl10

- To calculate the R, C and net area of a net (Assume use tsmc18_wl10):

Cwire = (fanout=3 ➔ length =66.667x3) x capacitance coefficent (1.5×10^{-4}) =0.03 pf

Rwire = (fanout=3 ➔ length =66.667x3) x resistance coefficient (8.5×10^{-8}) = 0.017 ohm

Net area = (fanout=3 ➔ length =66.667x3) x area coefficient (0.7) = 140 um^2



slow.lib header

```
/* unit attributes */
time_unit : "1ns";
voltage_unit : "1V";
current_unit : "1mA";
pulling_resistance_unit : "1kohm";
leakage_power_unit : "1pW";
capacitive_load_unit (1.0,pf);

/* operation conditions */
nom_process      : 1;
nom_temperature  : 125;
nom_voltage      : 1.62;
operating_conditions(slow) {
  process      : 1;
  temperature  : 125;
  voltage      : 1.62;
  tree_type    : balanced_tree
}
default_operating_conditions : slow;
```

slow.lib

```
/* wire-loads */
wire_load("tsmc18_wl10") {
  resistance  : 8.5e-8;
  capacitance : 1.5e-4;
  area        : 0.7;
  slope       : 66.667;
  fanout_length       (1,66.667);
}
wire_load("tsmc18_wl20") {
  resistance  : 8.5e-8;
  capacitance : 1.5e-4;
  area        : 0.7;
  slope       : 133.334;
  fanout_length       (1,133.334);
}
wire_load("tsmc18_wl30") {
  resistance  : 8.5e-8;
  capacitance : 1.5e-4;
  area        : 0.7;
  slope       : 200.001;
  fanout_length       (1,200.001);
}
wire_load("tsmc18_wl40") {
  resistance  : 8.5e-8;
  capacitance : 1.5e-4;
  area        : 0.7;
  slope       : 266.668
  fanout_length       (',266.668);
}
wire_load("tsmc18_wl50") {
  resistance  : 8.5e-8;
  capacitance : 1.5e-4;
  area        : 0.7;
  slope       : 333.335
  fanout_length       (',333.335);
}
/* QA wire-load */
wire_load("ForQA") {
  resistance  : 0;
  capacitance : 1;
  area        : 1;
  slope       : 1;
  fanout_length(1,0);
  fanout_length(10,0);
}
/* additional header data */
```

```
set_units  -time ns  -resistance kOhm  -capacitance pF  -voltage V  -current mA
```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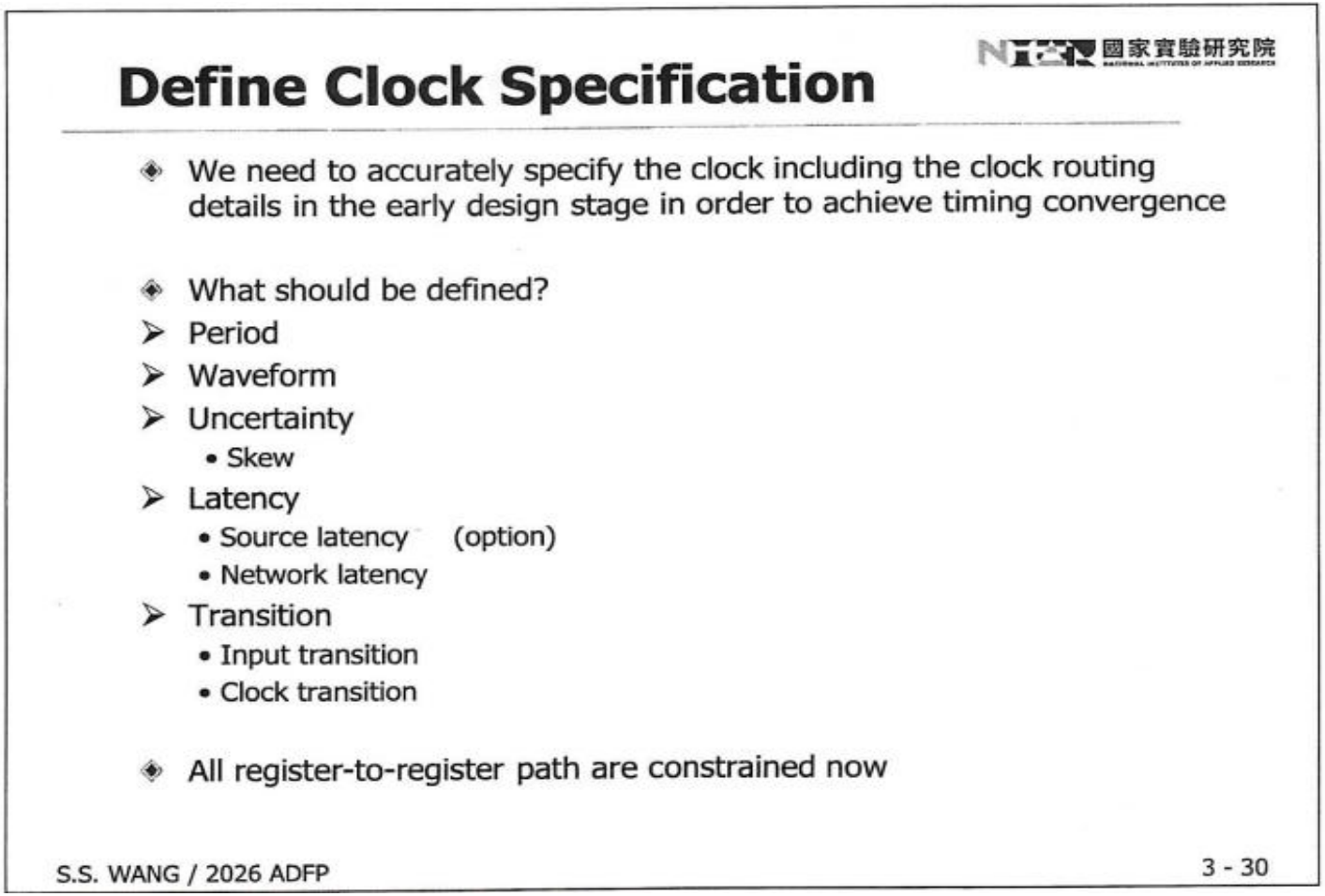
Define Clock Specification



Define Clock Specification

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- We need to accurately specify the clock including the clock routing details in the early design stage in order to achieve timing convergence

- What should be defined?
  - Period
  - Waveform
  - Uncertainty
    - Skew
  - Latency
    - Source latency (option)
    - Network latency
  - Transition
    - Input transition
    - Clock transition

- All register-to-register path are constrained now

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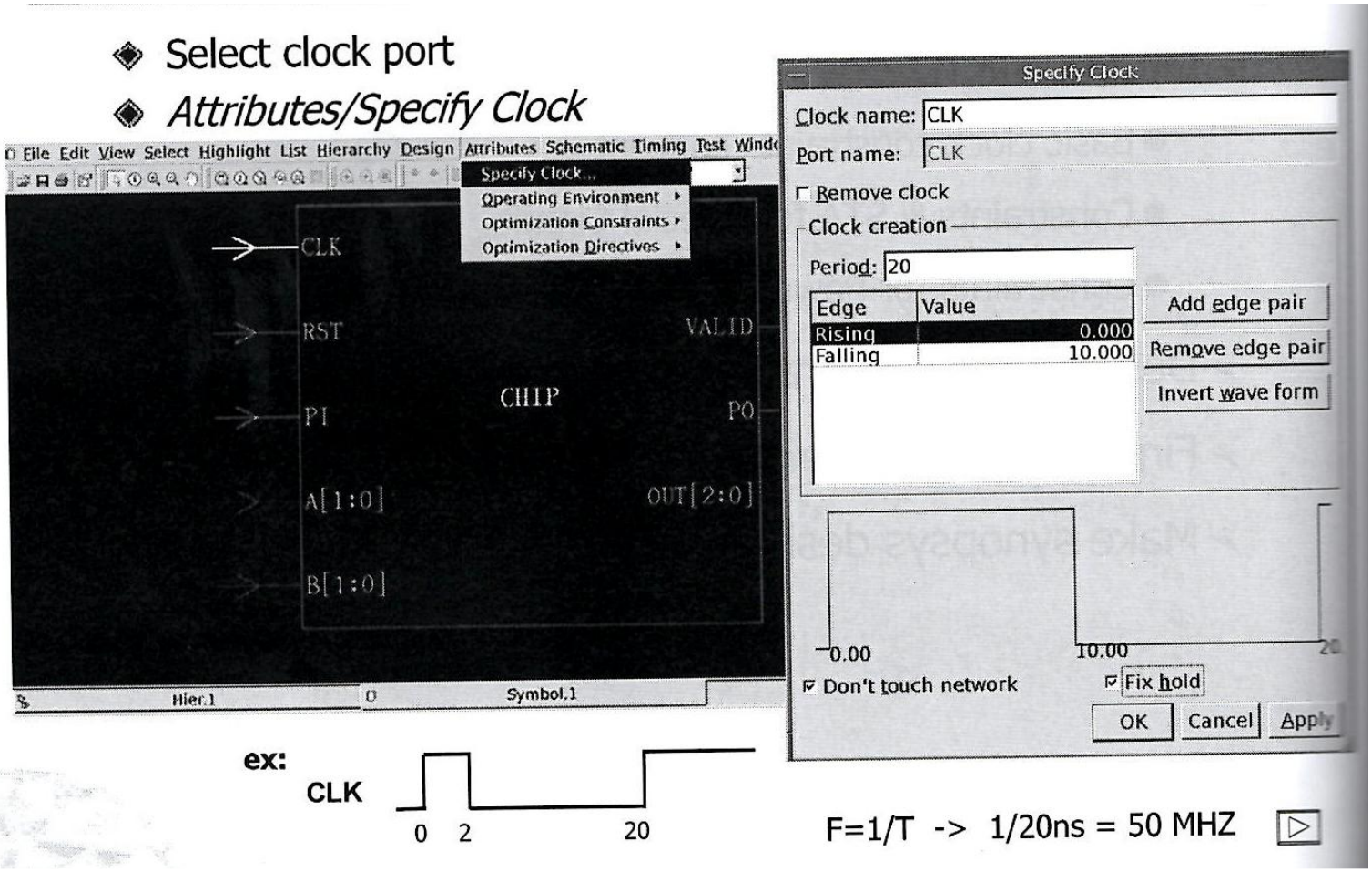
Sequential Circuit → Specify Clock(1/2)

- Select clock port
- *Attributes/Specify Clock*



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Specify Clock(2/2)

- *create_clock* : define your clock's waveform & respect the set-up time requirements of all clocked flip-flops

```
dc_shell-xg-t> create_clock -period 20 -waveform {0 10} [get_ports CLK]
```

- *set_fix_hold* : respect the hold time requirement of all clocked flip-flops

```
dc_shell-xg-t> set_fix_hold [get_clocks CLK]
```

- *set_dont_touch_network* : do not re-buffer the clock network

```
dc_shell-xg-t> set_dont_touch_network [get_clocks CLK]
```


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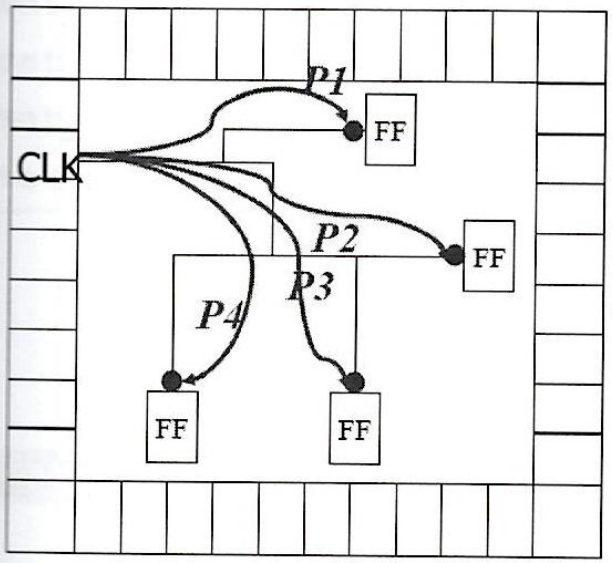
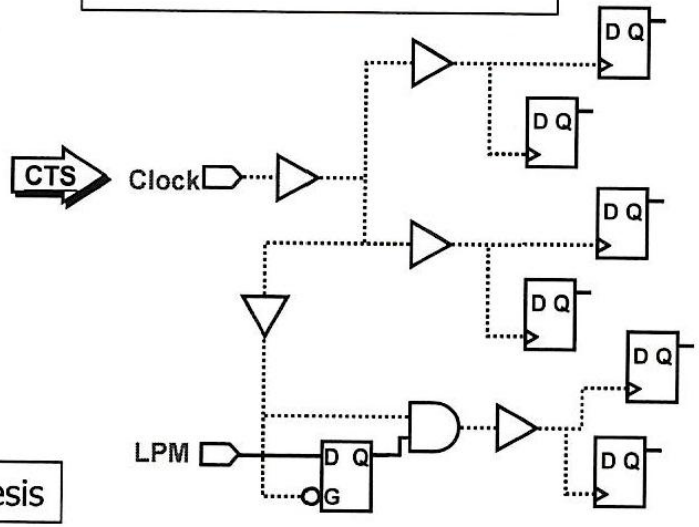
Clock Slew(1/2)

- **The maximum difference between the arrival of clock signals at sequential cells in one clock domain or between domains**




```
Arrival(P1) = 0.5ns
Arrival(P2) = 1ns
Arrival(P3) = 1.2ns
Arrival(P4) = 1.3ns

uncertainty = 1.3 - 0.5
= 0.8ns
```



CTS: Clock Tree Synthesis

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Clock Slew(2/2)

```
dc_shell> set_clock_uncertainty 0.8 [get_clocks CLK]
```

Point	Incr	Path
clock CLK (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
u1/CLK (fdef1a15)	0.00	0.00 r
u1/Q (fdef1a15)	0.50	0.50 r
u2/Y (buf1a27)	0.11	0.61 r
u3/Y (buf1a27)	0.11	0.72 r
u4/D (fdef1a15)	0.00	0.72 r
data arrival time		0.72
clock CLK (rise edge)	20.00	20.00
clock network delay (ideal)	0.00	20.00
clock uncertainty	-0.80	19.20
u4/CLK (fdef1a15)		19.20 r
library setup time	-0.21	18.99
data required time		18.99
data required time		18.99
data arrival time		-0.72
slack (MET)		18.27

Skew = 0.8

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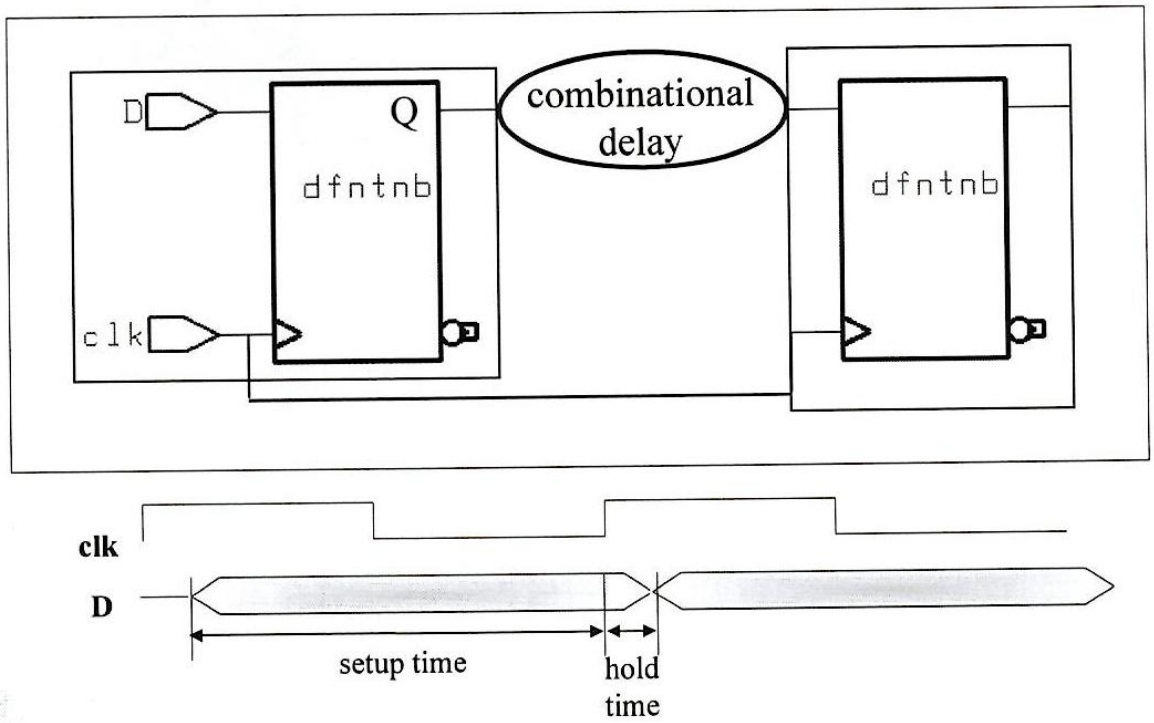
Sequential Circuits

- Sequential circuits are usually constrained by clock specify
- clock cycle >= $DFF_{clk\text{-}Qdelay}$ + combinational delay + DFF_{setup}



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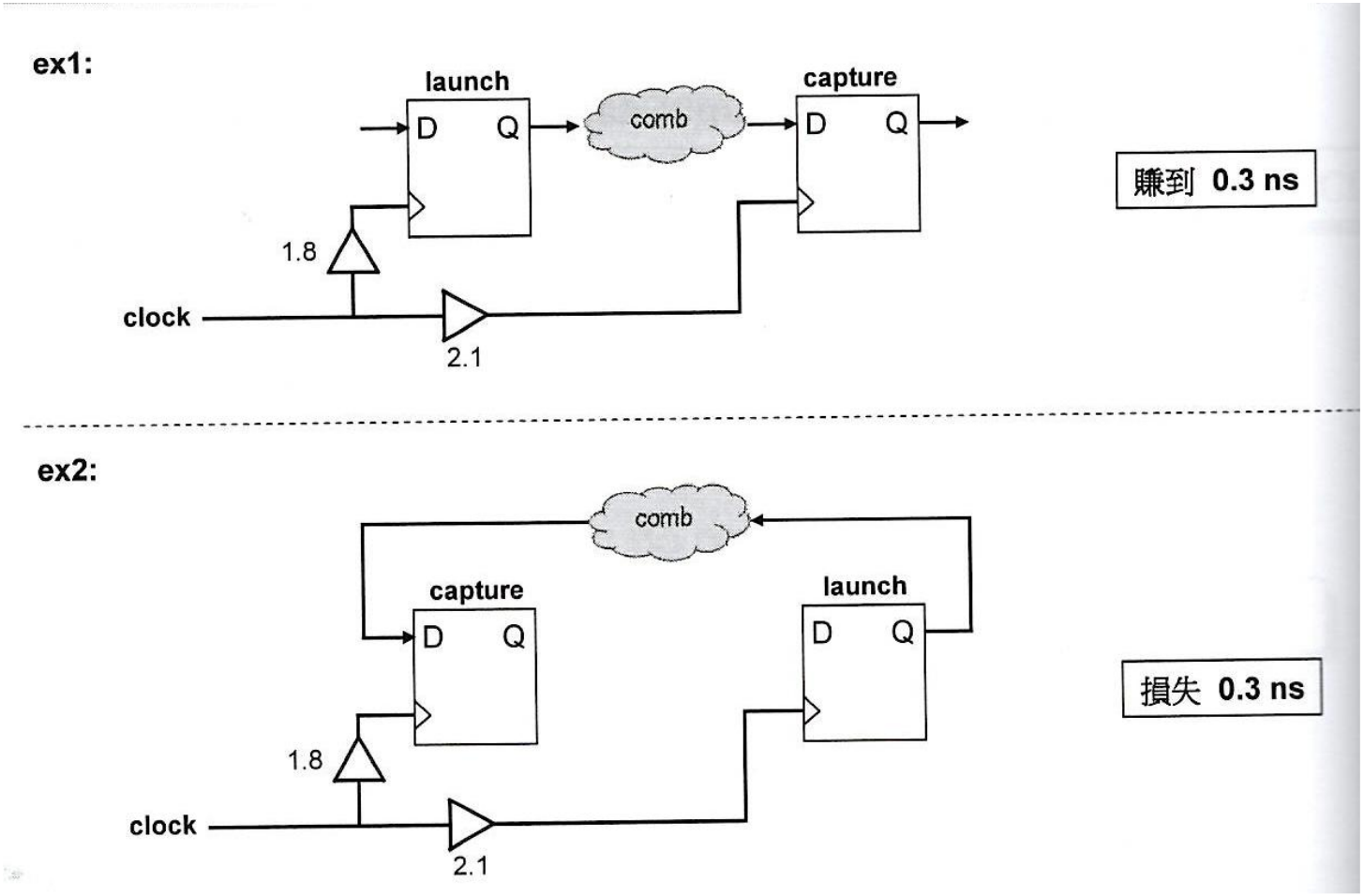
Clock Skew(補充)



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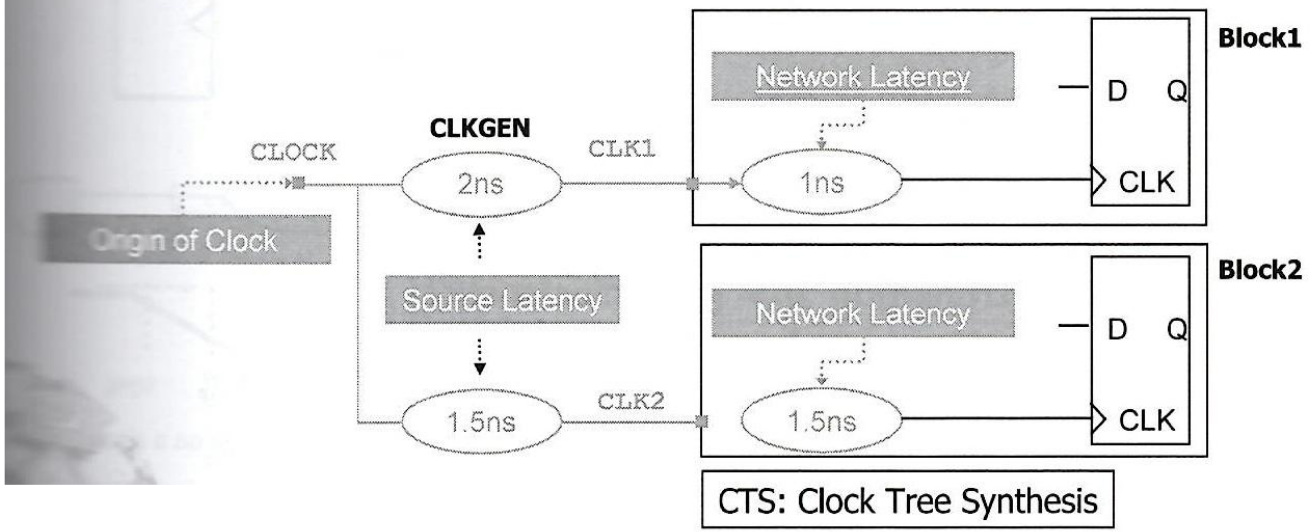
Model Source Latency

- *Source latency* is the propagation time from the actual clock origin to the clock definition point in the design

```
create_clock -name CLK1 -period 20 [get_ports CLOCK]
set_clock_latency -source 2   [get_clocks CLK1]
set_clock_latency -source 1.5 [get_clocks CLK2]
set_clock_latency 1   [get_clocks CLK1]
set_clock_latency 1.5 [get_clocks CLK2]
```



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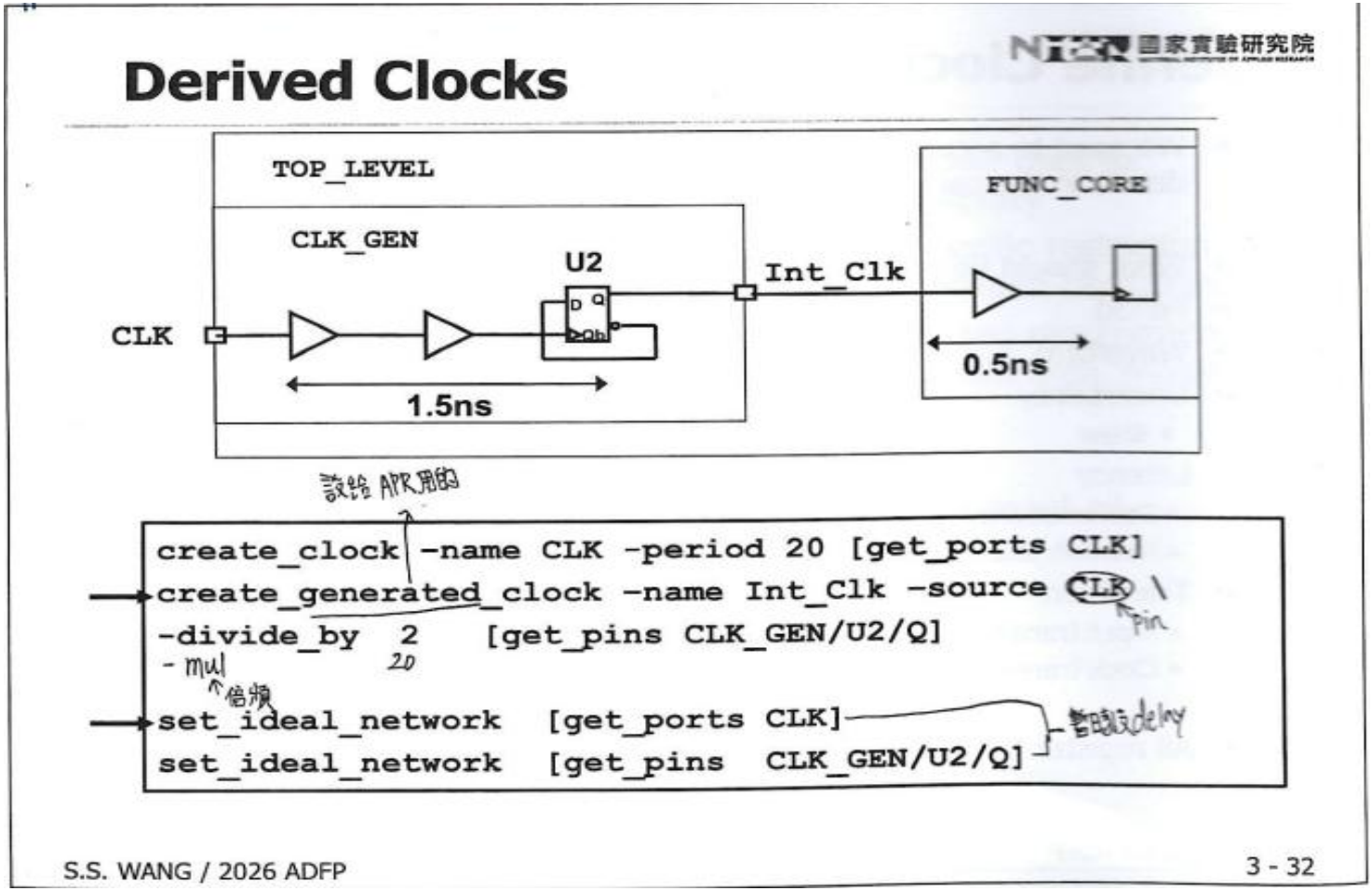
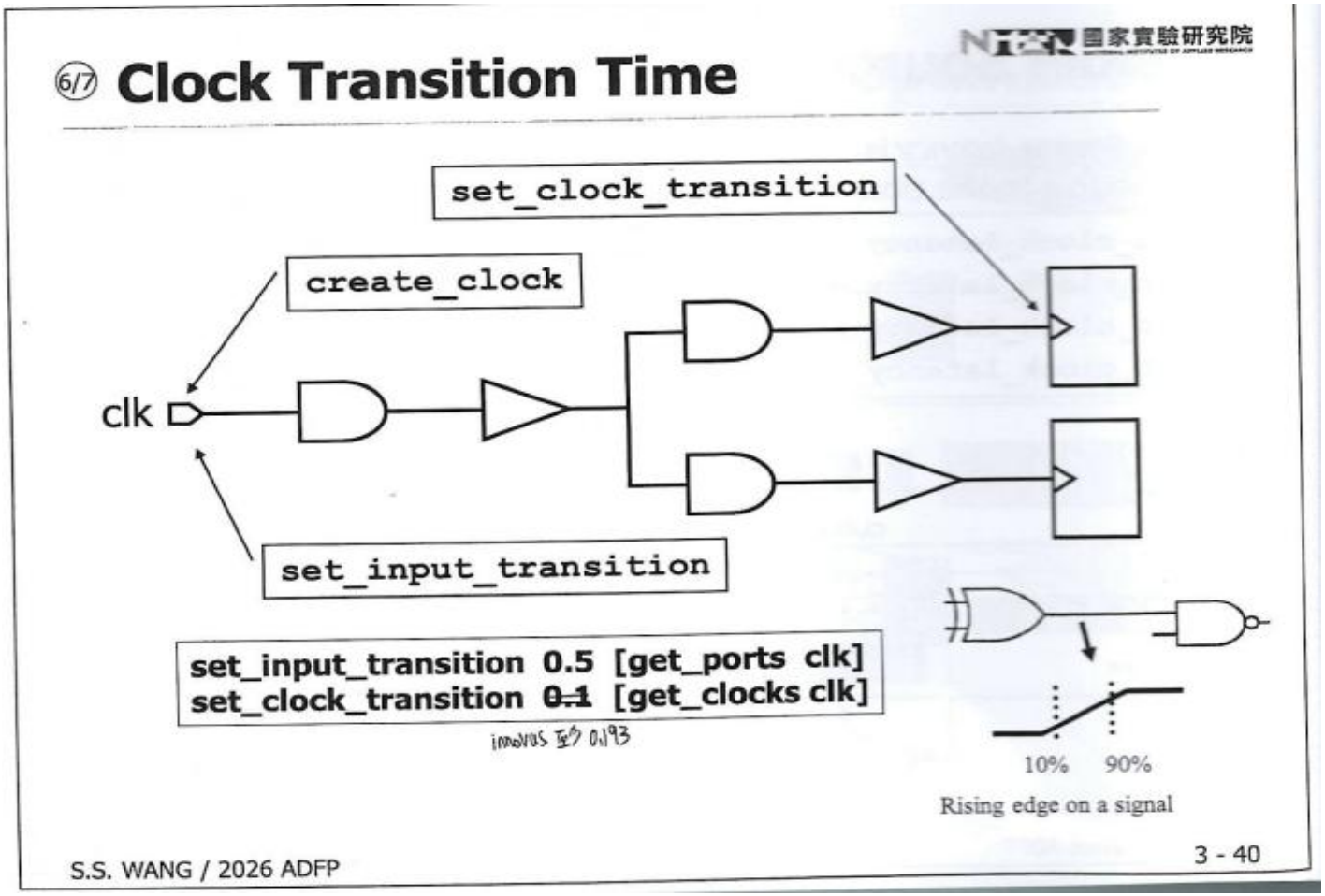
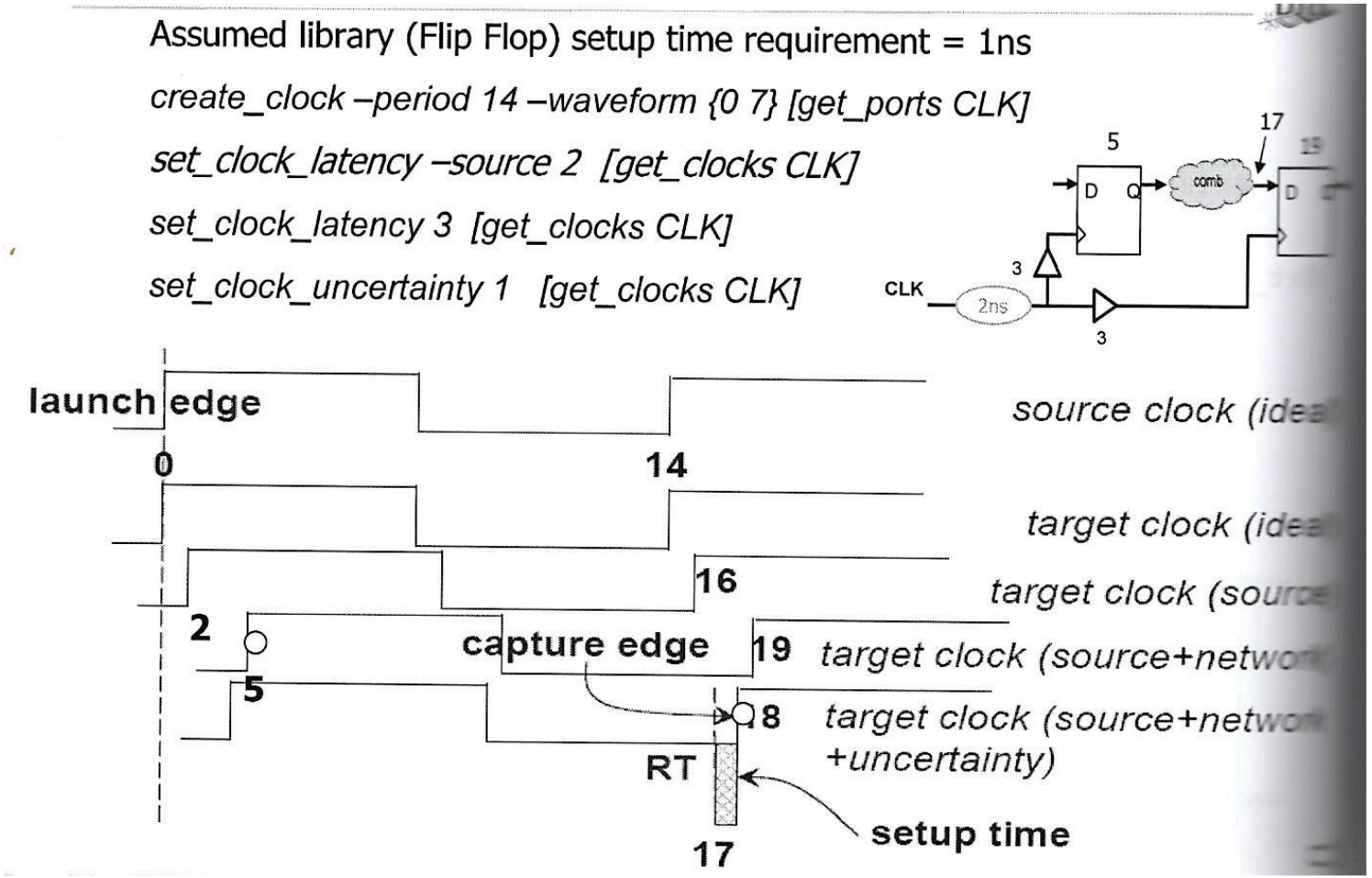
Derived Clocks



Clock Transition Time



Effect of Clock Tree Modeling on Setup Time



An Example for Setup Time Report

```
Startpoint: test (input port clocked by clk)
Endpoint: timer/sec_lsb_next_reg[0]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
top                G5K                   fsd0a_a_generic_core_wc

Point                                                    Incr       Path
-----------------------------------------------------------------------------
clock clk (rise edge)                                    0.0000     0.0000
clock network delay (ideal)                              1.0000     1.0000
input external delay                                     1.0000     2.0000 f
test (in)                                                0.0129     2.0129 f
...
timer/sec_lsb_next_reg[0]/D (QDFFX1)                     2.6016     4.6145 f
data arrival time                                                   4.6145

clock clk (rise edge)                                    4.0000     4.0000
clock network delay (ideal)                              1.0000     5.0000
clock uncertainty                                       -0.1000     4.9000
timer/sec_lsb_next_reg[0]/CK (QDFFX1)                    0.0000     4.9000 r
library setup time                                      -0.0820     4.8180
data required time                                                  4.8180
-----------------------------------------------------------------------------
data required time                                                  4.8180
data arrival time                                                  -4.6145
-----------------------------------------------------------------------------
slack (MET)                                                         0.2035
```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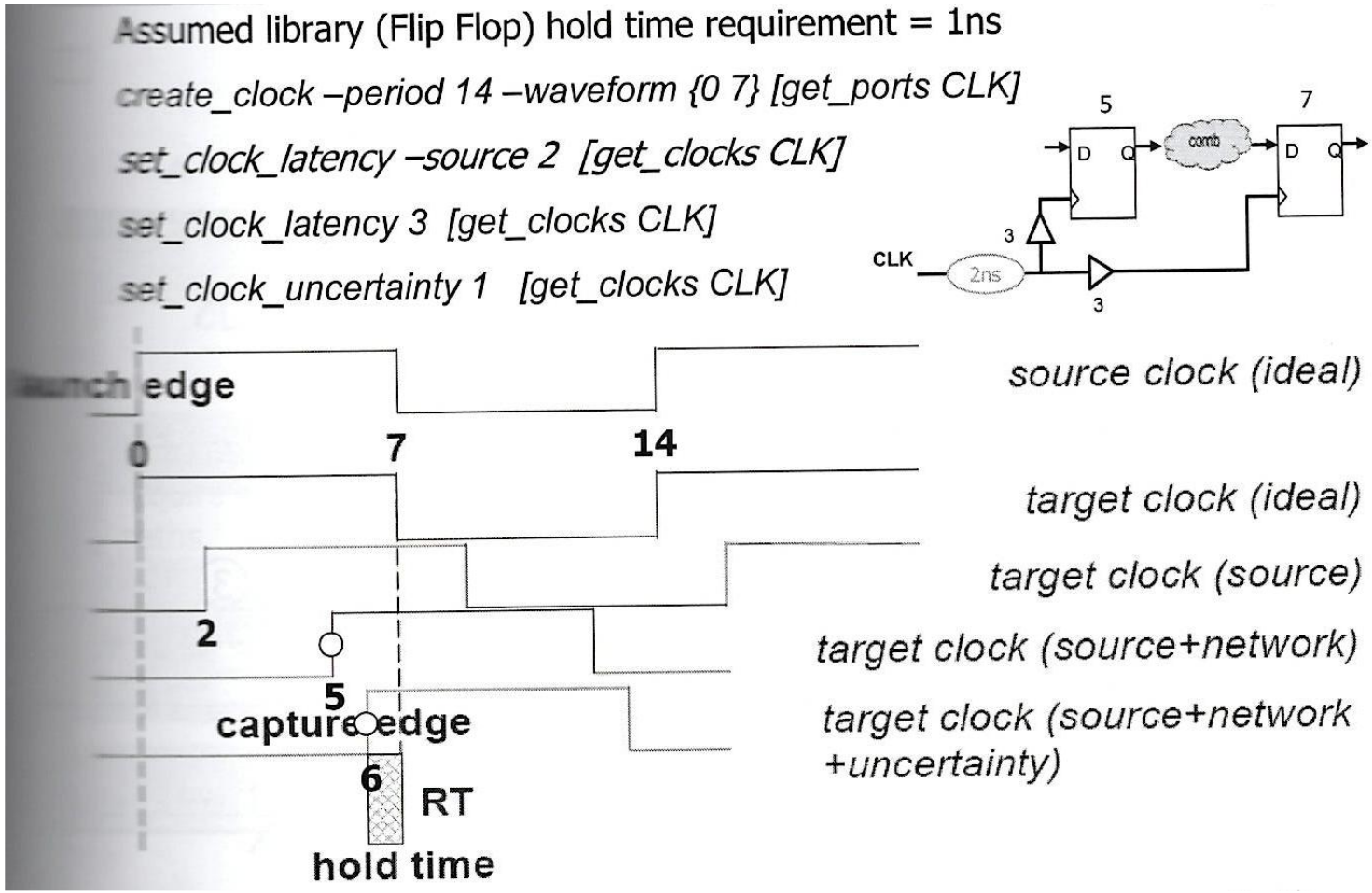
Effect of Clock Tree Modeling on Hold Time



An Example for Hold Time Report

```
Startpoint: microwave/micro_st/state_reg[1]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: microwave/micro_st/state_reg[2]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Des/Clust/Port      Wire Load Model       Library
------------------------------------------------
top                 G5K                   fsd0a_a_generic_core_wc

Point                                                    Incr       Path
-----------------------------------------------------------------------------
clock clk (rise edge)                                    0.0000     0.0000
clock network delay (ideal)                              1.0000     1.0000
microwave/micro_st/state_reg[1]/CK (QDFFRBX1)            0.0000     1.0000 r
microwave/micro_st/state_reg[1]/Q (QDFFRBX1)             0.0975     1.0975 f
...
microwave/micro_st/state_reg[2]/D (QDFFRBX1)             0.0344     1.1319 f
data arrival time                                                   1.1319

clock clk (rise edge)                                    0.0000     0.0000
clock network delay (ideal)                              1.0000     1.0000
clock uncertainty                                        0.1000     1.1000
microwave/micro_st/state_reg[2]/CK (QDFFRBX1)            0.0000     1.1000 r
library hold time                                        0.0016     1.1016
data required time                                                  1.1016
-----------------------------------------------------------------------------
data required time                                                  1.1016
data arrival time                                                  -1.1319
-----------------------------------------------------------------------------
slack (MET)                                                         0.0304
```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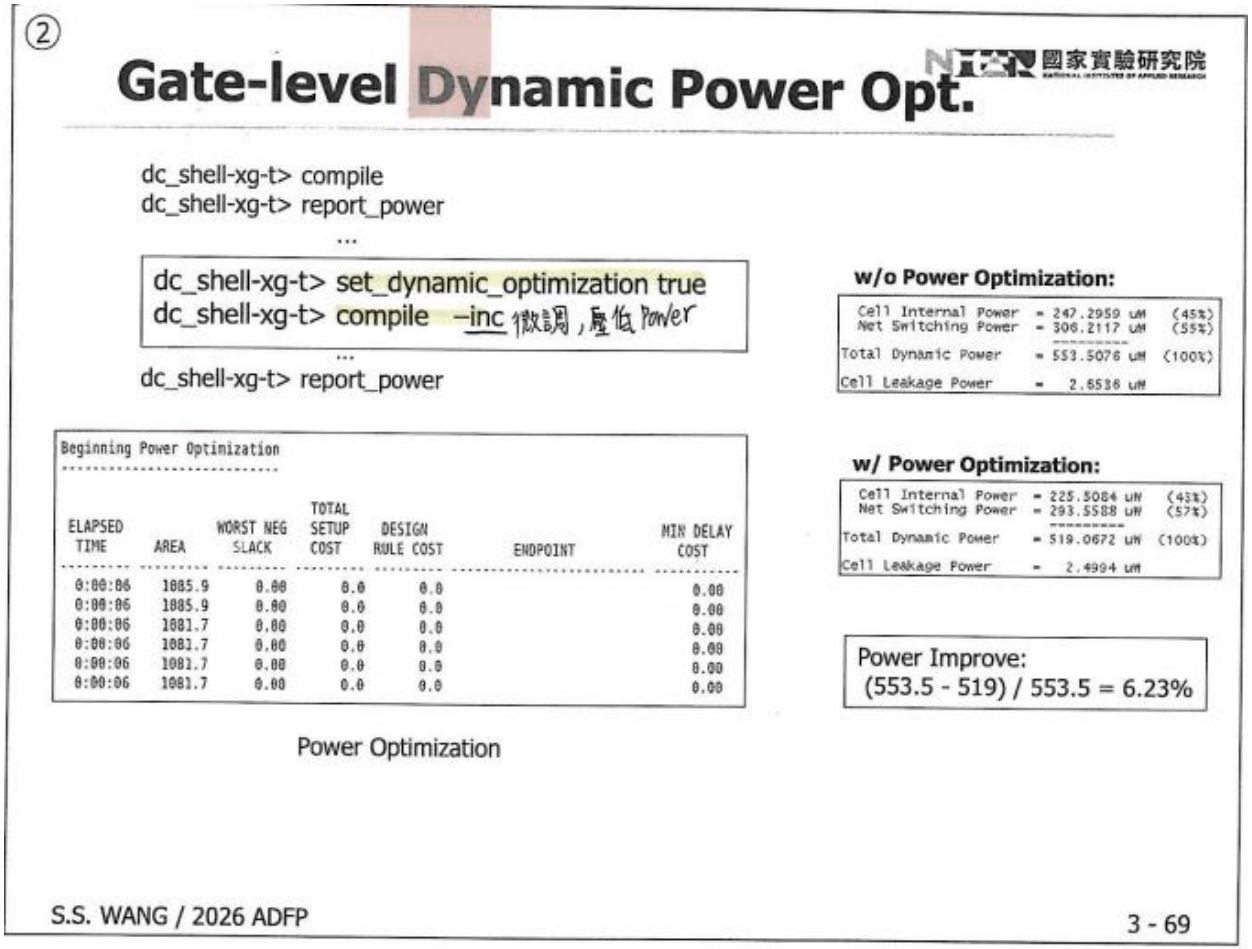
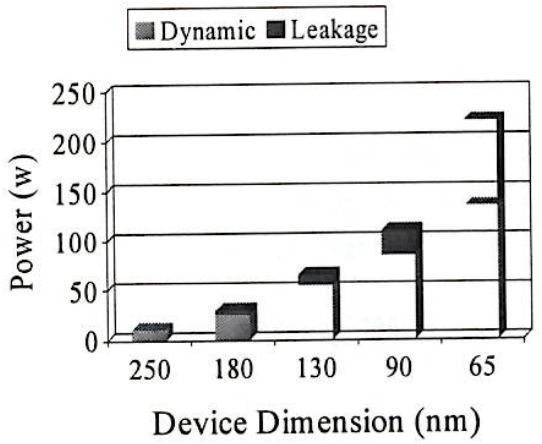
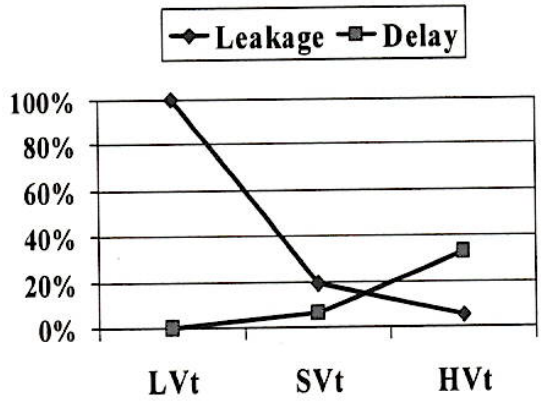
Gate-level Dynamic Power Opt.



Leakage Power optimization:
Vt Affects Leakage Power Significantly





- Vt affects sub-threshold leakage exponentially
- Vt vs. Leakage and delay:
  - Higher Vt cells
    - Lower leakage power
    - Longer delay
  - Lower Vt cells
    - Higher leakage power
    - Shorter delay
- Multi-Vt libraries enable low leakage design

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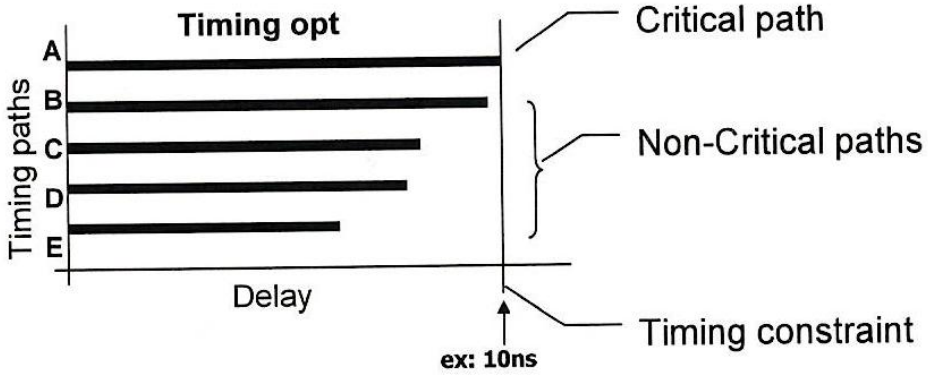
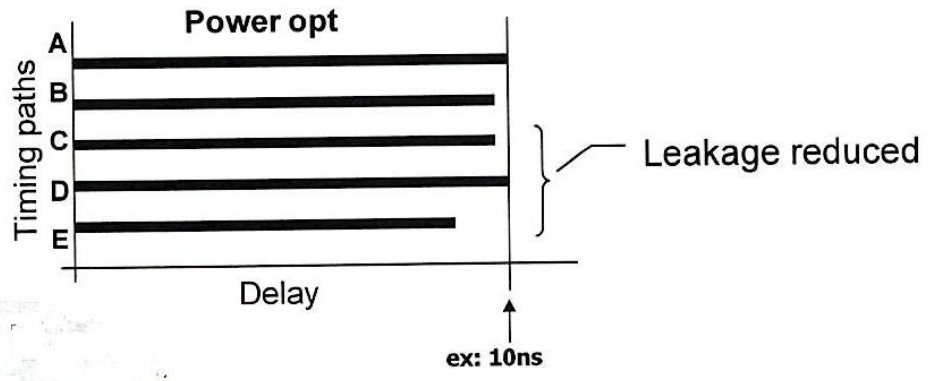
Saving Power On Non-critical Paths





- Use low V_t cells on critical paths to improve timing
- Use high V_t cells on non-critical paths to save power
- Low leakage and high performance design with MVt libraries

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How to reduce our Leakage Power

- Usually optimize leakage power after timing optimization
- It is OK to optimize timing and power at the same time
- Multi-Vt libraries are effective in reducing leakage power
- An example: We use 90nm TSMC/ARM library

```
#Set your .synopsys_dc.setup File:
set target_library  "slow.db            fast.db            \
                     slow_hvt.db        fast_hvt.db        \
                     tpzn90gv3wc.db     tpzn90gv3lt.db     \
                    "
set link_library    "* $target_library  dw_foundation.sldb"

#In DC script:
read_file –format verilog your_design.v
source  constraints.tcl
.....
compile

set_max_leakage_power 0 nw
compile -inc
```

TSMC90 Setting Condition:

```
set_operating_conditions -max_library slow.db:slow   -max slow
                         -min_library fast.db:fast   -min fast
```

slow.lib

```
* The copyright notice(s) in this file does not indicat
* publication of this file.
*
* artix(v4.30.20)-artichar(v3.60.59)-syntech(1.63.2.13)
*
* process (ss), voltage (0.9v), temp (125)
*
* corner: com1_ss_h_l
*/
/* table model: cell-delay */
library(slow) {

  /* general attributes */
  delay_model : table_lookup;
  in_place_swap_mode : match_footprint;
  library_features(report_delay_calculation);

  /* documentation attributes */
  revision : 1.3;
  date : "Wed Jun 22 12:31:54 2005";
  comment : "Copyright (c) 2005 Artisan Components, Inc.

  /* unit attributes */
  time_unit : "1ns";
  voltage_unit : "1V";
  current_unit : "1mA";
  pulling_resistance_unit : "1kohm";
  leakage_power_unit : "1pW";
  capacitive_load_unit (1.0,pf);

  /* operation conditions */
  nom_process      : 1;
  nom_temperature : 125;
  nom_voltage      : 0.9;
  operating_conditions(slow) {
    process : 1;
    temperature : 125;
    voltage : 0.9;
    tree_type    : balanced_tree
  }
```

An Leakage Power Opt. Example

Before Leakage Power Opt.

```
Cell Internal Power  =  78.5204 uW   (89%)
Net Switching Power  =   9.9821 uW   (11%)
                        ---------
Total Dynamic Power  =  88.5025 uW   (100%)

Cell Leakage Power   =   3.4123 uW
```

After Leakage Power Opt.

```
Cell Internal Power  =  75.5040 uW   (90%)
Net Switching Power  =   8.1082 uW   (10%)
                        ---------
Total Dynamic Power  =  83.6122 uW   (100%)

Cell Leakage Power   = 827.0530 nW
```

using leakage power opt. technique:

Leakage Power Improve:

(3.4123 uw –827.0530nw) / 3.4123 uw = 75.76%

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Example for Report Multi-Vt Ratio

Sample script:

```
set RVt_lib  "slow.db:slow"
set HVt_lib  "slow_hvt.db:slow"

set_attribute [get_libs $HVt_lib] default_threshold_voltage_group  HVt  -type string
set_attribute [get_libs $RVt_lib] default_threshold_voltage_group  RVt  -type string
```

Report:

```
design_vision> report_threshold_voltage_group
****************************************
                Threshold Voltage Group Report

Vth Group       All                 Blackbox            Non-blackbox
Name            cells               cells               cells
****************************************
HVt             205  (89.13%)       0    (0.00%)        205  (89.13%)
RVt              25  (10.87%)       0    (0.00%)         25  (10.87%)
****************************************
Total           230 (100.00%)       0    (0.00%)        230 (100.00%)

Vth Group       All                 Blackbox            Non-blackbox
Name            cell area           cell area           cell area
****************************************
HVt             953.97  (90.01%)    0.00    (0.00%)     953.97  (90.01%)
RVt             105.84   (9.99%)    0.00    (0.00%)     105.84   (9.99%)
****************************************
Total          1059.81 (100.00%)    0.00    (0.00%)    1059.81 (100.00%)
```


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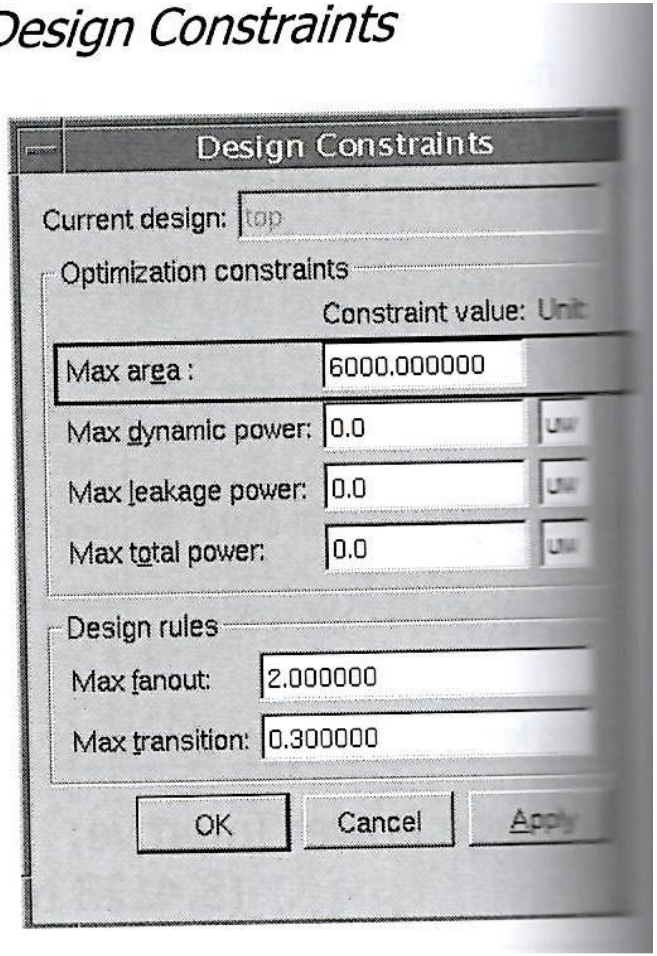
Setting Area Constraint(1/2)

- *Attributes/Optimization Constraints/Design Constraints*
- If you only want to concern the area, but don't care the timing. You can use the following constraint script
  - *remove constraints –all*
  - *set max_area 0*
  - *compile –effort medium*

Area unit :
(1) equivalent gate counts
(2) um x um
(3) Transistors



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Setting Area Constraint(2/2)

tsmc18.pdf

Process Technology:
TSMC CL018G

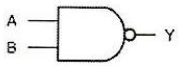
NAND2

Cell Description

The NAND2 cell provides the logical NAND of two inputs (A, B). The output (Y) is represented by the logic equation:

$Y = \overline{(A \bullet B)}$

Logic Symbol



Functions

A	B	Y
0	x	1
x	0	1
1	1	0

Cell Size

Drive Strength	Height (µm)	Width (µm)
NAND2XL	5.04	1.98
NAND2X1	5.04	1.98
NAND2X2	5.04	3.30
NAND2X4	5.04	4.62

gate counts= Chip Report Area / NAND2 Area

ex: Report area = 6000 um^2

⇒ gate counts = 6000 / (5.04x1.98) = 601 gates

補充:

Instance Count: 261 cells
design_vision> sizeof_collection [get_cells -hier *]
261

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Setting Design Constraint

- Optimization Constraints
  - Basic clock constraints concept
  - Constraints & STA for Special Circuit
  - Constraints for Power & Area
- Design Rule Constraint
- Final check constraints before compile
- Make synopsys design constraint for APR tool

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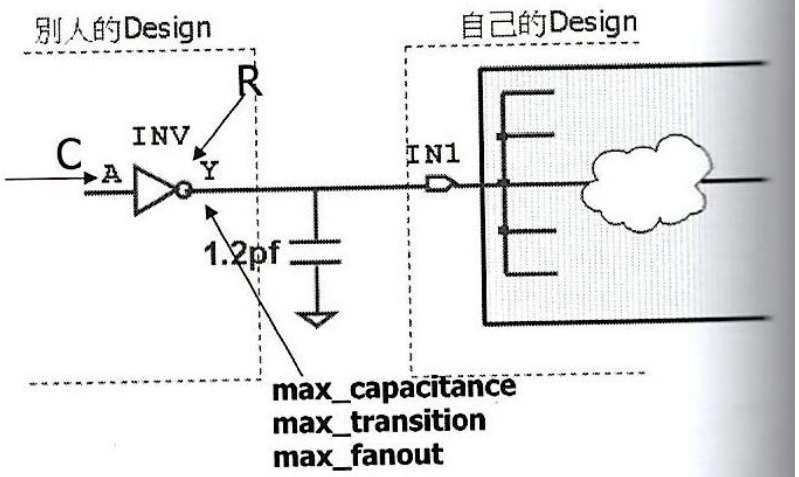
Design Rule Constraints

- Vendors impose design rules that restrict how many cells are connected to one another based on capacitance, transition and fanout
- You may apply more conservative design rules to:
  - Anticipate the interface environment your block will see
  - Prevent the design from operating cells close to their limits, where performance degrades rapidly

DRC Syntax:

```
set_max_capacitance 2 [get_ports IN1]
set_max_transition 0.2 [get_ports IN1]
set_max_fanout 6 [get_ports IN1]
```



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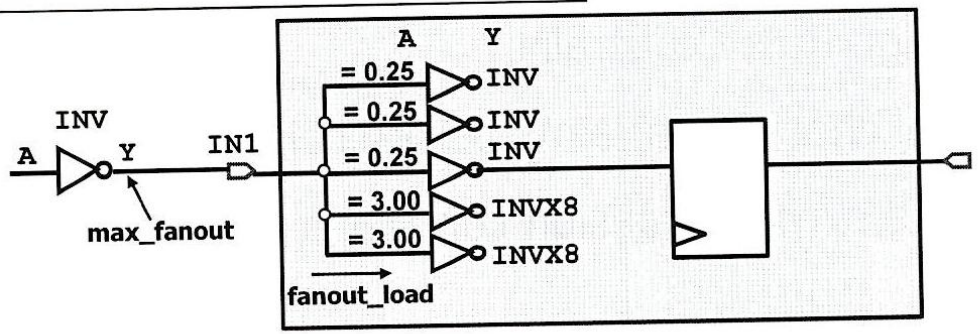
Set_max_fanout

`set_max_fanout` uses `fanout_load`, NOT absolute fanout number!

Sum of `fanout_load` on a port must be less than `max_fanout` design rule constraint

```
set_max_fanout 6 [get_ports IN1]
```



Some cell/pins have no `fanout_load` attribute:

- DC then checks library for `default_fanout_load` attribute
- If neither exists, DC assumes a value of zero which is equal to unconstrained

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Setting Design Constraint

- Optimization Constraints
  - Basic clock constraints concept
  - Constraints & STA for Special Circuit
  - Constraints for Power & Area
- Design Rule Constraint
- Final check constraints before compile
- Make synopsys design constraint for APR tool

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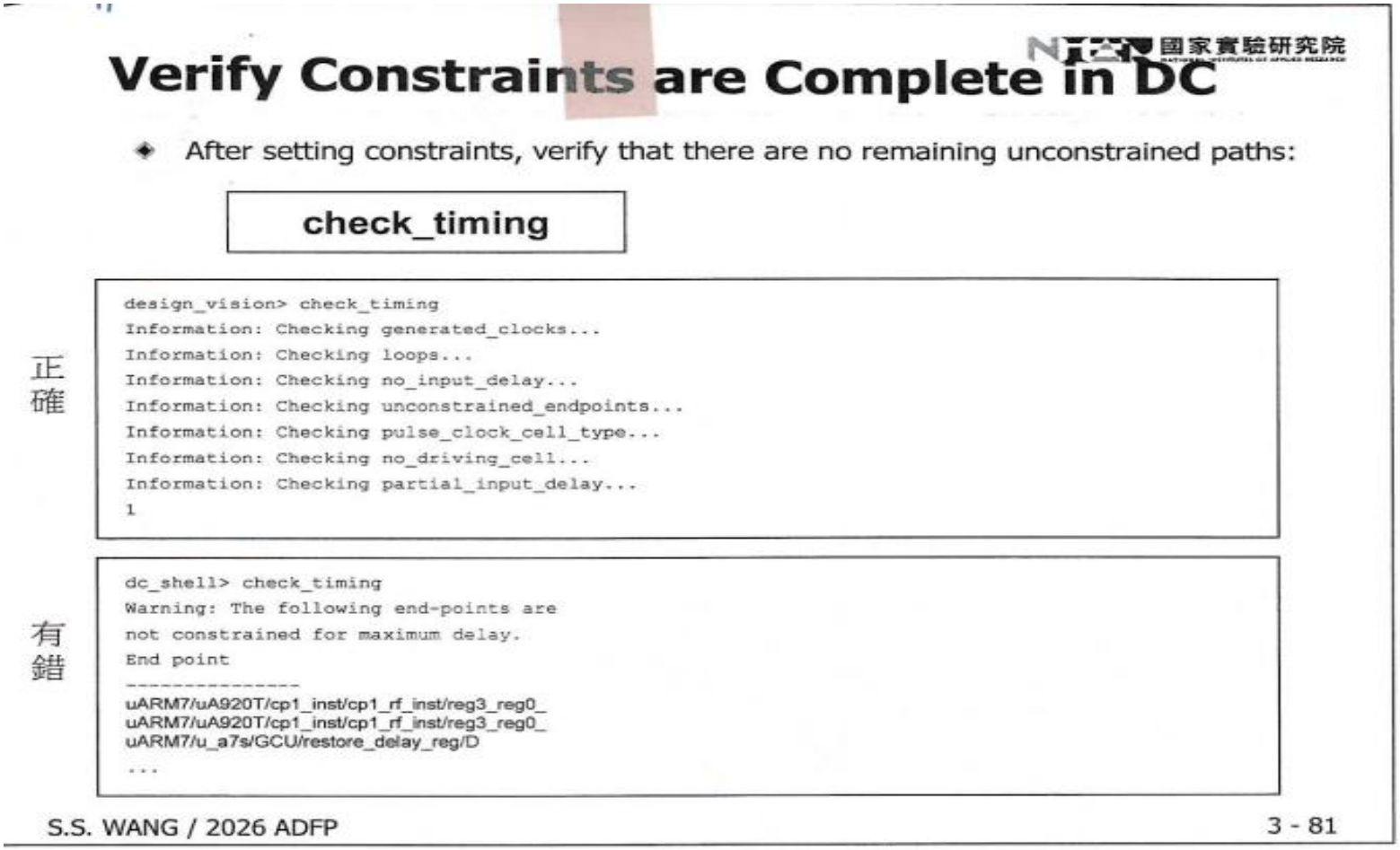
Verify Constraints are Complete in DC



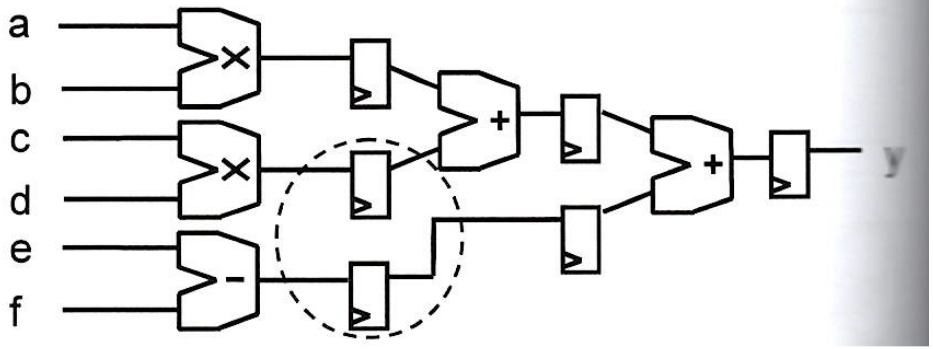
Check Design

- After you set up the deign attributes & design constraints, we recommend the next step is to check design
- *Design/Check Design*
- Maybe you will meet the warning message as follow

```
module display( ….. );
…
conveter  U1( … );
conveter  U2( … );
conveter  U3( … );
conveter  U4( … );
…
endmodule
```

```
Warning: Design 'converter' is instantiated 4 times.  (LINT-45)
         Cell 'convml' in design 'display'
         Cell 'convmm' in design 'display'
         Cell 'convsl' in design 'display'
         Cell 'convsm' in design 'display'
design_vision-t>
```

Example: multiple instance



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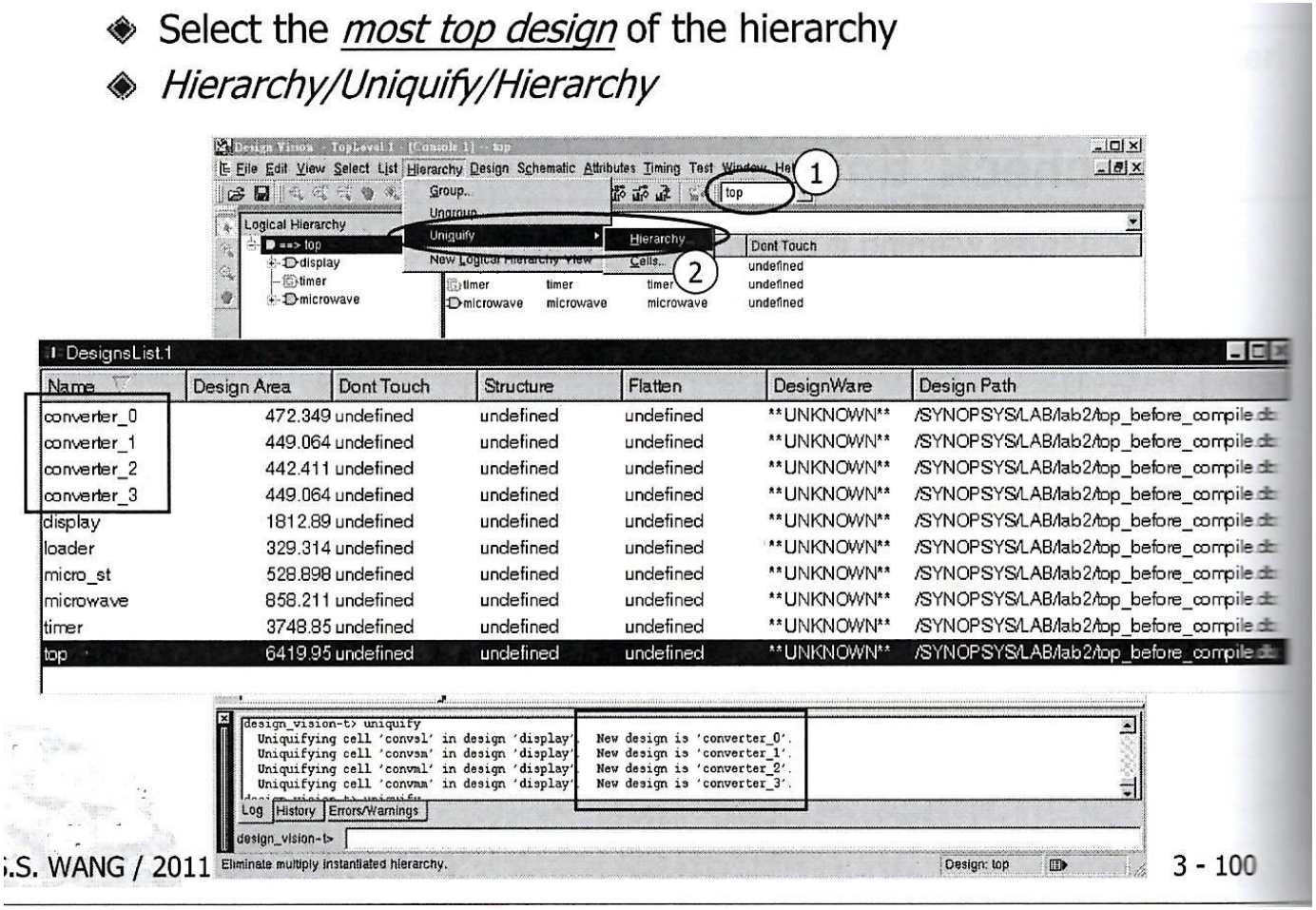
Solved the Multiple Instance:uniquify

- Select the *most top design* of the hierarchy
- *Hierarchy/Uniquify/Hierarchy*



Constraints Priority

- During the optimization, there exists a constraint priority relationship
  1. Design Rule Constraint
     (max_transition, max_fanout, max_capacitance)
  2. Timing constraint (max_delay, min_delay)
  3. Power constraint
  4. Area constraint
- Use *set_cost_priority* command to modify the order

  set_cost_priority [-default] [-delay] [cost_list]

  - -default : the compile uses its default priority.
  - -delay : Specifies that max_delay has higher priority than the max design rules.
  - cost_list : Specifies a list of costs from the following: max_delay, min_delay, max_transition, max_fanout, max_design_rules.
- ex:

  set_cost_priority {max_fanout max_capacitance max_delay}

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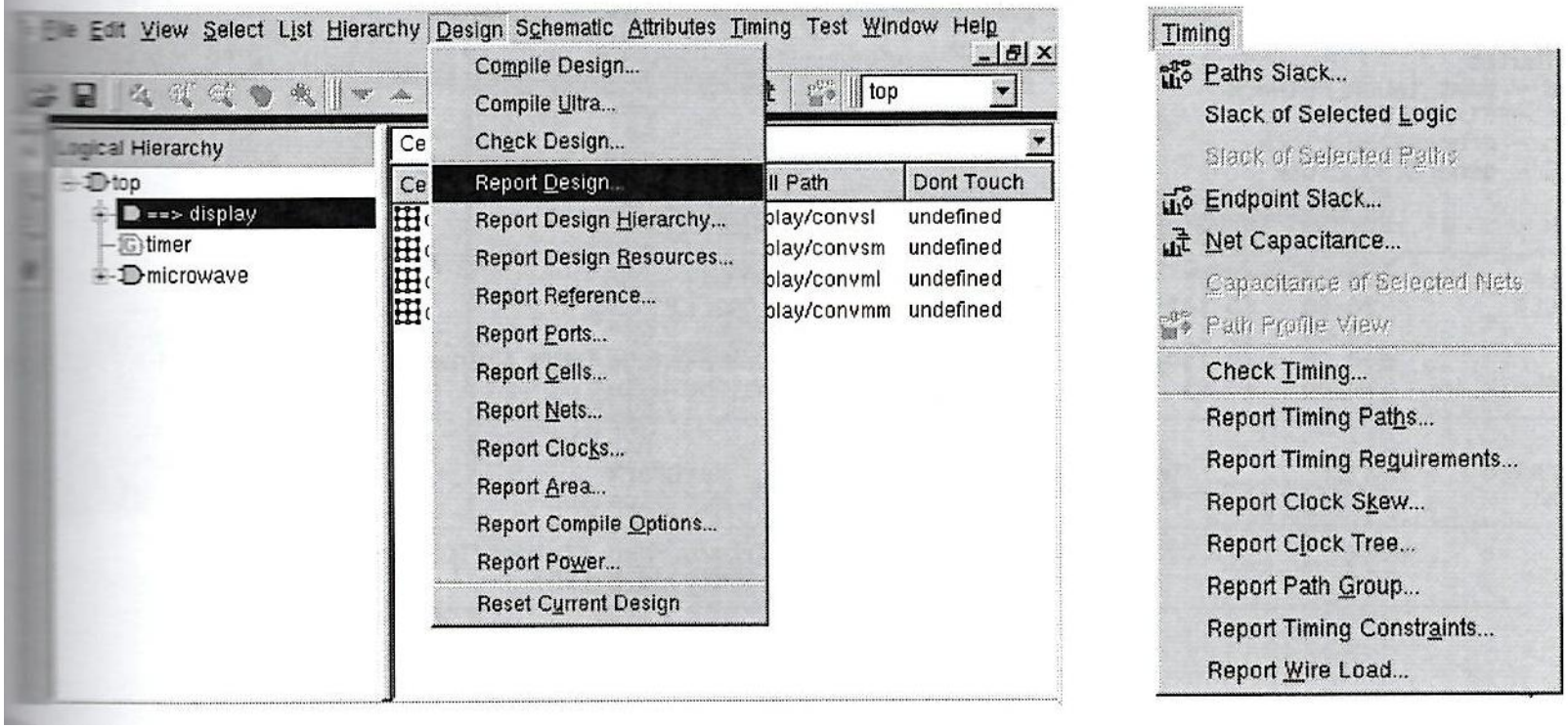
Check Constraints

- Use the following reports to check constraints before compiling
- *Design/Report*



Report Design

In this report you can check the operating condition and wire load model

```
****************************************
Report : design
Design : top
Version: D-2010.03-SP5
Date   : Tue Nov 30 16:25:02 2010
****************************************
Operating Conditions:

    Operating Condition Name   slow
    Library : slow
    Process :   1.00
    Temperature : 125.00
    Voltage :   0.90
    Interconnect Model : balanced_tree

Min Operating Conditions:

    Operating Condition Name   fast
    Library : fast
    Process :   1.00
    Temperature : -40.00
    Voltage :   1.10
    Interconnect Model : balanced_tree

Wire Loading Model:

    Selected manually by the user.

Name           :   tsmc090_wl10
Location       :   slow
Resistance     :   8.5e-08
Capacitance    :   0.00015
Area           :   0.7
Slope          :   66.667
Fanout   Length   Points Average Cap Std Deviation
------------------------------------------------------------
```


TAIWAN TECH

Report Clock

```
****************************************
Report : clocks
Design : NorthBridgePad
Version: C-2009.06-SP2
Date   : Thu Dec 10 18:35:37 2009
****************************************

Attributes:
    d - dont_touch_network
    f - fix_hold
    p - propagated_clock
    G - generated_clock

Clock          Period   Waveform        Attrs     Sources
-------------------------------------------------------------------------------
ARMCLK          10.00   {0 5}           d f       {ARMCLK}
CLCDCLK         20.00   {0 10}          d f       {CLCDCLK}
HCLK            10.00   {0 5}           d f       {HCLK}
ICLK           500.00   {0 250}         G d f     {uNorthBridge/uMDM_APB/uapb2i2c/u_freq_divider/ICLK_reg
dmc_fbclk_in    10.00   {0 5}           d f       {dmc_fbclk_in}
-------------------------------------------------------------------------------

Generated      Master         Generated      Master         Waveform
Clock          Source         Source         Clock          Modification
-------------------------------------------------------------------------------
ICLK           HCLK           {uNorthBridge/uMDM_APB/uapb2i2c/u_freq_divider/ICLK_reg/Q}
                                              HCLK           divide_by(50)
-------------------------------------------------------------------------------
```

Duty Cycle

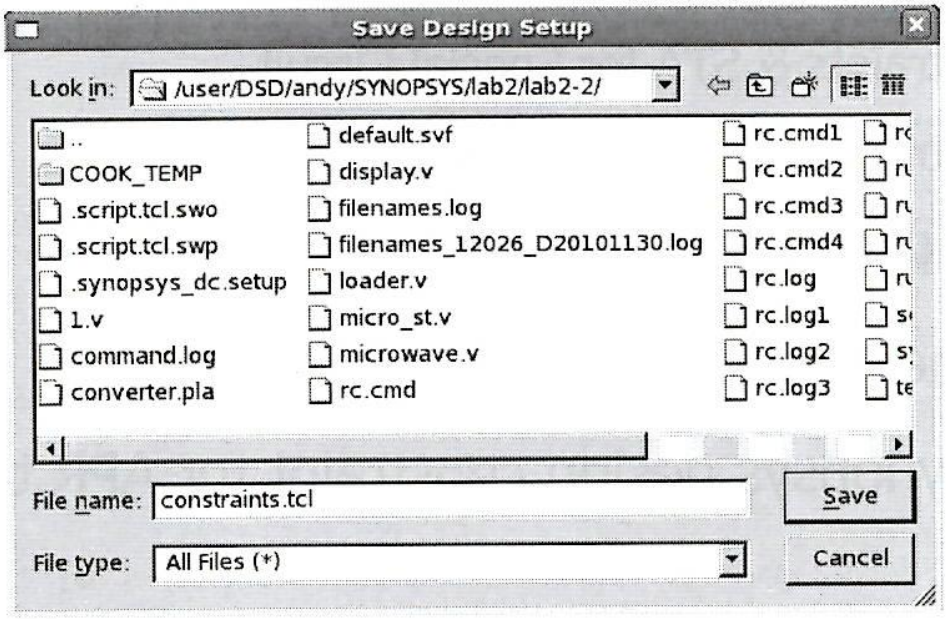
複習: Clock Gating只能用此指令才看得出來

```
report_clock_gating  -gating_elements
```


TAIWAN TECH

Save Constraints & Attributes

- Save attributes & constraints setting as the design setup file in dc_shell-t command format, use *File/Save Info/Design Setup*

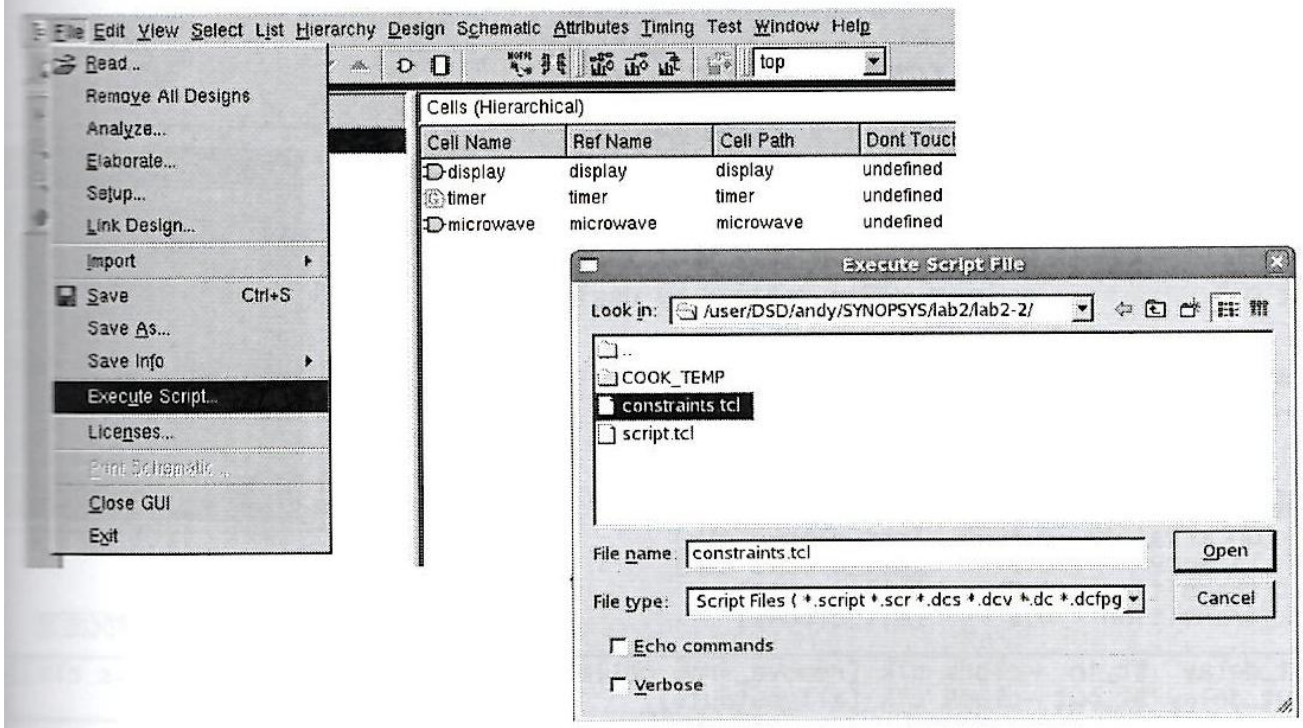



```
dc_shell-xg-t> write_script  -output  constraints.tcl
```


TAIWAN TECH

Execute Script File

- Execute dc_shell command script file, use *Setup/Execute Script* (GUI)



- Or use *"source your_script.tcl"* in design_vision command line

TAIWAN TECH

Setting Design Constraint

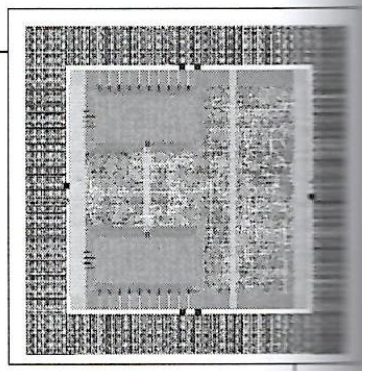
- Optimization Constraints
  - Basic clock constraints concept
  - Constraints & STA for Special Circuit
  - Constraints for Power & Area
- Design Rule Constraint
- Final check constraints before compile
- Make synopsys design constraint for APR tool

TAIWAN TECH

Synopsys Design Constraint for APR

- Synopsys Design Constraint: CHIP.sdc can be used under:
  - IC Compiler
  - SOC Encounter
  - Design Compiler (CHIP Level Report)
    - write_sdc -version 1.7 CHIP.sdc

```
# CHIP Level Constraint

set cycle 20            ; #clock period defined by designer
set t_in  10            ; #input delay defined by designer
set t_out 0.5           ; #output delay defined by designer


set_operating_conditions -min fast  -max slow
set_wire_load_model -name tsmc18_wl10 -library slow

create_clock -period $cycle     [get_ports clk]
set_dont_touch_network          [get_clocks clk]
set_fix_hold                    [get_clocks clk]
set_clock_uncertainty  0.1      [get_clocks clk]
set_clock_latency      0.5      [get_clocks clk]
set_input_transition   0.5      [all_inputs]
set_clock_transition   0.5      [all_clocks]

set_input_delay  $t_in  -clock clk [remove_from_collection [all_inputs] [get_clocks clk]]
set_output_delay $t_out -clock clk [all_outputs]

set_load    40     [all_outputs]

set_case_analysis 0 [get_ports se]
```



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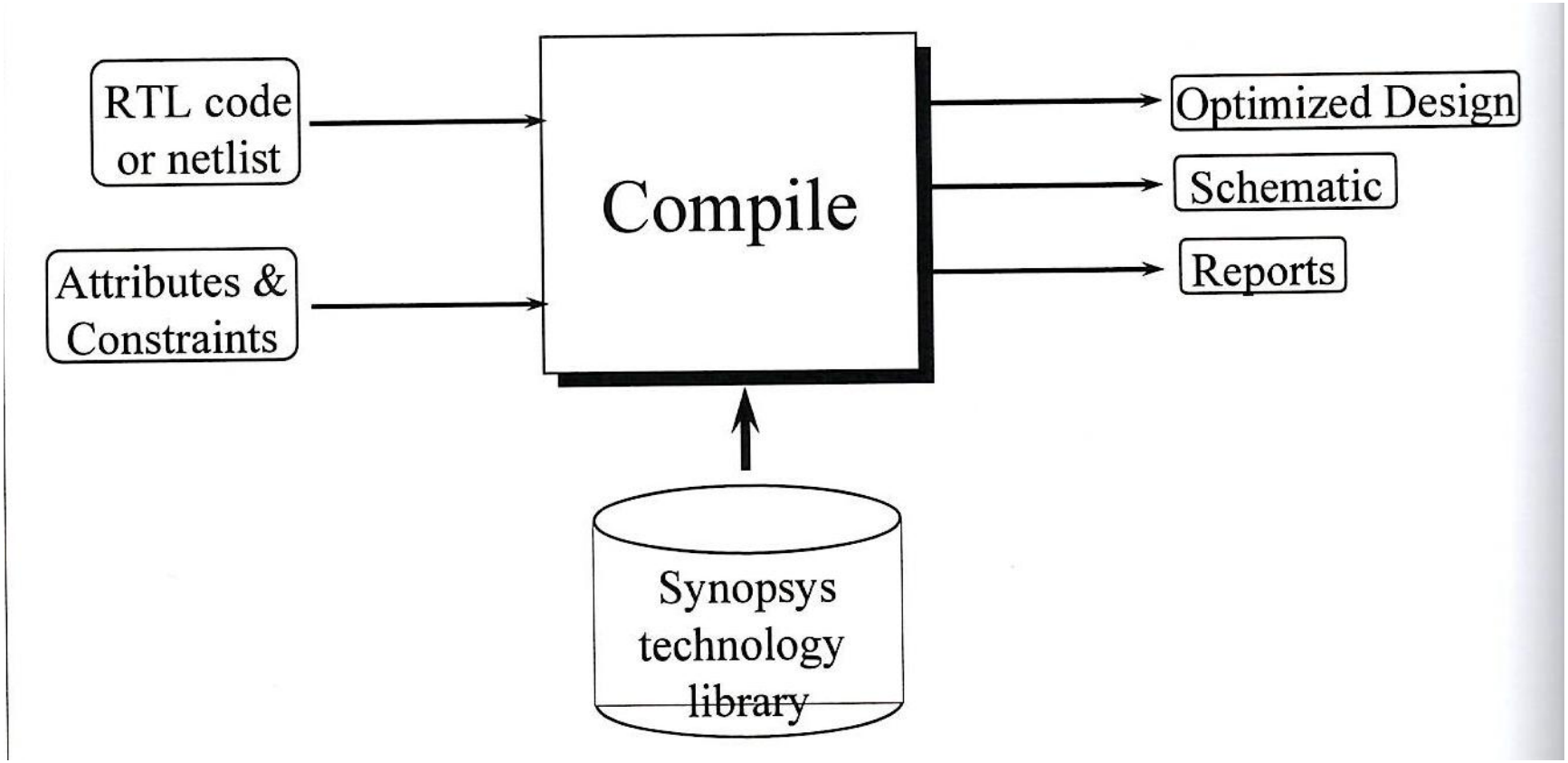
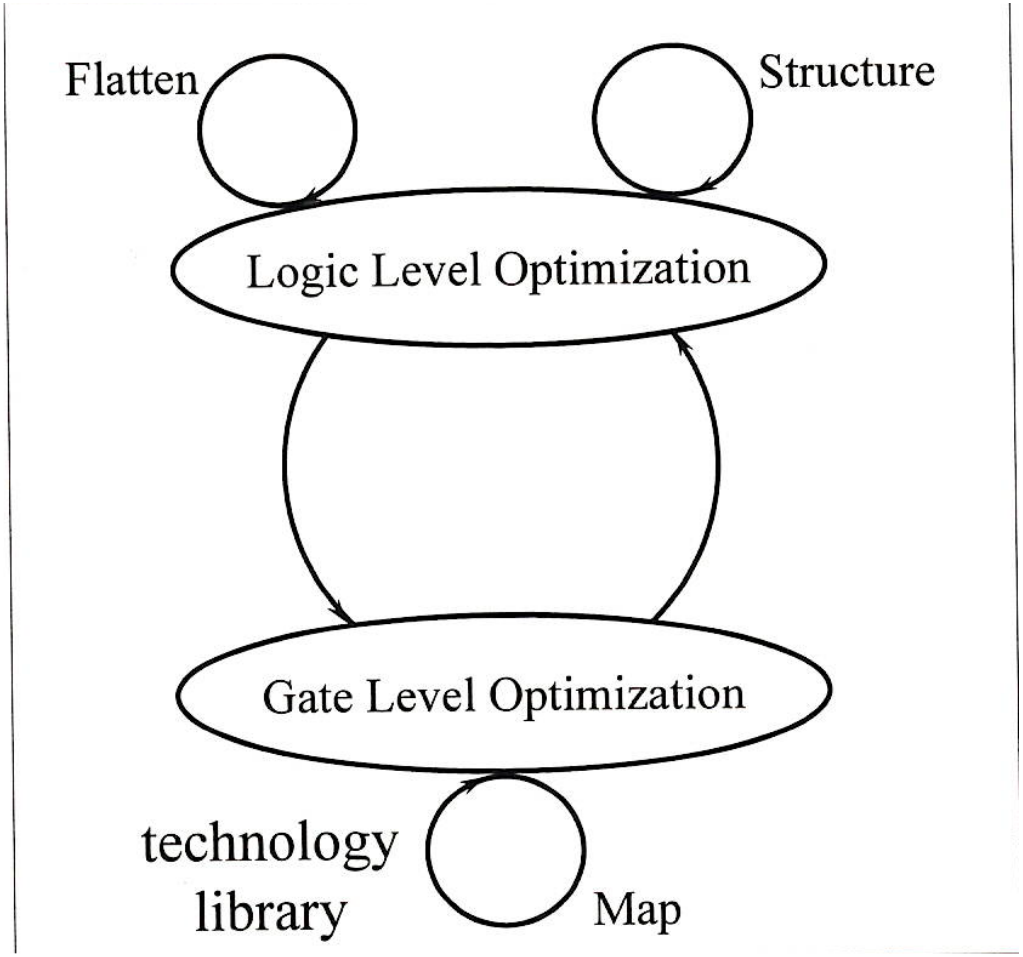
Function Mode

TAIWAN TECH

Compiler(1/2)

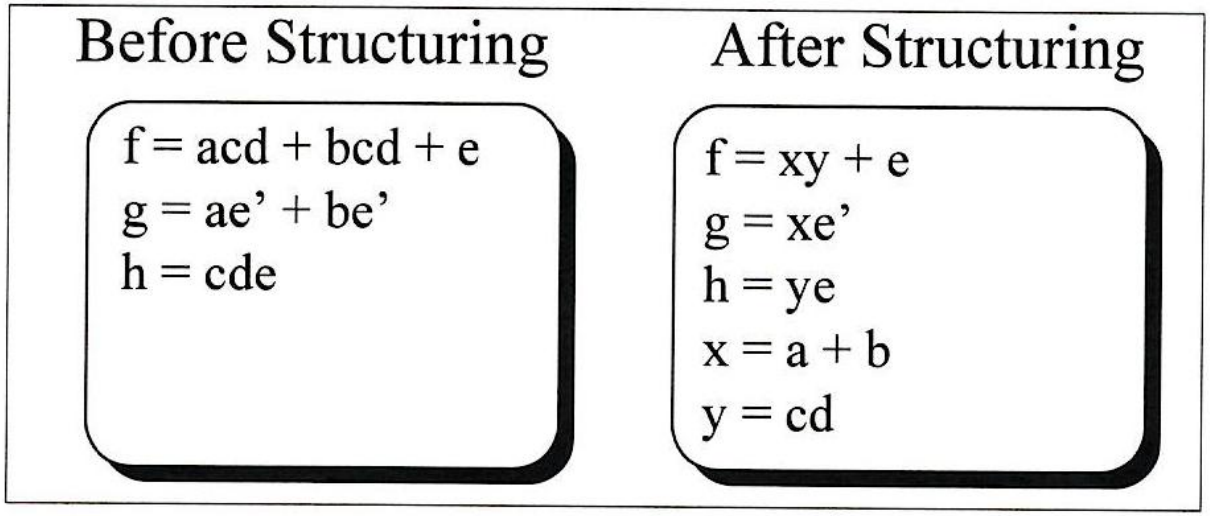

RTL code or netlist
Attributes & Constraints
Compile
Optimized Design
Schematic
Reports
Synopsys technology library

Compiler(2/2)


Flatten
Structure
Logic Level Optimization
Gate Level Optimization
technology
library
Map

Structure

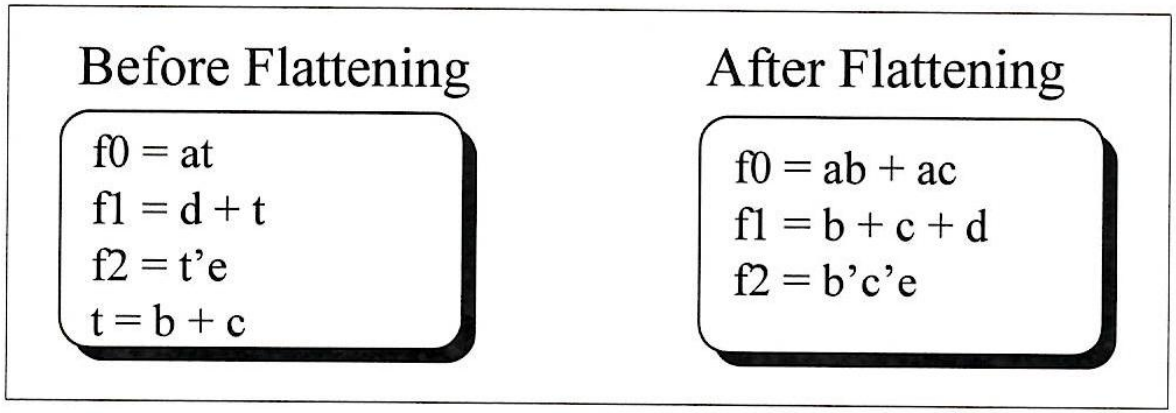
- Factors out common sub-expression as intermediate variable
- Useful for *speed* optimization as well as *area* optimization
- The *default* logic-level optimization strategy; suitable for structured circuits (e.g. adders and ALU's)
- Example:



TAIWAN TECH

Flatten

- Flatten is default OFF
- Remove all intermediate variable
- Result a two-level sum-of-product (SOP) form
  - Note: it doesn't mean that you will have a 2-level hardware due to *library limitations*
- *Flatten is default OFF* ;Use when you have a timing goal and have don't cares(x) in your HDL code.
- Example:

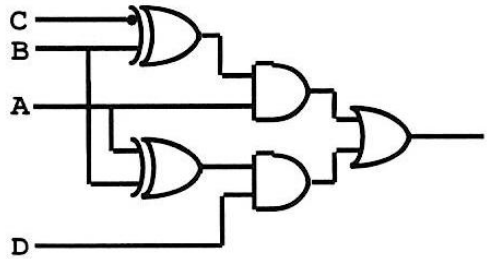


TAIWAN TECH

Structuring & Flattening(review)

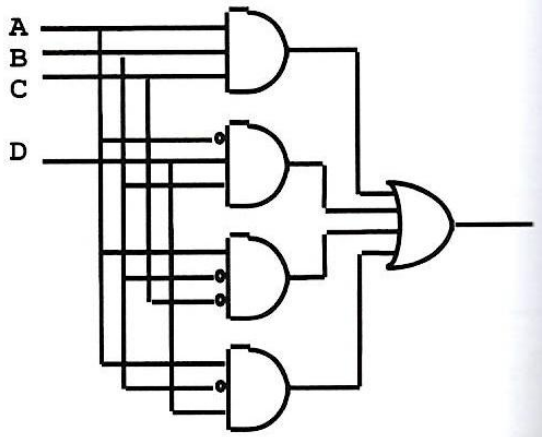
Structuring

- Structure is default ON
- Factors out common subexpression as intermediate variable
- Can help both area and speed of a design
- Suitable for structured circuits



Flattening

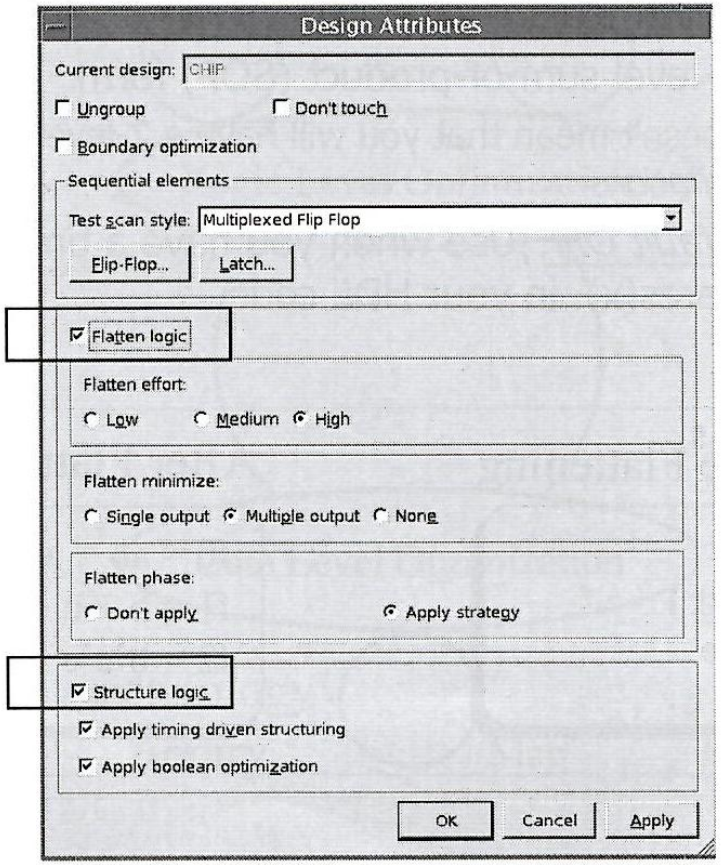
- Flatten is default OFF
- Remove all intermediate variable
- Result a two-level sum-of-product form
- Suitable for unstructured circuits



TAIWAN TECH

Apply Structure & Flatten

- *Attributes/Optimization Directives/Design*



TAIWAN TECH

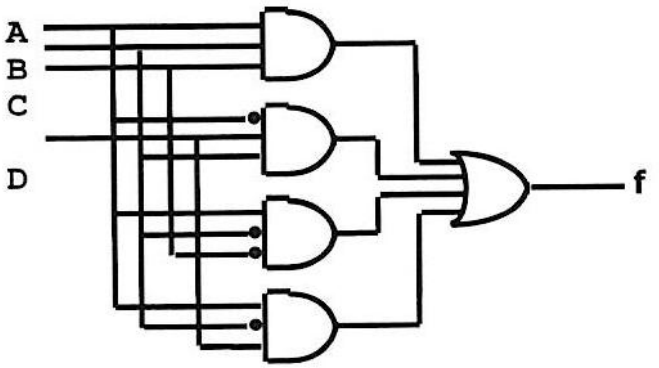
Structure Options

- Two options can be used:
  - *set_structure -timing flag*
    - flag can be true or false
    - true (*by default*) considers timing constraints during structuring
  - *set_structure –boolean flag*
    - flag can be true or false
    - *Default is "false"*, true indicates to use boolean algebra to reduce size of a design. (a)(~a) = 0, (a) + (~a) =1, (a) + (a) = a

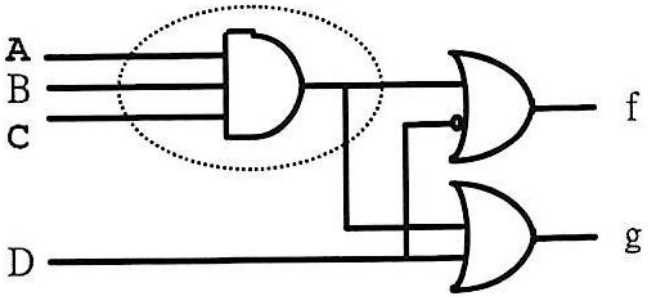
TAIWAN TECH

Flatten Options

- Three options can be used:
  - *set_flatten -effort effort*
    - effort can be low, medium or high
  - *set_flatten –minimize method*
    - method can be none, single, or multiple
    - Single: Single output minimization works on each output , it's generally faster but increases area and loading on inputs
    - Multiple: Multiple output shares product terms between outputs. It's generally more area efficient
  - *set_flatten –phase flag*
    - flag can be true or false
    - true indicates flatten will also evaluates the phase solution of a karnaugh map.



(single)



(multiple)

TAIWAN TECH

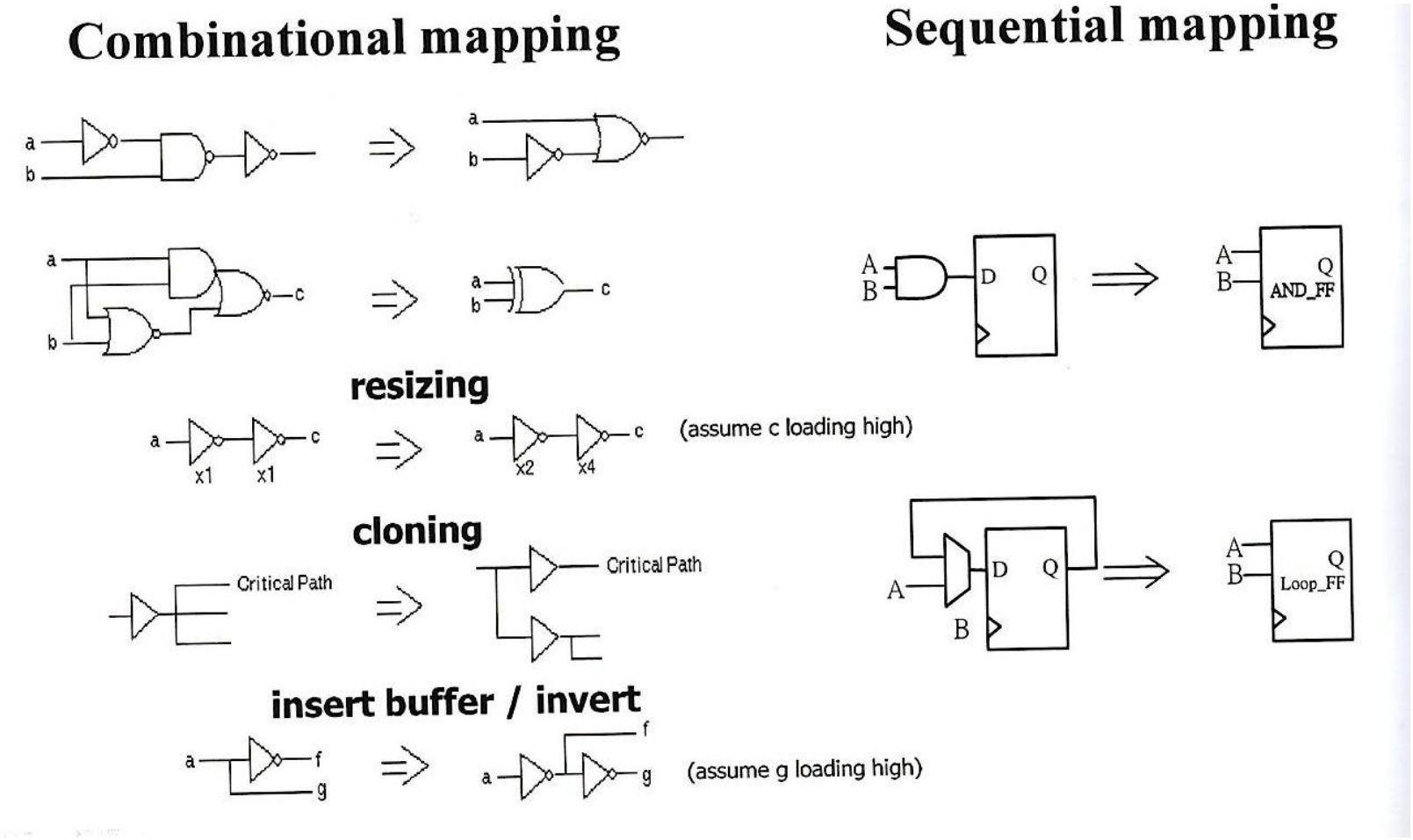
Gate Level Optimization - Mapping

- **Combination mapping**
  - Mapping rearranges components, combining and re-combining logic into different components
  - May use different algorithms such as cloning, resizing or buffering
  - Try to meet the design rule constraints and and timing/area goals

- **Sequential Mapping**
  - Optimize the mapping to sequential cells from technology library
  - Analyze combinational surrounding a sequential cell to see if it can absorb the logic attribute with HDL
  - Try to save speed and area by using a more complex sequential cell

TAIWAN TECH

Mapping

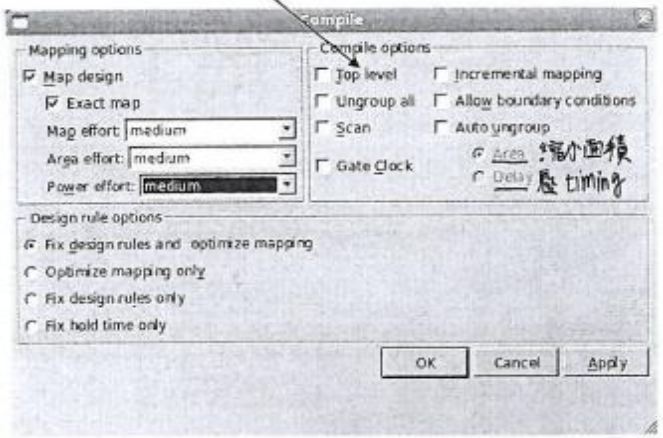

Combinational mapping
Sequential mapping
a
b
c
D
Q
A
B
AND_FF
resizing
x1
x2
x4
(assume c loading high)
cloning
Critical Path
Loop_FF
insert buffer / invert
f
g
(assume g loading high)

Mapping Effort

Mapping Effort

- Three effort levels, *low*, *medium*, *high*, determine relative amount of CPU time spent during mapping phase of compile
  - low - quicker synthesis, does not do all algorithms
  - medium - default, good for many design
  - high - it does *critical path re-synthesis*, but it will use more CPU time; in some cases the action of compile will not terminate

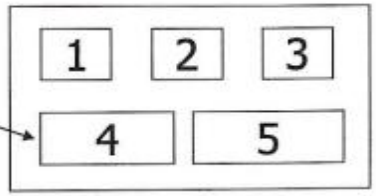
set compile_top_all_paths true




```
set_optimize_registers  true -designs u_4
compile_ultra  -retime  -no_autoungroup
```

retiming

ex:



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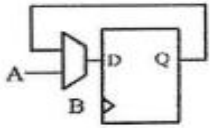
High Performance Designs(DC-Ultra)

set_cost_priority -min_delay ← 優先 hold time、setup time

國家實驗研究院

High Performance Designs (DC-Ultra)

Use **compile_ultra** { -timing 20%↓ / -area 4%↓
The full strength of DC Ultra in a single command

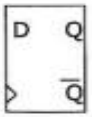


- A push-button solution for timing critical, ultra high performance designs
- Total has 8 options:
  - -scan # For test ready compile
  - -no_autoungroup # For turning off the auto-ungrouping feature
  - -no_boundary_optimization # For turning off the boundary_optimization feature

默認 →

Advanced Feature: 20|X Ver

2020↑ Ver



- -area_high_effort_script # golden solution for area (**optimize_netlist -area**)
- -timing_high_effort_script # golden solution for timing (**set compile_timing_high_effort true**)
- -no_seq_output_inversion #supress sequential output inversion
- -incremental_mapping = -inc #incremental compile
- -retime #support retiming engine

- User can define the maximum block size for auto-ungroup:
  - **set compile_auto_ungroup_area_num_cells 50 (default = 30)**
  - **set compile_auto_ungroup_delay_num_cells 100 (default = 500)**

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TAIWAN TECH

Start of Optimization

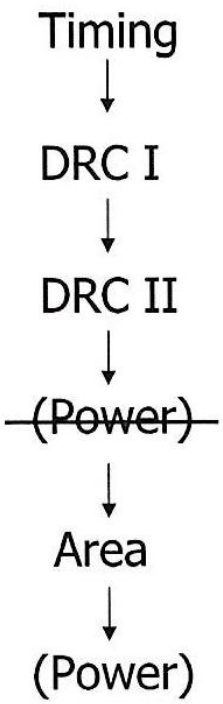
```
design_vision-xg-t> compile -boundary_optimization  -map_effort high -area_effort high

Loading target library 'slow'
Loading target library 'fast'
Loading design 'top'

Beginning Pass 1 Mapping
------------------------
Structuring 'converter_0'
Mapping 'converter_0'
Structuring 'converter_1'
Mapping 'converter_1'
Structuring 'converter_2'
Mapping 'converter_2'
Structuring 'converter_3'
Mapping 'converter_3'
Structuring 'timer'
Mapping 'timer'
Structuring 'loader'
Mapping 'loader'
Structuring 'micro_st'
Mapping 'micro_st'
```


TAIWAN TECH

Mapping Optimzation(1/2)



Beginning Delay Optimization Phase

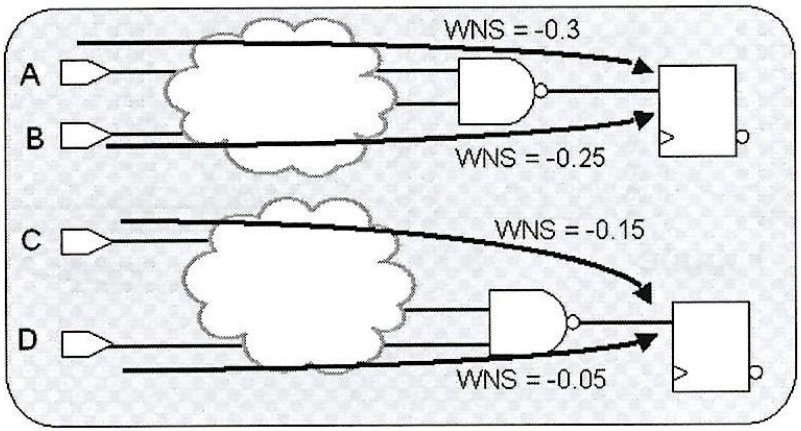
ELAPSED TIME	AREA	(WNS) WORST NEG SLACK	(TNS) TOTAL NEG SLACK	DESIGN RULE COST	ENDPOINT
0:00:31	10265.6	0.09	0.5	179.2	timer/sec_lsb_next_reg[0]/SE
0:00:31	10265.6	0.09	0.5	179.2	timer/sec_lsb_next_reg[0]/SE
0:00:32	10278.9	0.09	0.5	179.2	timer/sec_lsb_next_reg[0]/SE
0:00:32	10202.4	0.09	0.5	178.7	timer/sec_lsb_next_reg[0]/SE
0:00:32	10185.8	0.09	0.5	178.7	timer/sec_lsb_next_reg[0]/SE
0:00:33	10209.1	0.04	0.2	176.7	timer/sec_lsb_next_reg[0]/SE
0:00:33	10209.1	0.04	0.2	176.7	timer/min_msb_next_reg[0]/SE
0:00:33	10195.8	0.04	0.2	176.7	timer/sec_lsb_next_reg[3]/SE
0:00:33	10195.8	0.04	0.2	176.7	timer/sec_lsb_next_reg[3]/SE
0:00:33	10212.4	0.03	0.2	174.7	timer/sec_lsb_next_reg[3]/SE
0:00:34	10185.8	0.03	0.1	174.7	timer/min_lsb_next_reg[2]/SE
0:00:34	10182.5	0.03	0.1	174.7	timer/sec_msb_next_reg[2]/D
0:00:35	10235.7	0.02	0.1	174.7	timer/min_msb_next_reg[2]/SE
0:00:35	10235.7	0.02	0.1	173.7	timer/sec_lsb_next_reg[3]/SE
0:00:35	10288.9	0.02	0.1	173.7	timer/min_lsb_next_reg[2]/SE
0:00:35	10229.0	0.02	0.1	173.7	timer/min_lsb_next_reg[2]/SE
0:00:35	10242.3	0.01	0.0	173.7	timer/min_msb_next_reg[2]/SE
0:00:36	10222.4	0.01	0.0	173.4	timer/sec_msb_next_reg[2]/D
0:00:36	10222.4	0.01	0.0	173.4	timer/sec_msb_next_reg[2]/D
0:00:36	10215.7	0.01	0.0	173.4	timer/sec_lsb_next_reg[0]/SE
0:00:36	10215.7	0.00	0.0	173.4	timer/sec_lsb_next_reg[0]/SE

Timing

TAIWAN TECH

Cost Function

- Constraint difference: $\triangle = actual - target$
- Design Compiler attempts to reduce the cost functions to 0
- Design Rule Constraint cost function
  - $\mathbf{Cost_{DRC}} = \Sigma\triangle_{max_fanout} + \Sigma\triangle_{max_transition} + \Sigma\triangle_{max_cap}$
- Worst NEG Slack (WNS) refer to the worst timing in the single timing path. DC will do its best to optimize all of the timing path. So, you can see the WNS gradually close to "0".
- Total Negative Slack (TNS) refer to the total timing in the timing path.



TNS=-0.55

TNS=-0.2

NG / 2011.07

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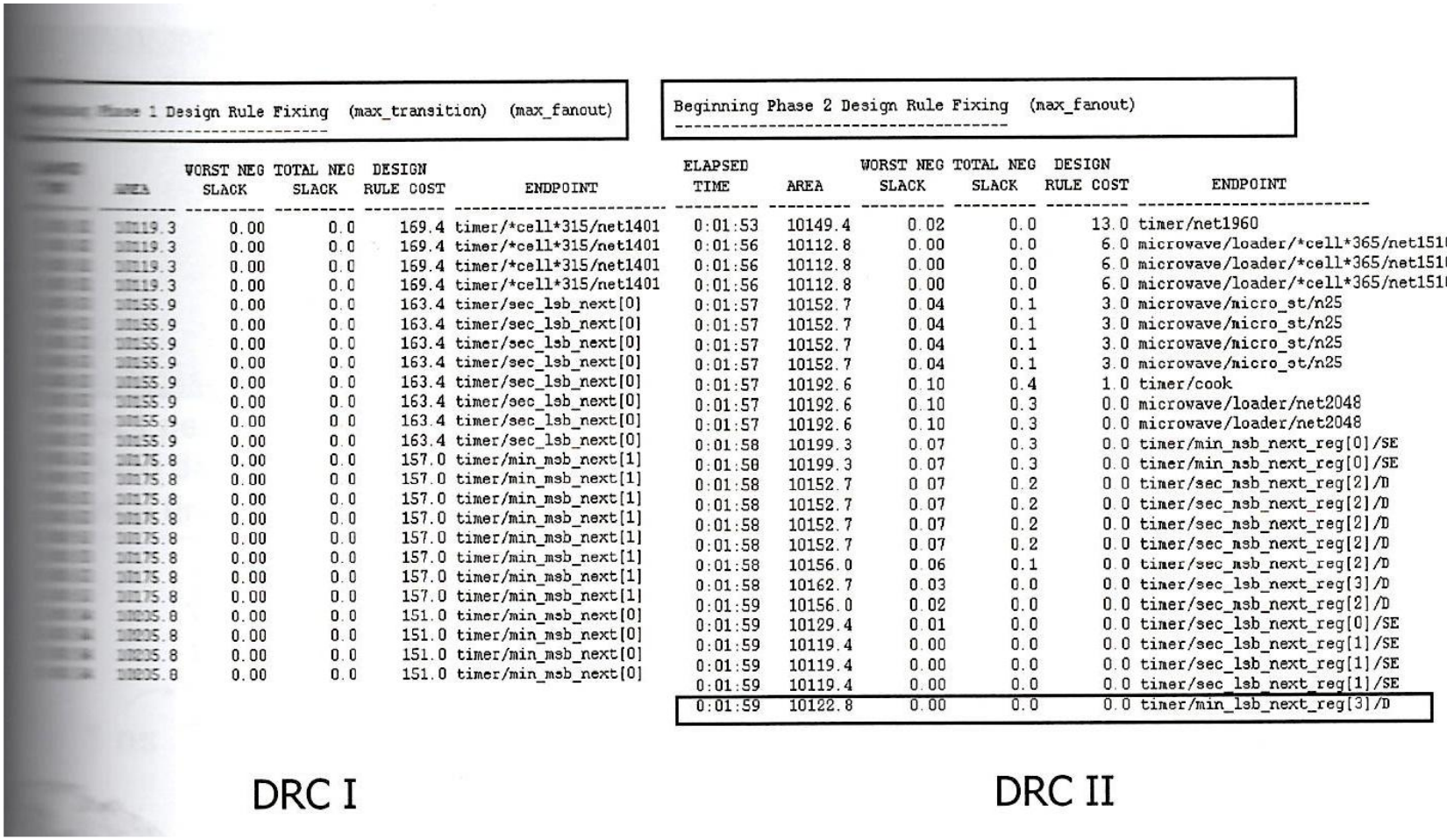
Mapping Optimzation(2/2)



Phase 1 Design Rule Fixing (max_transition) (max_fanout)

AREA	WORST NEG SLACK	TOTAL NEG SLACK	DESIGN RULE COST	ENDPOINT
10119.3	0.00	0.0	169.4	timer/*cell*315/net1401
10119.3	0.00	0.0	169.4	timer/*cell*315/net1401
10119.3	0.00	0.0	169.4	timer/*cell*315/net1401
10119.3	0.00	0.0	169.4	timer/*cell*315/net1401
10155.9	0.00	0.0	163.4	timer/sec_lsb_next[0]
10155.9	0.00	0.0	163.4	timer/sec_lsb_next[0]
10155.9	0.00	0.0	163.4	timer/sec_lsb_next[0]
10155.9	0.00	0.0	163.4	timer/sec_lsb_next[0]
10155.9	0.00	0.0	163.4	timer/sec_lsb_next[0]
10155.9	0.00	0.0	163.4	timer/sec_lsb_next[0]
10155.9	0.00	0.0	163.4	timer/sec_lsb_next[0]
10155.9	0.00	0.0	163.4	timer/sec_lsb_next[0]
10175.8	0.00	0.0	157.0	timer/min_msb_next[1]
10175.8	0.00	0.0	157.0	timer/min_msb_next[1]
10175.8	0.00	0.0	157.0	timer/min_msb_next[1]
10175.8	0.00	0.0	157.0	timer/min_msb_next[1]
10175.8	0.00	0.0	157.0	timer/min_msb_next[1]
10175.8	0.00	0.0	157.0	timer/min_msb_next[1]
10175.8	0.00	0.0	157.0	timer/min_msb_next[1]
10175.8	0.00	0.0	157.0	timer/min_msb_next[1]
10205.8	0.00	0.0	151.0	timer/min_msb_next[0]
10205.8	0.00	0.0	151.0	timer/min_msb_next[0]
10205.8	0.00	0.0	151.0	timer/min_msb_next[0]
10205.8	0.00	0.0	151.0	timer/min_msb_next[0]

Beginning Phase 2 Design Rule Fixing (max_fanout)

ELAPSED TIME	AREA	WORST NEG SLACK	TOTAL NEG SLACK	DESIGN RULE COST	ENDPOINT
0:01:53	10149.4	0.02	0.0	13.0	timer/net1960
0:01:56	10112.8	0.00	0.0	6.0	microwave/loader/*cell*365/net1510
0:01:56	10112.8	0.00	0.0	6.0	microwave/loader/*cell*365/net1510
0:01:56	10112.8	0.00	0.0	6.0	microwave/loader/*cell*365/net1510
0:01:57	10152.7	0.04	0.1	3.0	microwave/micro_st/n25
0:01:57	10152.7	0.04	0.1	3.0	microwave/micro_st/n25
0:01:57	10152.7	0.04	0.1	3.0	microwave/micro_st/n25
0:01:57	10152.7	0.04	0.1	3.0	microwave/micro_st/n25
0:01:57	10192.6	0.10	0.4	1.0	timer/cook
0:01:57	10192.6	0.10	0.3	0.0	microwave/loader/net2048
0:01:57	10192.6	0.10	0.3	0.0	microwave/loader/net2048
0:01:58	10199.3	0.07	0.3	0.0	timer/min_msb_next_reg[0]/SE
0:01:58	10199.3	0.07	0.3	0.0	timer/min_msb_next_reg[0]/SE
0:01:58	10152.7	0.07	0.2	0.0	timer/sec_msb_next_reg[2]/D
0:01:58	10152.7	0.07	0.2	0.0	timer/sec_msb_next_reg[2]/D
0:01:58	10152.7	0.07	0.2	0.0	timer/sec_msb_next_reg[2]/D
0:01:58	10152.7	0.07	0.2	0.0	timer/sec_msb_next_reg[2]/D
0:01:58	10156.0	0.06	0.1	0.0	timer/sec_msb_next_reg[2]/D
0:01:58	10162.7	0.03	0.0	0.0	timer/sec_lsb_next_reg[3]/D
0:01:59	10156.0	0.02	0.0	0.0	timer/sec_msb_next_reg[2]/D
0:01:59	10129.4	0.01	0.0	0.0	timer/sec_lsb_next_reg[0]/SE
0:01:59	10119.4	0.00	0.0	0.0	timer/sec_lsb_next_reg[1]/SE
0:01:59	10119.4	0.00	0.0	0.0	timer/sec_lsb_next_reg[1]/SE
0:01:59	10119.4	0.00	0.0	0.0	timer/sec_lsb_next_reg[1]/SE
0:01:59	10122.8	0.00	0.0	0.0	timer/min_lsb_next_reg[3]/D

DRC I

DRC II

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Optimization Conclusion

Beginning Area-Recovery Phase

ELAPSED TIME	AREA	WORST NEG SLACK	TOTAL NEG SLACK	DESIGN RULE COST
0:02:00	10086.2	0.00	0.0	0.0
0:02:00	10066.2	0.00	0.0	0.0
0:02:00	10029.6	0.00	0.0	0.0
0:02:00	10019.6	0.00	0.0	0.0
0:02:00	10006.3	0.00	0.0	0.0
0:02:00	9973.0	0.00	0.0	0.0
0:02:00	9949.8	0.00	0.0	0.0
0:02:00	9923.1	0.00	0.0	0.0
0:02:01	9869.9	0.00	0.0	0.0
0:02:01	9859.9	0.00	0.0	0.0
0:02:01	9820.0	0.00	0.0	0.0
0:02:01	9820.0	0.00	0.0	0.0
0:02:01	9806.7	0.00	0.0	0.0
0:02:01	9793.4	0.00	0.0	0.0
0:02:01	9780.1	0.00	0.0	0.0
0:02:01	9753.5	0.00	0.0	0.0
0:02:01	9740.2	0.00	0.0	0.0
0:02:01	9710.2	0.00	0.0	0.0
0:02:01	9683.6	0.00	0.0	0.0
0:02:02	9657.0	0.00	0.0	0.0
0:02:02	9630.4	0.00	0.0	0.0
0:02:02	9603.8	0.00	0.0	0.0
0:02:02	9590.5	0.00	0.0	0.0
0:02:02	9563.8	0.00	0.0	0.0
0:03:08	7784.2	0.00	0.0	0.0
0:03:22	7770.9	0.00	0.0	0.0
0:03:22	7770.9	0.00	0.0	0.0
0:03:23	7764.2	0.00	0.0	0.0
0:03:25	7757.5	0.00	0.0	0.0
0:03:25	7757.5	0.00	0.0	0.0
0:03:27	7747.6	0.00	0.0	0.0
0:03:30	7724.3	0.00	0.0	0.0
0:03:31	7714.3	0.00	0.0	0.0
0:03:31	7711.0	0.00	0.0	0.0
0:03:31	7694.3	0.00	0.0	0.0
0:03:32	7691.0	0.00	0.0	0.0
0:03:32	7684.4	0.00	0.0	0.0
0:03:35	7667.7	0.00	0.0	0.0
0:03:44	7664.4	0.00	0.0	0.0
0:03:44	7651.1	0.00	0.0	0.0
0:03:48	7644.4	0.00	0.0	0.0
0:03:51	7627.8	0.00	0.0	0.0
0:03:56	7621.1	0.00	0.0	0.0
0:04:00	7611.2	0.00	0.0	0.0
0:04:00	7604.5	0.00	0.0	0.0
0:04:00	7601.2	0.00	0.0	0.0
0:04:02	7597.8	0.00	0.0	0.0
0:04:05	7587.9	0.00	0.0	0.0
0:04:07	7577.9	0.00	0.0	0.0
0:04:07	7564.6	0.00	0.0	0.0
0:04:07	7551.3	0.00	0.0	0.0

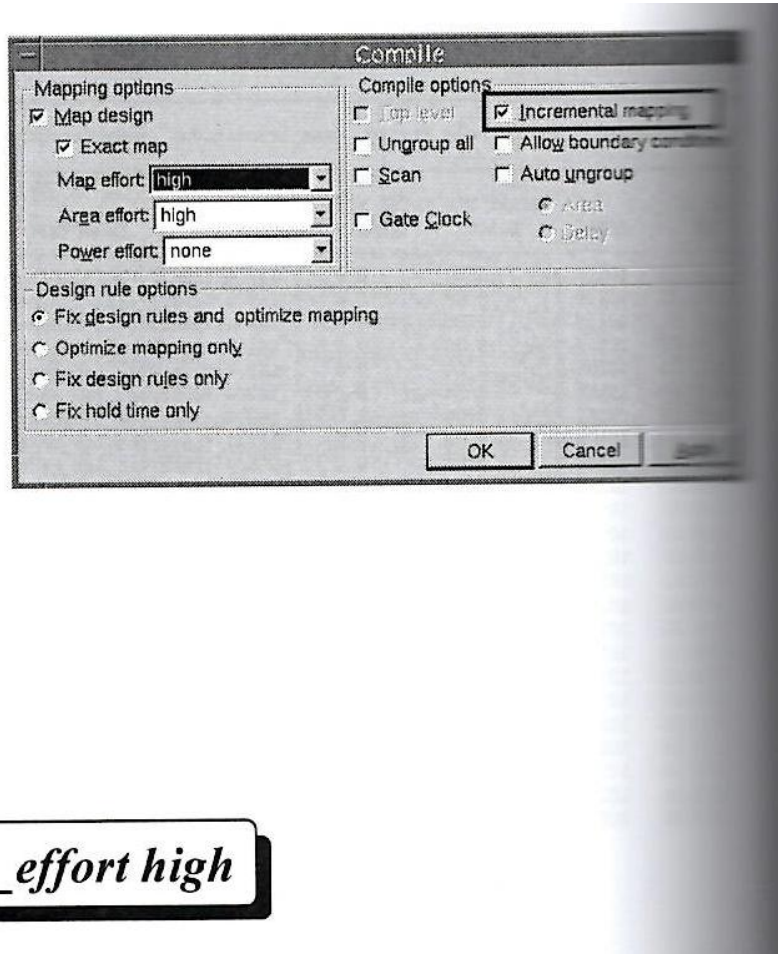
Optimization Complete

Area

TAIWAN TECH

How to Solve the Setup Time Violation

- Compile design with map effort *high*
  - it does *critical path re-synthesis*, but it will use more CPU time; in some cases the action of compile will not terminate
  - ***compile -map_effort high***



- Perform incremental gate level optimization but *no logic level* optimization

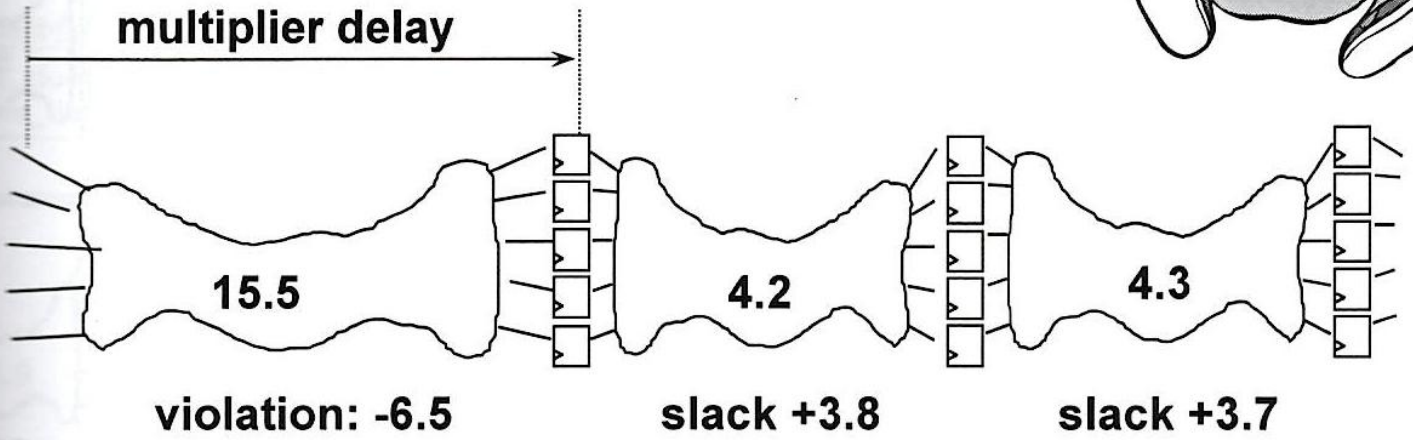
compile -incremental_mapping –map_effort high

TAIWAN TECH

Solved Critical Path Too Long:Retiming

- My Worst Negative Slack is still too large!
- My pipeline paths either
  - have plenty of slack
  - or violate timing!

clock period = 10 ns
clk->q delay = 1 ns
setup time = 1 ns



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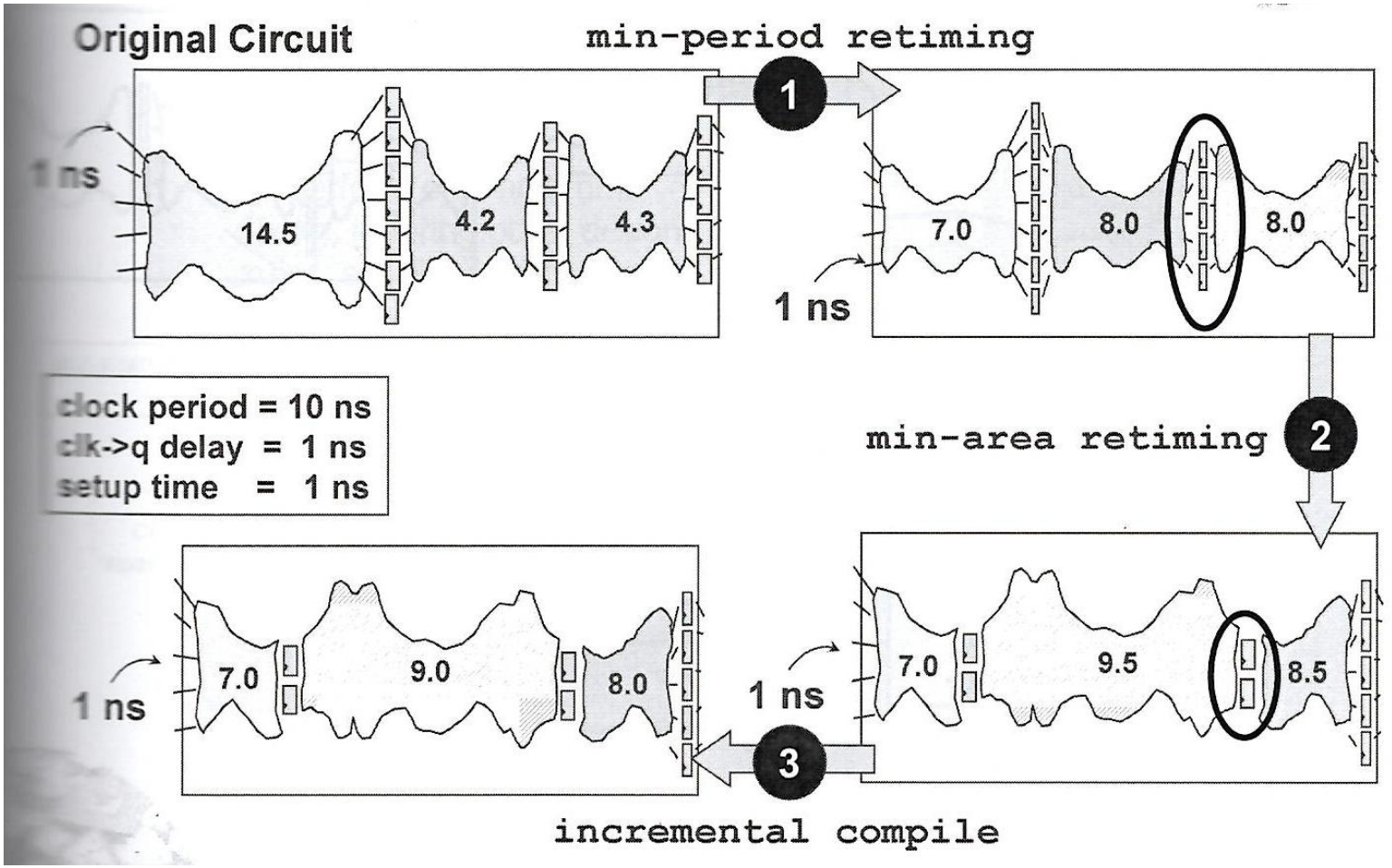
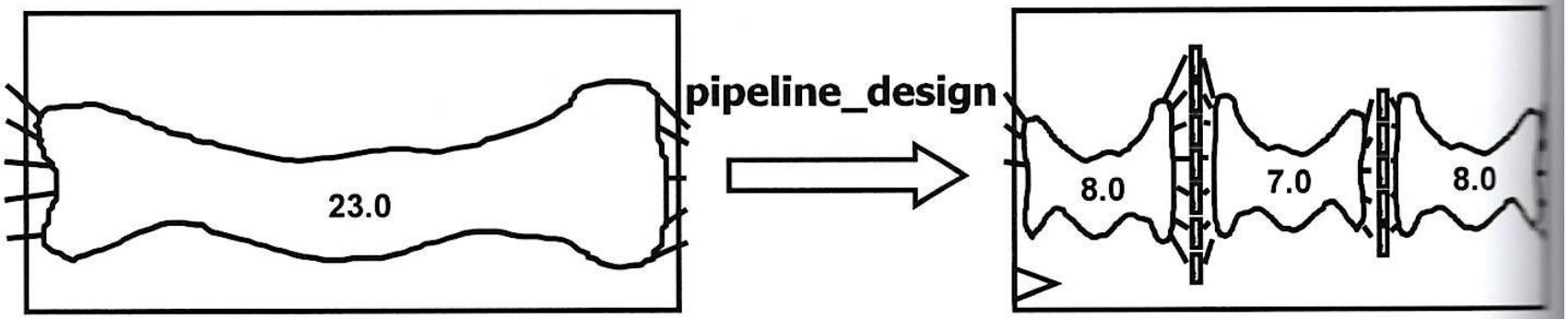
Retiming: optimize_registers



TAIWAN TECH

Solved Critical Path Too Long:Pipeline

- Command to create high-performance, pipelined designs
  - Extension of **`optimize_registers`**
  - Takes a combinational design and adds pipe stages
  - Easily experiment with clock rates and pipe lengths



- Other features:
  - Automatically inserts sync or async set or reset
  - Supports complex pipelines (e.g., enables for pipe stalls)

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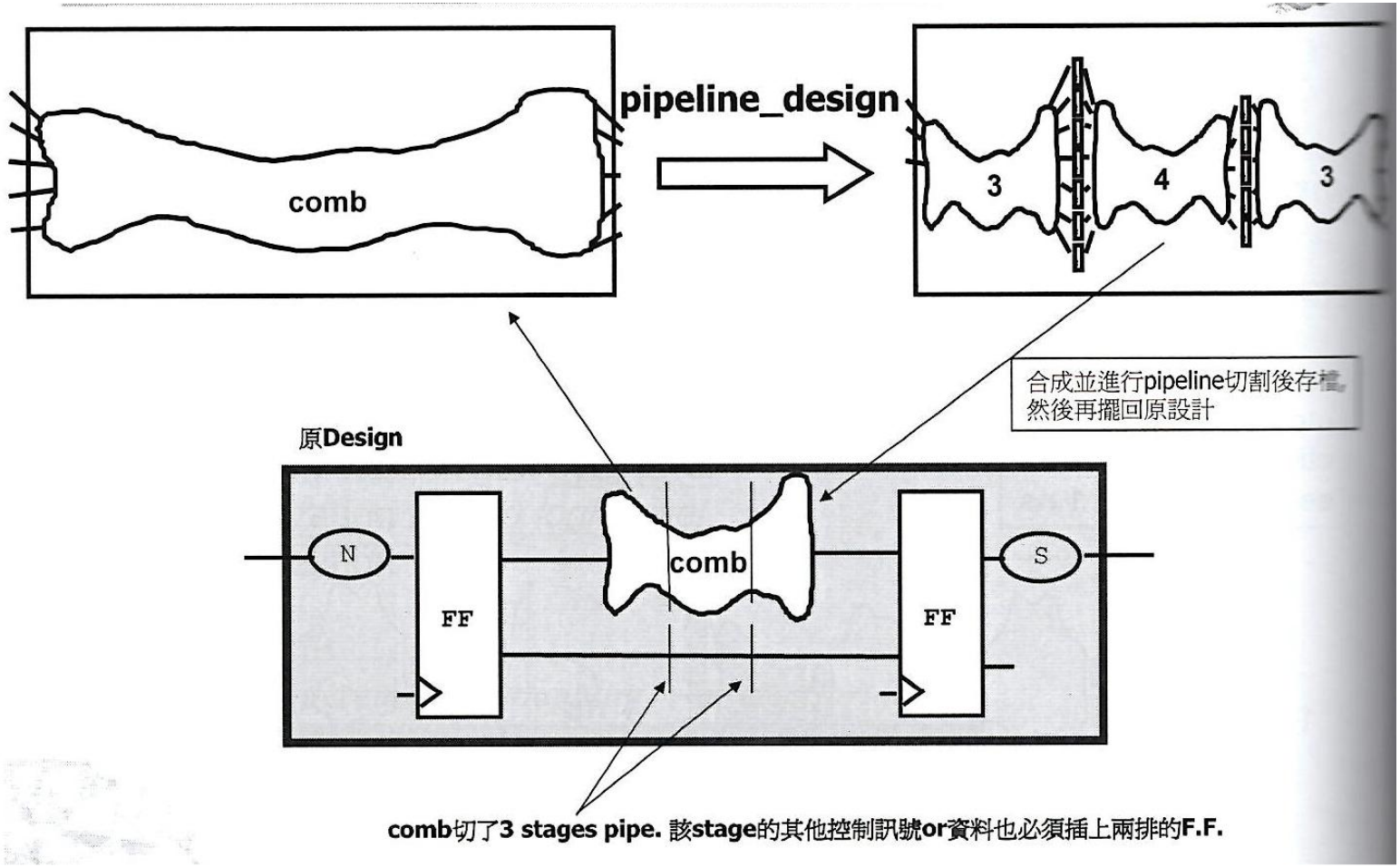
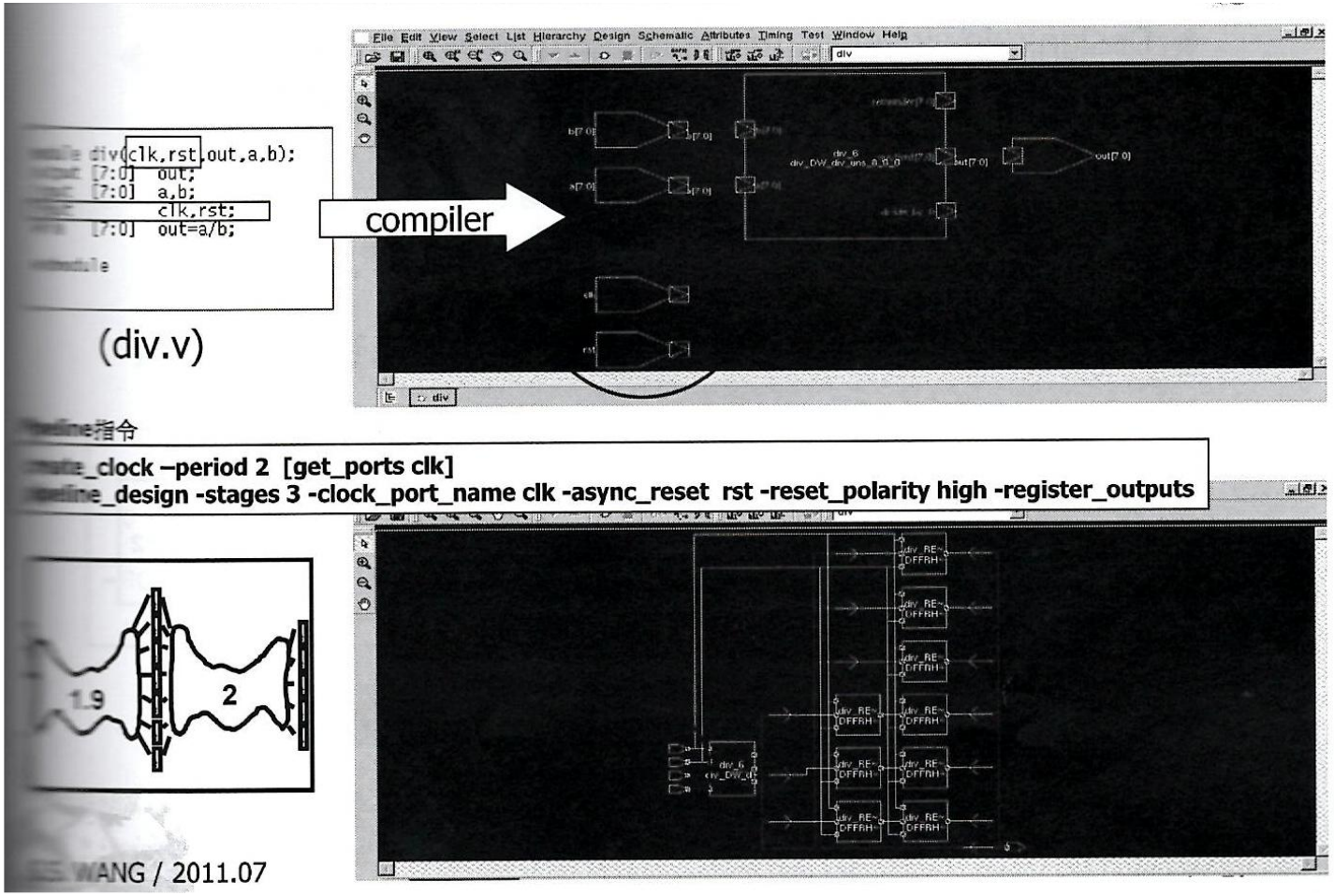
Pipeline 使用時機

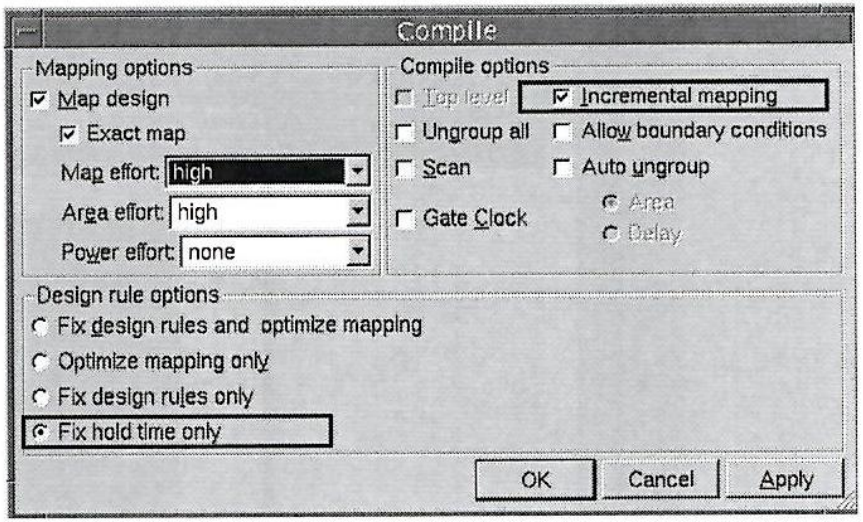


Pipelining Example:pipeline_design



Solved the Hold Time Violation

- How to solve this problem for best case :
  - **-only_hold_time**
    to perform only hold time fixing, ignoring other design rules. set_fix_hold command must be specified for hold time fixing to be performed.



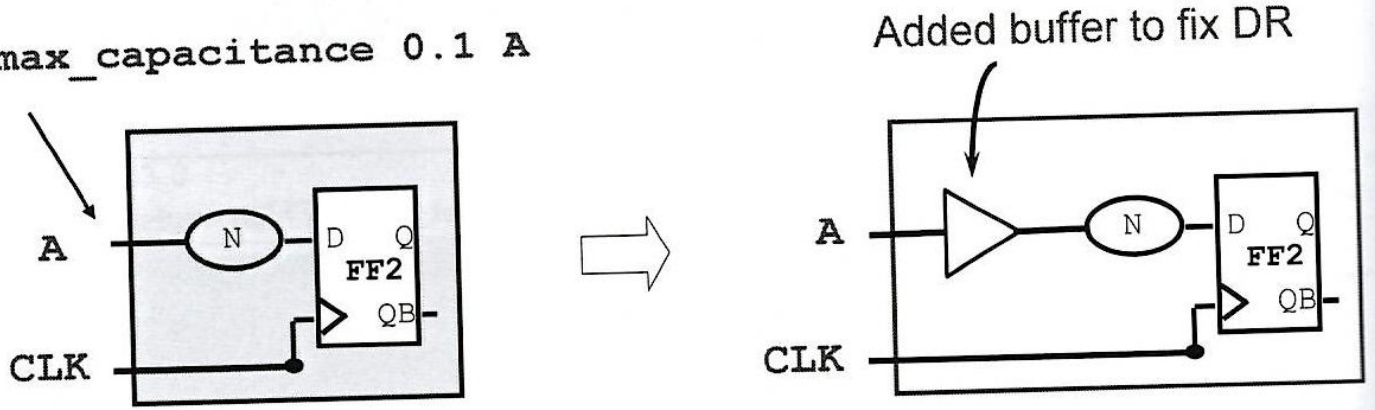

```
compile -inc -only_hold_time
```


TAIWAN TECH

When You Have Design Rule Violations

- Design rule violations may cause timing violations
- Design Compiler inserts buffers and resizes cells to correct design rule violations
- `report_constraint -all_violators`
  - Reports all constraints which have been violated in the design
  - Includes design rules, setup, hold and area

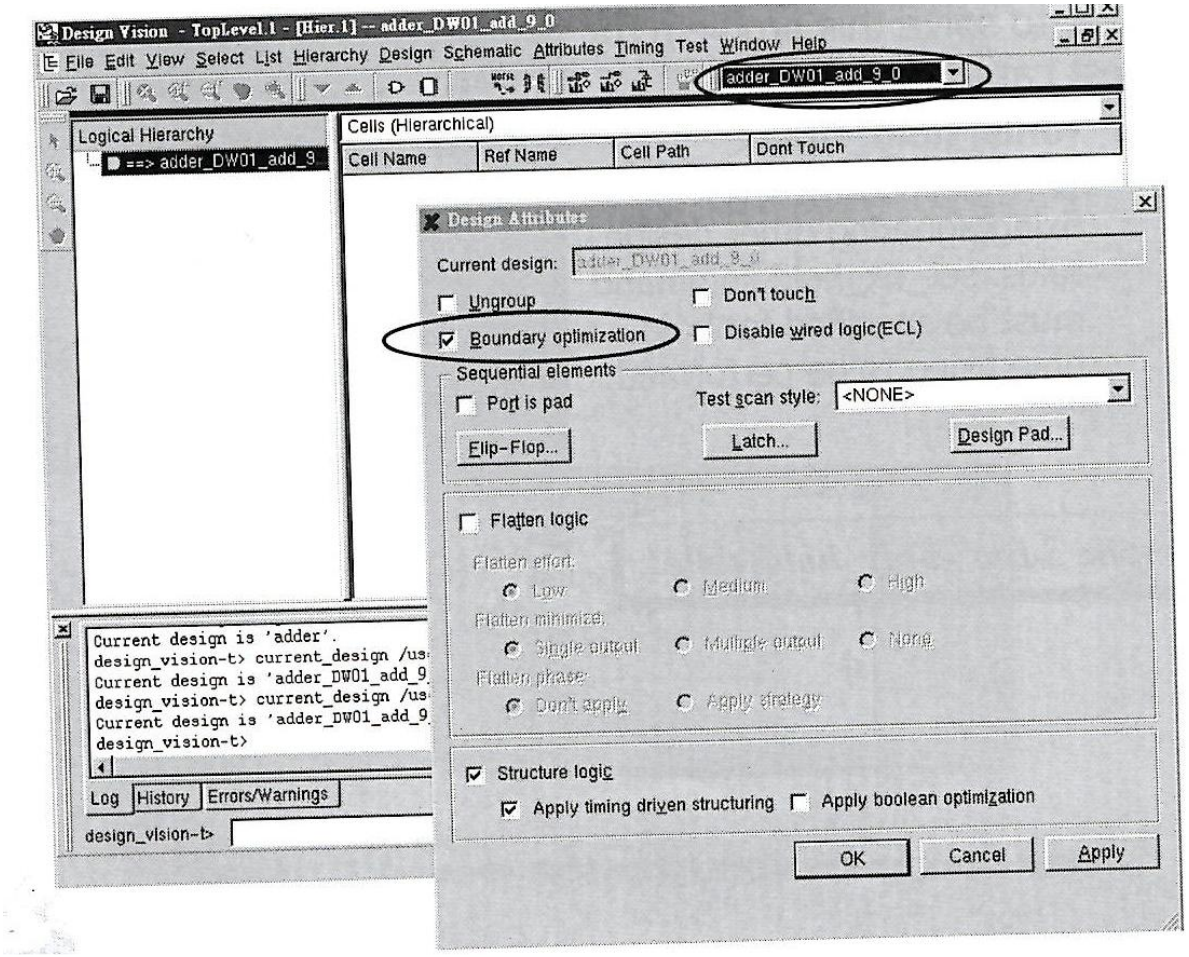
`set_max_capacitance 0.1 A`




```
compile -incremental -only_design_rule
```


TAIWAN TECH

Boundary Optimization(1/2)



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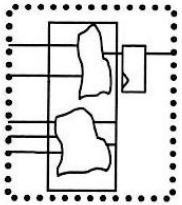
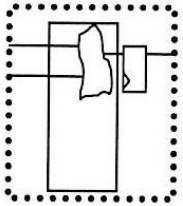
Boundary Optimization(2/2)

- DC can do some optimization across boundaries
  1. Removes logic driving unconnected output ports

compile

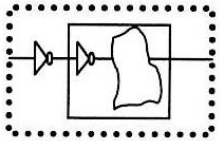


compile –boundary_optimization

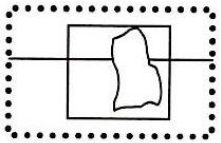


  2. Removes redundant inverters across boundaries

compile

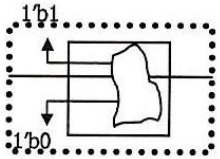


compile –boundary_optimization

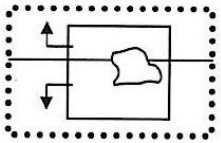


  3. Propagates constants to reduce logic

compile



compile –boundary_optimization



補充:

```
set_boundary_optimization [get_cells PWR_CTRL] false
```

NG / 2011.07

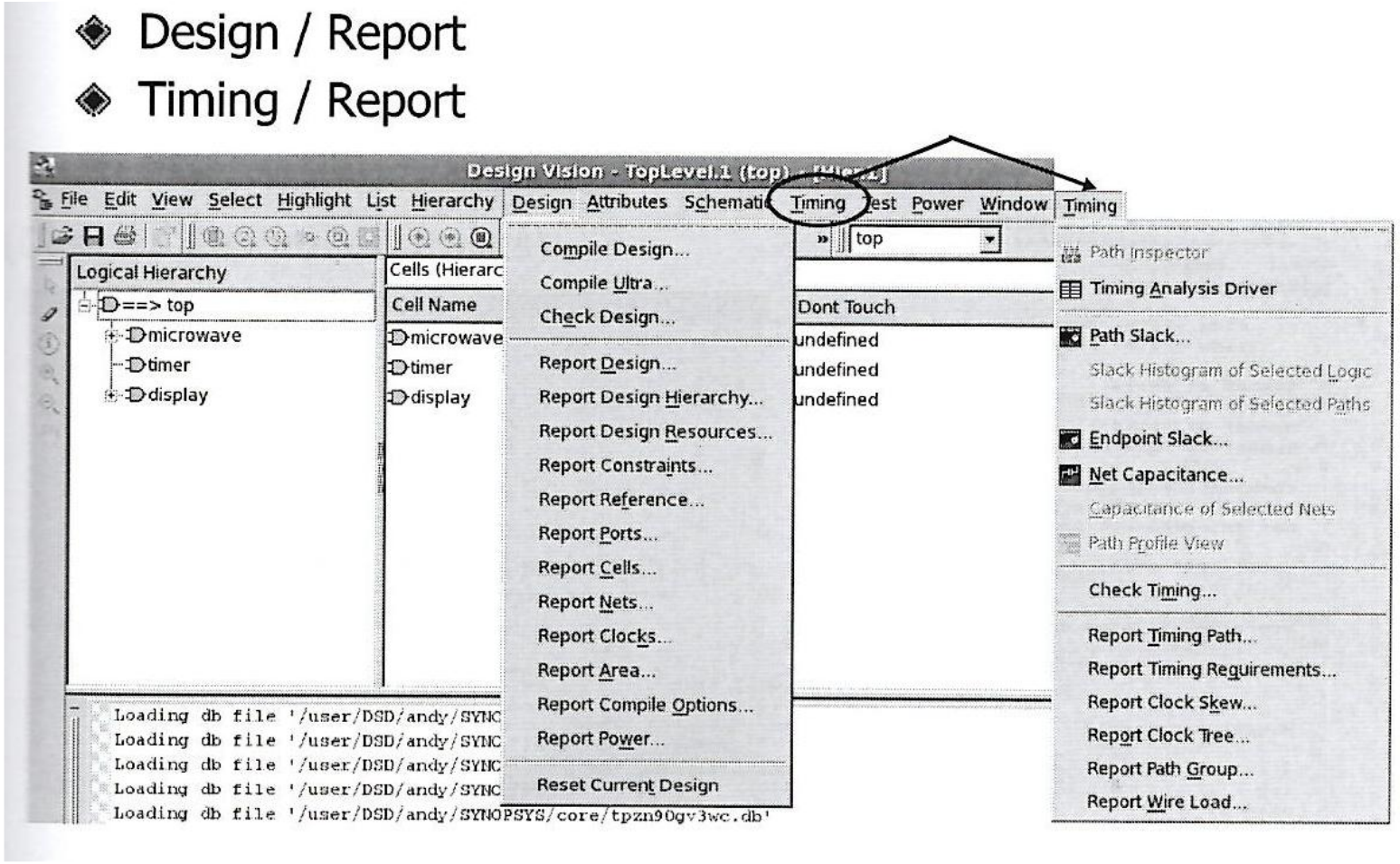
TAIWAN TECH

Synthesis Report and Analysis

TAIWAN TECH

Report

- Design / Report
- Timing / Report



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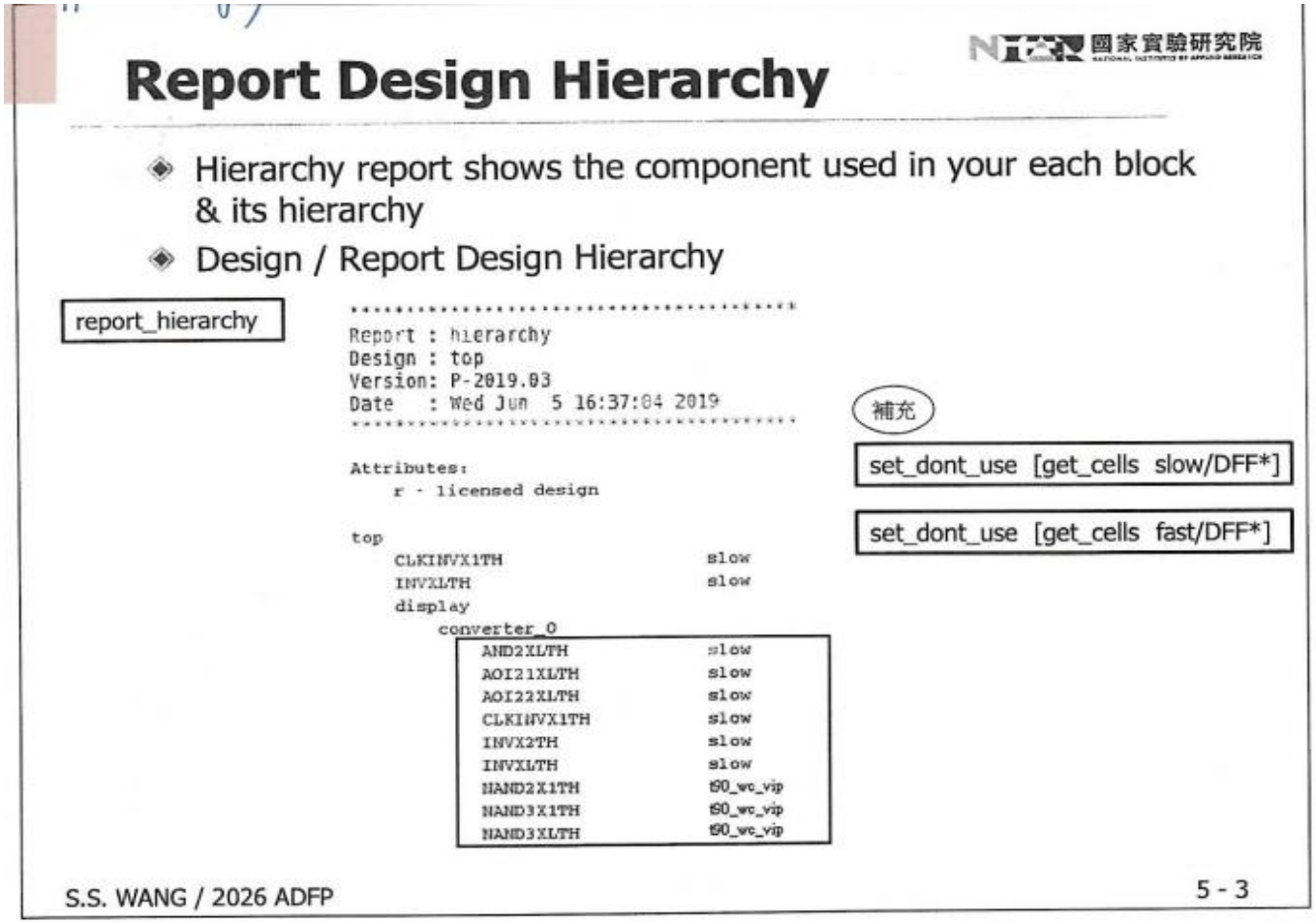
Report Design Hierarchy



Report Area

- Area report shows the um^2 of the design
- Design / Report Area

report_area -hier

```
****************************************
Report : area
Design : top
Version: D-2010.03-SP5
Date   : Wed Dec  1 13:57:38 2010
****************************************

Library(s) Used:

    slow (File: /user/DSD/andy/SYNOPSYS/core/slow_hvt.db)
    slow (File: /user/DSD/andy/SYNOPSYS/core/slow.db)

Number of ports:                 49
Number of nets:                  86
Number of cells:                  5
Number of references:             5

Combinational area:       1306.065621
Noncombinational area:     342.215997
Net Interconnect area:   31873.492676

Total cell area:          1648.281618
Total area:              33521.774294
```

don't care net area

Hierarchical area distribution

Hierarchical cell	Global cell area: Absolute Total	Global cell area: Percent Total	Local cell area: Combinational	Local cell area: Noncombinational	Local cell area: Black boxes	Design
top	1648.2815	100.0	4.2336	0.0000	0.0000	top
display	330.2209	20.0	0.0000	0.0000	0.0000	display
display/convml	82.5552	5.0	82.5552	0.0000	0.0000	converter_2
display/convmm	82.5552	5.0	82.5552	0.0000	0.0000	converter_3
display/convsl	82.5552	5.0	82.5552	0.0000	0.0000	converter_0
display/convsm	82.5552	5.0	82.5552	0.0000	0.0000	converter_1
microwave	299.1744	18.2	0.0000	0.0000	0.0000	microwave
microwave/loader	112.1904	6.8	112.1904	0.0000	0.0000	loader
microwave/micro_st	186.9840	11.3	129.1248	57.8592	0.0000	micro_st
timer	1014.6529	61.6	730.2960	284.3568	0.0000	timer
Total			1306.0657	342.2160	0.0000	

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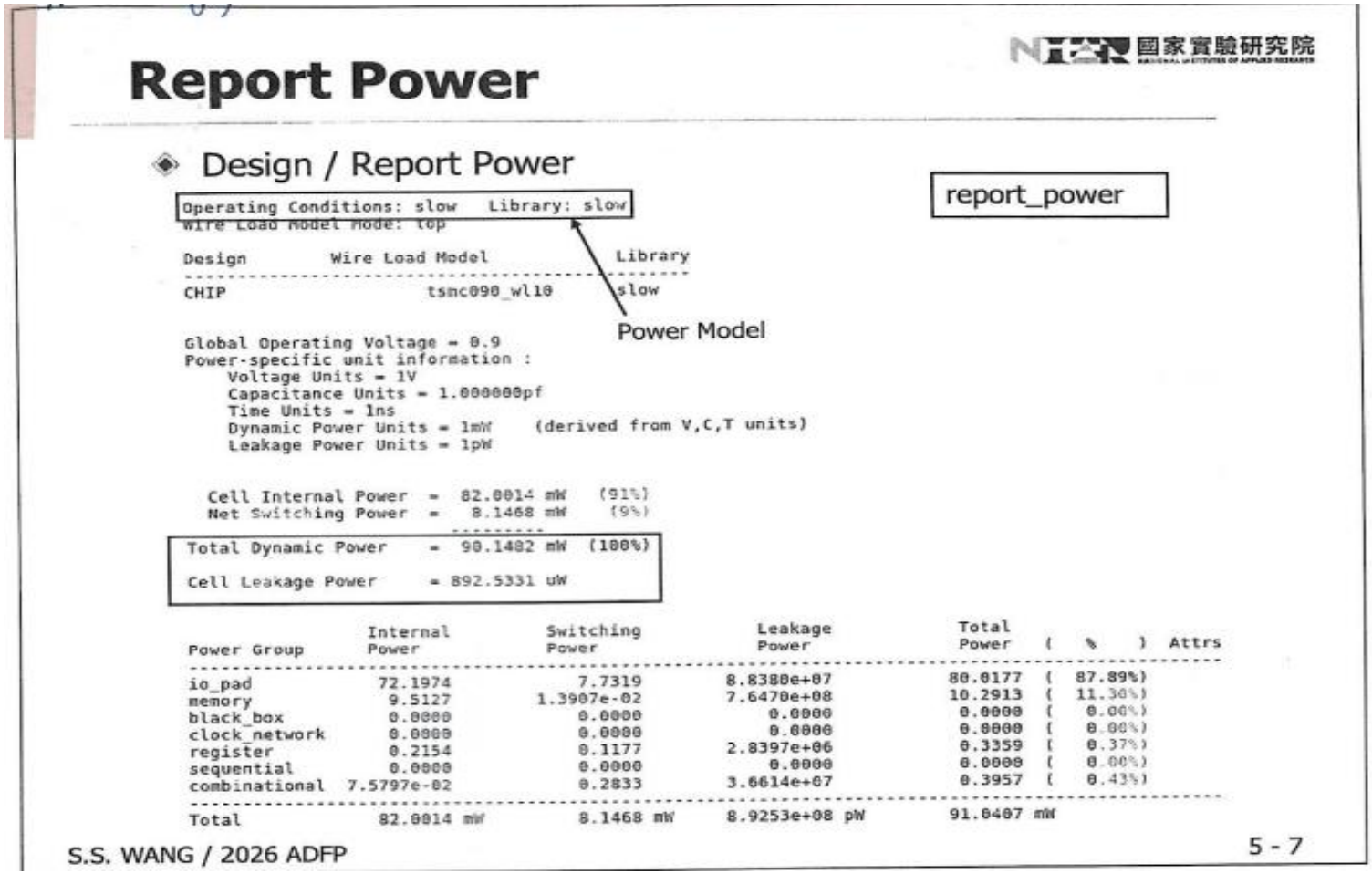
Report Power



Report Power

國家實驗研究院

Design / Report Power

report_power

```
Operating Conditions: slow   Library: slow
Wire Load Model Mode: top

Design          Wire Load Model            Library
------------------------------------------------
CHIP                    tsmc090_wl10       slow

Global Operating Voltage = 0.9
Power-specific unit information :
    Voltage Units = 1V
    Capacitance Units = 1.000000pf
    Time Units = 1ns
    Dynamic Power Units = 1mW    (derived from V,C,T units)
    Leakage Power Units = 1pW


  Cell Internal Power  =  82.0014 mW   (91%)
  Net Switching Power  =   8.1468 mW    (9%)
                         ---------
Total Dynamic Power    =  90.1482 mW  (100%)

Cell Leakage Power     = 892.5331 uW
```

Power Model

Power Group	Internal Power	Switching Power	Leakage Power	Total Power	(%)	Attrs
io_pad	72.1974	7.7319	8.8380e+07	80.0177	(87.89%)	
memory	9.5127	1.3907e-02	7.6470e+08	10.2913	(11.30%)	
black_box	0.0000	0.0000	0.0000	0.0000	(0.00%)	
clock_network	0.0000	0.0000	0.0000	0.0000	(0.00%)	
register	0.2154	0.1177	2.8397e+06	0.3359	(0.37%)	
sequential	0.0000	0.0000	0.0000	0.0000	(0.00%)	
combinational	7.5797e-02	0.2833	3.6614e+07	0.3957	(0.43%)	
Total	82.0014 mW	8.1468 mW	8.9253e+08 pW	91.0407 mW		

S.S. WANG / 2026 ADFP

5 - 7

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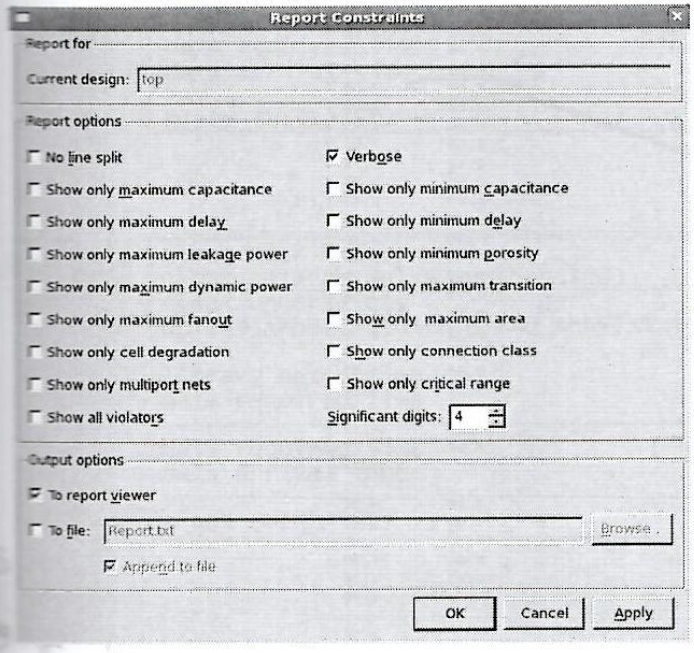
Report - Constraints

- Constraints report shows whether compiled design meets your constraints
- dc_shell-xg-t> *report_constraint –all_violators*

(Design / Report Constraints)



Group (max_delay/setup)	Cost	Weight	Weighted Cost
clk	0.0000	1.0000	0.0000
default	0.0000	1.0000	0.0000
max_delay/setup			0.0000

Group (critical_range)	Total Neg Slack	Critical Endpoints	Cost
clk	0.0000	0	0.0000
default	0.0000	0	0.0000
critical_range			0.0000

Group (min_delay/hold)	Cost	Weight	Weighted Cost
clk	0.0000	1.0000	0.0000
default	0.0000	1.0000	0.0000
min_delay/hold			0.0000

Constraint	Cost
max_transition	0.0000 (MET)
max_fanout	0.0000 (MET)
max_capacitance	0.0000 (MET)
max_delay/setup	0.0000 (MET)
critical_range	0.0000 (MET)
min_delay/hold	0.0000 (MET)
max_dynamic_power	0.1492 (VIOLATED)
max_leakage_power	2557557.5000 (VIOLATED)
max_area	33521.7734 (VIOLATED)

TAIWAN TECH

Report Timing(1/2)

- Path information produced by report_timing command you should know
- Timing / Report Timing Path

```
****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : top
Version: D-2010.03-SP5
Date   : Wed Dec  1 14:33:28 2010
****************************************

Operating Conditions: slow   Library: slow
Wire Load Model Mode: top

  Startpoint: set_time (input port clocked by clk)
  Endpoint: timer/min_msb_next_reg[2]
            (rising edge-triggered flip-flop clocked by clk)
  Path Group: clk
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  top                tsmc090_wl10          slow

  Point                                                   Incr       Path
  ------------------------------------------------------------------------
  clock clk (rise edge)                                   0.00       0.00
  clock network delay (ideal)                             1.00       1.00
  input external delay                                    1.00       2.00 f
  set_time (in)                                           0.10       2.10 f
  microwave/set_time (microwave)                          0.00       2.10 f
  microwave/micro_st/set_time (micro_st)                  0.00       2.10 f
  microwave/micro_st/U4/Y (BUFX10TH)                      0.13       2.24 f
  microwave/micro_st/U5/Y (CLKNAND2X8TH)                  0.05       2.29 r
  microwave/micro_st/U20/Y (CLKINVX4TH)                   0.08       2.37 f
  microwave/micro_st/U19/Y (NAND2X6TH)                    0.09       2.46 r
```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Report Timing(2/2)

- Timing report shows maximum or minimum delay path of design, the default is to display one maximum delay path

start point

Net_delay + cell_delay are combined

end point

Point	Incr	Path
clock clk (rise edge)	0.00	0.00
clock network delay (ideal)	1.00	1.00
input external delay	1.00	2.00 f
set_time (in)	0.10	2.10 f
microwave/set_time (microwave)	0.00	2.10 f
microwave/micro_st/set_time (micro_st)	0.00	2.10 f
microwave/micro_st/U4/Y (BUFX10TH)	0.13	2.24 f
microwave/micro_st/U5/Y (CLKNAND2X8TH)	0.05	2.29 r
microwave/micro_st/U20/Y (CLKINVX4TH)	0.08	2.37 f
microwave/micro_st/U19/Y (NAND2X6TH)	0.09	2.46 r
microwave/micro_st/U21/Y (INVX3TH)	0.07	2.53 f
microwave/micro_st/U24/Y (OR2X8TH)	0.13	2.65 f
microwave/micro_st/load_clk (micro_st)	0.00	2.65 f
microwave/loader/load_clk (loader)	0.00	2.65 f
microwave/loader/U19/Y (NAND2BX8TH)	0.12	2.77 f
microwave/loader/load (loader)	0.00	2.77 f
microwave/load (microwave)	0.00	2.77 f
timer/load (timer)	0.00	2.77 f
timer/U51/Y (BUFX18TH)	0.11	2.88 f
timer/U50/Y (CLKINVX6TH)	0.08	2.96 r
timer/U43/Y (CLKINVX24TH)	0.13	3.09 f
timer/U23/Y (OAI22X1TH)	0.29	3.38 r
timer/U40/Y (BUFX6TH)	0.19	3.57 r
timer/U101/Y (NAND4X8TH)	0.21	3.79 f
timer/U64/Y (OR2X4TH)	0.21	4.00 f
timer/U25/Y (CLKINVX12TH)	0.07	4.07 r
timer/U53/Y (NAND4X8TH)	0.18	4.25 f
timer/U34/Y (NAND2BX4)	0.20	4.45 f
timer/U75/Y (OAI32X2TH)	0.31	4.77 r
timer/min_msb_next_reg[2]/D (DFFQX4)	0.00	4.77 r
data arrival time		4.77
clock clk (rise edge)	4.00	4.00
clock network delay (ideal)	1.00	5.00
clock uncertainty	-0.10	4.90
timer/min_msb_next_reg[2]/CK (DFFQX4)	0.00	4.90 r
library setup time	-0.12	4.78
data required time		4.78
data required time		4.78
data arrival time		-4.77
slack (MET)		0.01

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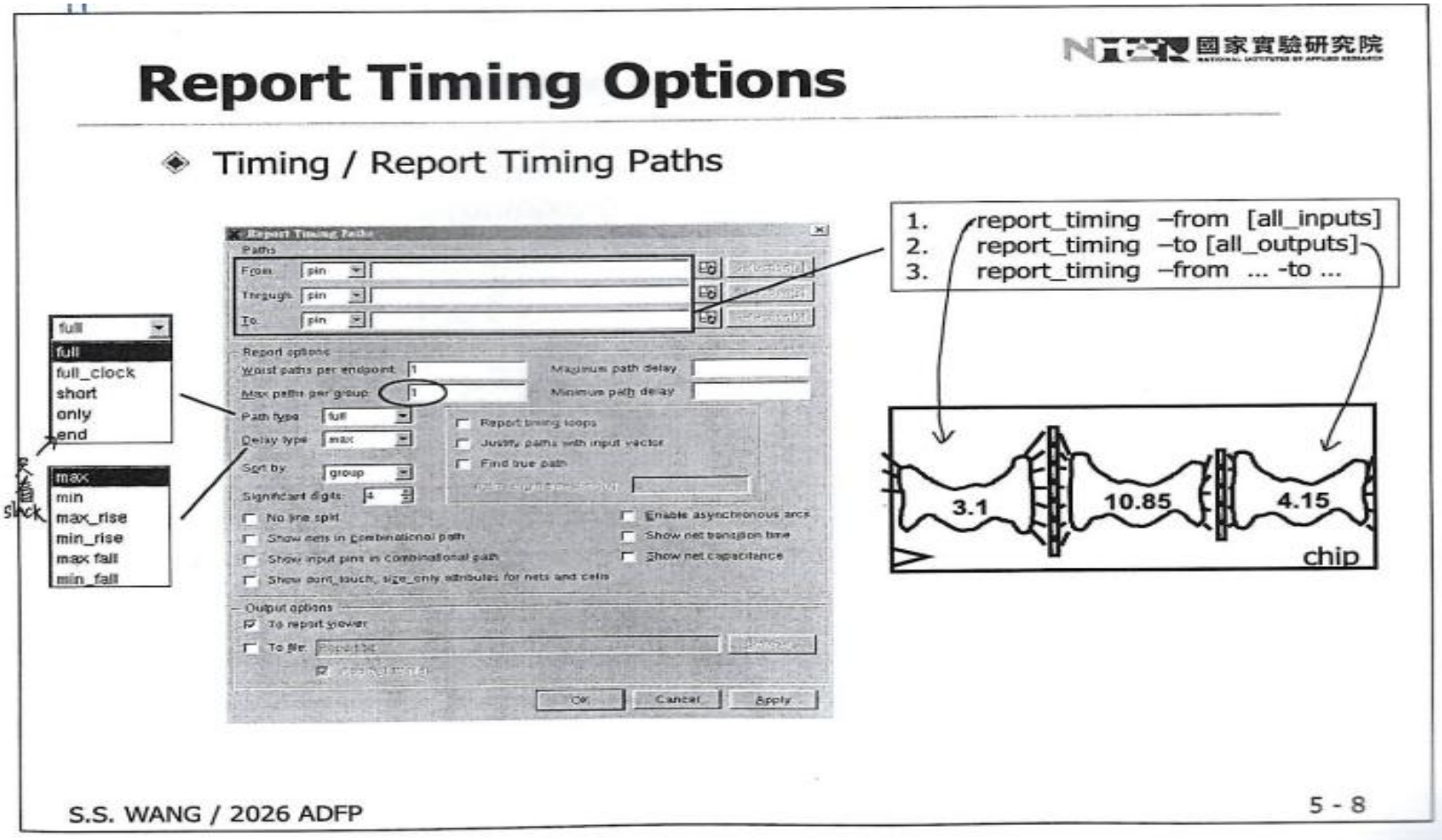
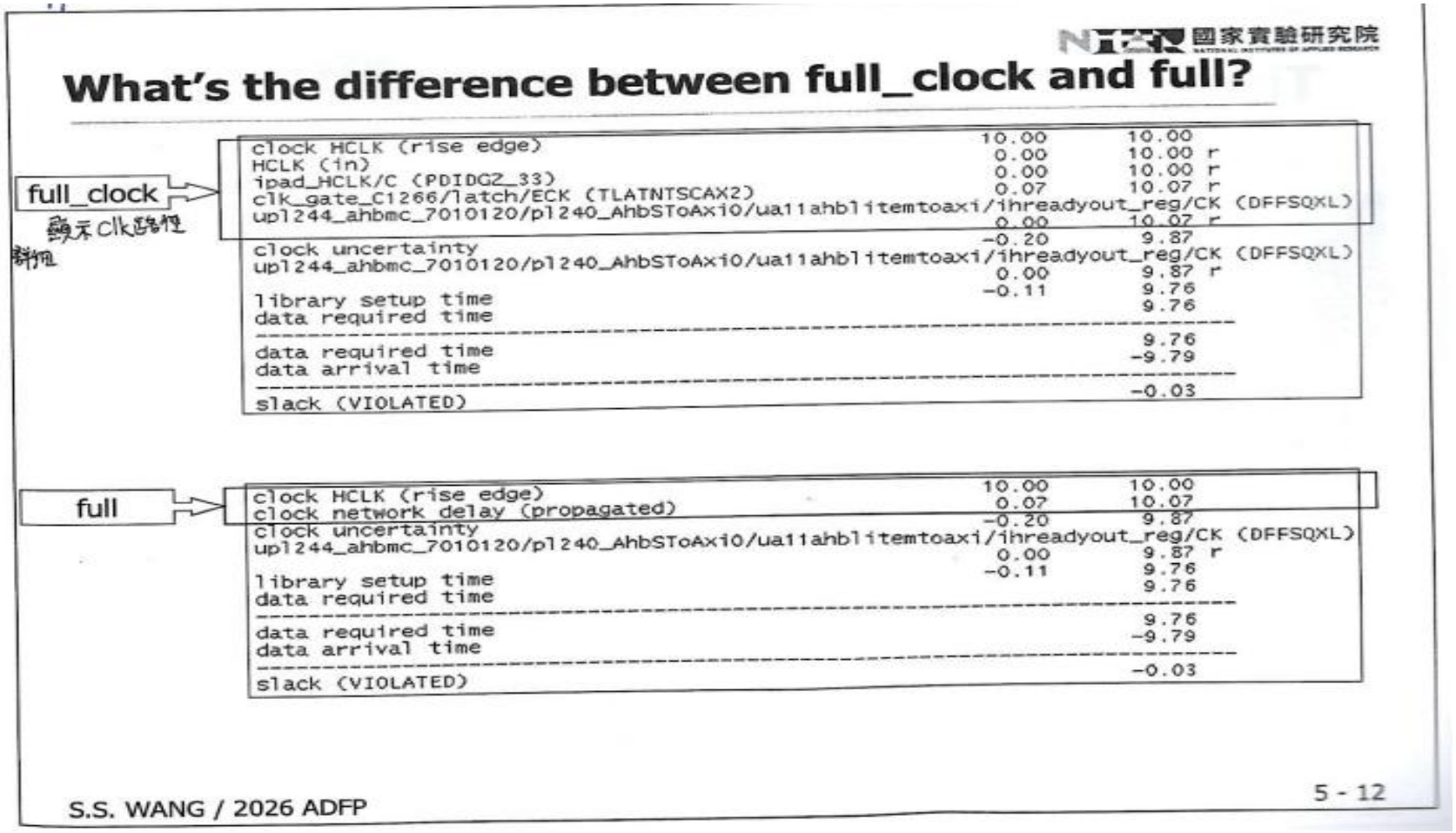
Report Timing Options



What's the difference between full_clock and full?



Path Type Option

```
Point                                                   Incr       Path
-----------------------------------------------------------------------
clock clk (rise edge)                                   0.0000     0.0000
clock network delay (ideal)                             1.0000     1.0000
input external delay                                    1.0000     2.0000 r
test (in)                                               0.0257     2.0257 r
...
timer/sec_lsb_next_reg[0]/D (QDFFX1)                    2.8098     4.8355 r
data arrival time                                                  4.8355

clock clk (rise edge)                                   4.0000     4.0000
clock network delay (ideal)                             1.0000     5.0000
clock uncertainty                                      -0.1000     4.9000
timer/sec_lsb_next_reg[0]/CK (QDFFX1)                   0.0000     4.9000 r
library setup time                                     -0.0476     4.8524
data required time                                                 4.8524
-----------------------------------------------------------------------
data required time                                                 4.8524
data arrival time                                                 -4.8355
-----------------------------------------------------------------------
slack (MET)                                                        0.0169
```

short

```
Point                                                   Incr       Path
-----------------------------------------------------------------------
input external delay                                    1.0000     1.0000 r
test (in)                                               0.0257     1.0257 r
microwave/test (microwave)                              0.0000     1.0257 r
microwave/micro_st/test (micro_st)                      0.0000     1.0257 r
microwave/micro_st/U157/O (INVXLP)                      0.0589     1.0846 f
microwave/micro_st/U370/O (AOI112X1)                    0.1785     1.2631 r
microwave/micro_st/load_8888 (micro_st)                 0.0000     1.2631 r
microwave/loader/load_8888 (loader)                     0.0000     1.2631 r
microwave/loader/U3/O (NR2X1)                           0.2062     1.4693 f
microwave/loader/U209/O (OR2B1XLP)                      0.0924     1.5617 r
microwave/loader/load (loader)                          0.0000     1.5617 r
microwave/load (microwave)                              0.0000     1.5617 r
timer/load (timer)                                      0.0000     1.5617 r
timer/U180/O (BUFXLP)                                   0.4054     1.9670 r
timer/U65/O (INVX1)                                     0.3096     2.2766 f
timer/U390/O (AOI22XLP)                                 0.2384     2.5150 r
timer/U41/O (INVXLP)                                    0.1009     2.6159 f
timer/U4/O (OR4B2XLP)                                   0.2690     2.8849 f
timer/U157/O (ND3X1)                                    0.1767     3.0616 r
timer/U17/O (AN2XLP)                                    0.1338     3.1954 r
timer/U81/O (AN2XLP)                                    0.1920     3.3874 r
timer/U183/O (AN2B1XLP)                                 0.1093     3.4967 f
timer/U9/O (AN2B1XLP)                                   0.1208     3.6174 f
timer/U8/O (AN2B1XLP)                                   0.1268     3.7442 f
timer/U30/O (OAI112XLP)                                 0.0914     3.8355 r
timer/sec_lsb_next_reg[0]/D (QDFFX1)                    0.0000     3.8355 r
data arrival time                                                  3.8355
```

only

Endpoint	Path Delay	Path Required	Slack
timer/sec_lsb_next_reg[0]/D (QDFFX1)	4.8355 r	4.8524	0.0169
timer/sec_lsb_next_reg[3]/D (QDFFX1)	4.8355 r	4.8524	0.0169
timer/sec_msb_next_reg[2]/D (QDFFX1)	4.7120 r	4.8537	0.1418
timer/min_msb_next_reg[2]/D (QDFFX1)	4.6216 f	4.8262	0.2046
timer/min_msb_next_reg[1]/D (QDFFX1)	4.6120 f	4.8183	0.2064
timer/min_msb_next_reg[3]/D (QDFFX1)	4.6120 f	4.8183	0.2064
timer/min_msb_next_reg[0]/D (QDFFX1)	4.6032 f	4.8263	0.2231
timer/min_lsb_next_reg[1]/D (QDFFX1)	4.5932 f	4.8183	0.2251
timer/min_lsb_next_reg[0]/D (QDFFX1)	4.5932 f	4.8198	0.2266
timer/min_lsb_next_reg[3]/D (QDFFX1)	4.5932 f	4.8198	0.2266

end

max path per group=10

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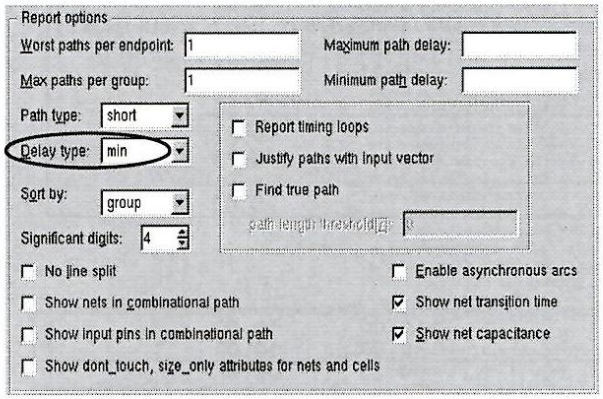
Delay Type Option



(Hold Time Check Report)

```
****************************************
Report : timing
         -path full
         -delay min
         -max_paths 1
Design : top
Version: D-2010.03-SP5
Date   : Wed Dec  1 14:42:17 2010
****************************************

Operating Conditions: fast  Library: fast
Wire Load Model Mode: top

  Startpoint: timer/min_lsb_next_reg[3]
              (rising edge-triggered flip-flop clocked by clk)
  Endpoint: microwave/micro_st/state_reg[2]
            (rising edge-triggered flip-flop clocked by clk)
  Path Group: clk
  Path Type: min

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  top                tsmc090_wl10          slow

  Point                                                   Incr
  --------------------------------------------------------------
  clock clk (rise edge)                                   0.00
  clock network delay (ideal)                             1.00
  timer/min_lsb_next_reg[3]/CK (DFFHQX8)                  0.00
  timer/min_lsb_next_reg[3]/Q (DFFHQX8)                   0.23
  timer/U22/Y (NOR4X8TH)                                  0.09
  timer/U10/Y (AND4X8TH)                                  0.14
  timer/done (timer)                                      0.00
  microwave/done (microwave)                              0.00
  microwave/micro_st/done (micro_st)                      0.00
  microwave/micro_st/U11/Y (AND2X6TH)                     0.12
  microwave/micro_st/state_reg[2]/D (DFFRX2TH)            0.00
  data arrival time

  clock clk (rise edge)                                   0.00
  clock network delay (ideal)                             1.00
  clock uncertainty                                       0.10
  microwave/micro_st/state_reg[2]/CK (DFFRX2TH)           0.00
  library hold time                                      -0.01
  data required time
  --------------------------------------------------------------
  data required time
  data arrival time
  --------------------------------------------------------------
  slack (MET)
```

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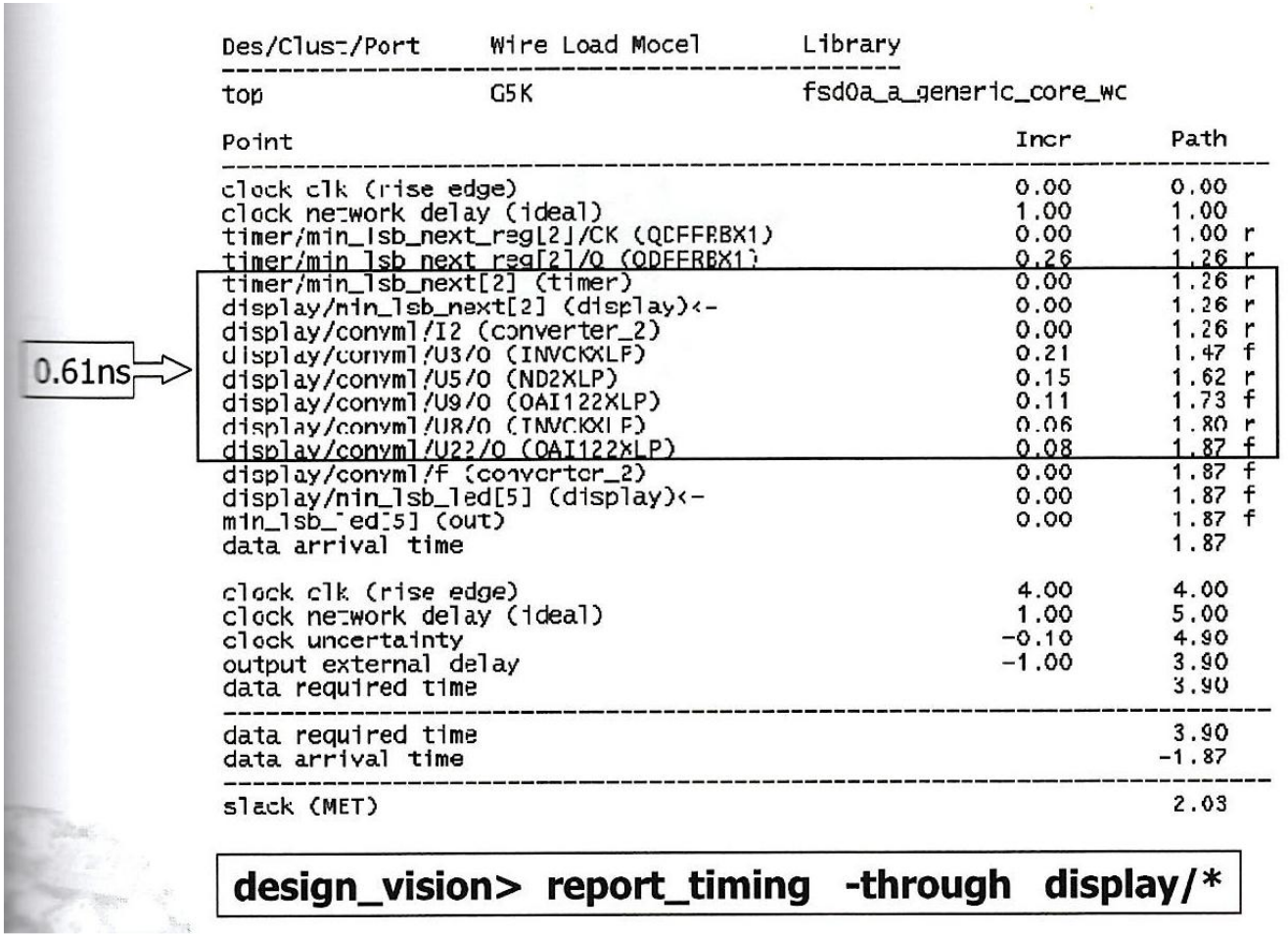
Report Timing for Interesting Block



```
Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
top                G5K                   fsd0a_a_generic_core_wc

Point                                                    Incr       Path
-----------------------------------------------------------------------
clock clk (rise edge)                                    0.00       0.00
clock network delay (ideal)                              1.00       1.00
timer/min_lsb_next_reg[2]/CK (QDFFRBX1)                  0.00       1.00 r
timer/min_lsb_next_reg[2]/Q (QDFFRBX1)                   0.26       1.26 r
timer/min_lsb_next[2] (timer)                            0.00       1.26 r
display/min_lsb_next[2] (display)<-                      0.00       1.26 r
display/conv_ml/I2 (converter_2)                         0.00       1.26 r
display/conv_ml/U3/O (INVCKXLP)                          0.21       1.47 f
display/conv_ml/U5/O (ND2XLP)                            0.15       1.62 r
display/conv_ml/U9/O (OAI122XLP)                         0.11       1.73 f
display/conv_ml/U8/O (INVCKXLP)                          0.06       1.80 r
display/conv_ml/U22/O (OAI122XLP)                        0.08       1.87 f
display/conv_ml/f (converter_2)                          0.00       1.87 f
display/min_lsb_led[5] (display)<-                       0.00       1.87 f
min_lsb_led[5] (out)                                     0.00       1.87 f
data arrival time                                                   1.87

clock clk (rise edge)                                    4.00       4.00
clock network delay (ideal)                              1.00       5.00
clock uncertainty                                       -0.10       4.90
output external delay                                   -1.00       3.90
data required time                                                  3.90
-----------------------------------------------------------------------
data required time                                                  3.90
data arrival time                                                  -1.87
-----------------------------------------------------------------------
slack (MET)                                                         2.03
```

0.61ns

```
design_vision> report_timing -through display/*
```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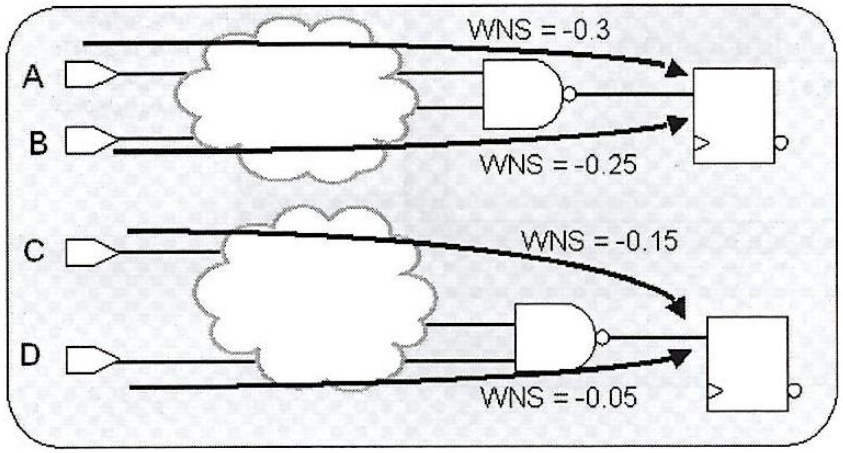
EndPoint V.S. Path Slack

- Path Slack will report all of the slack path in the clock domain.
- Endpoint slack will report all of the worst slack path of all of endpoint in the clock domain.

- Ex:
  - Path slack = {A, B, C, D}
  - EndPoint Slack = {A, C}



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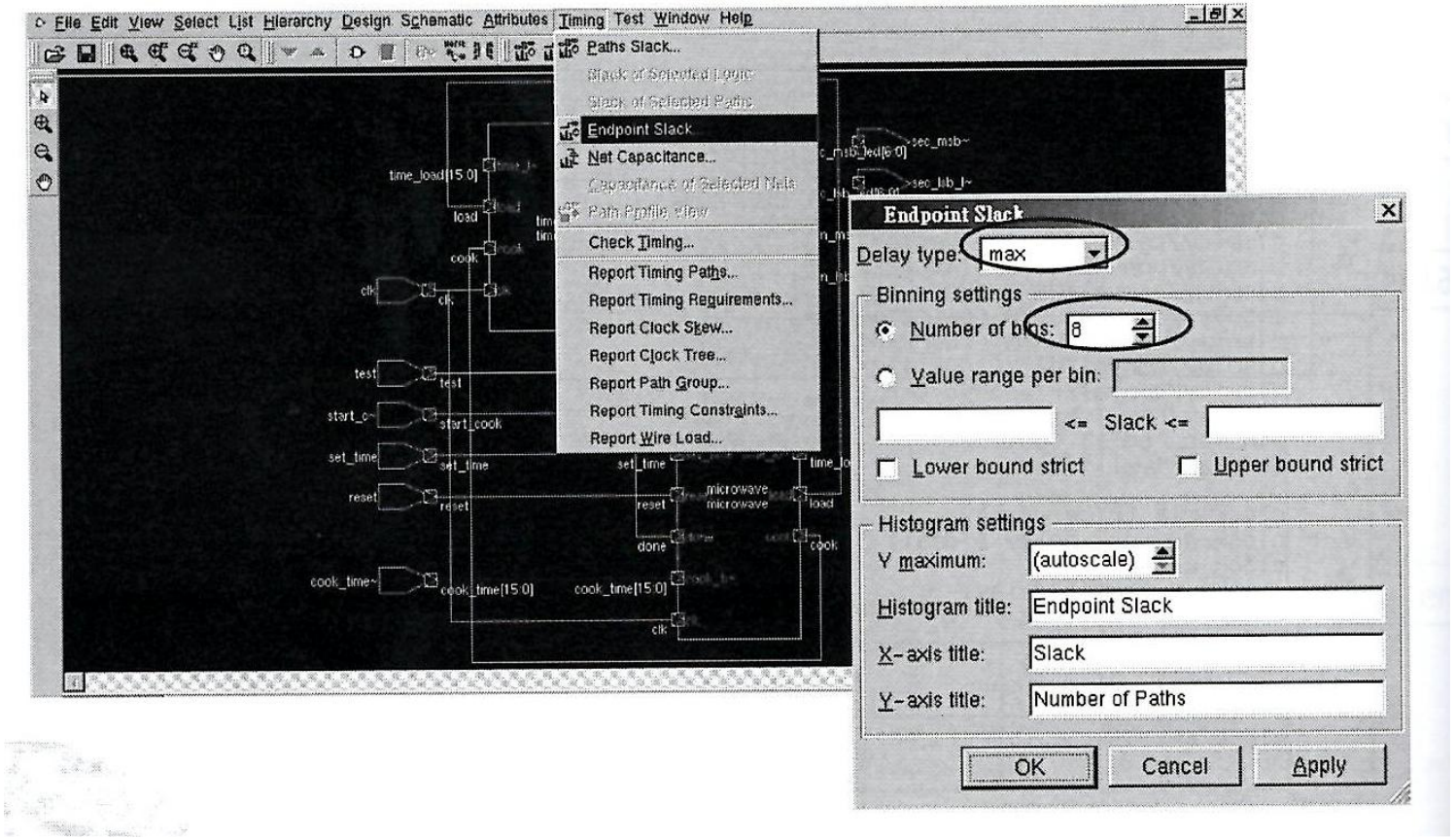
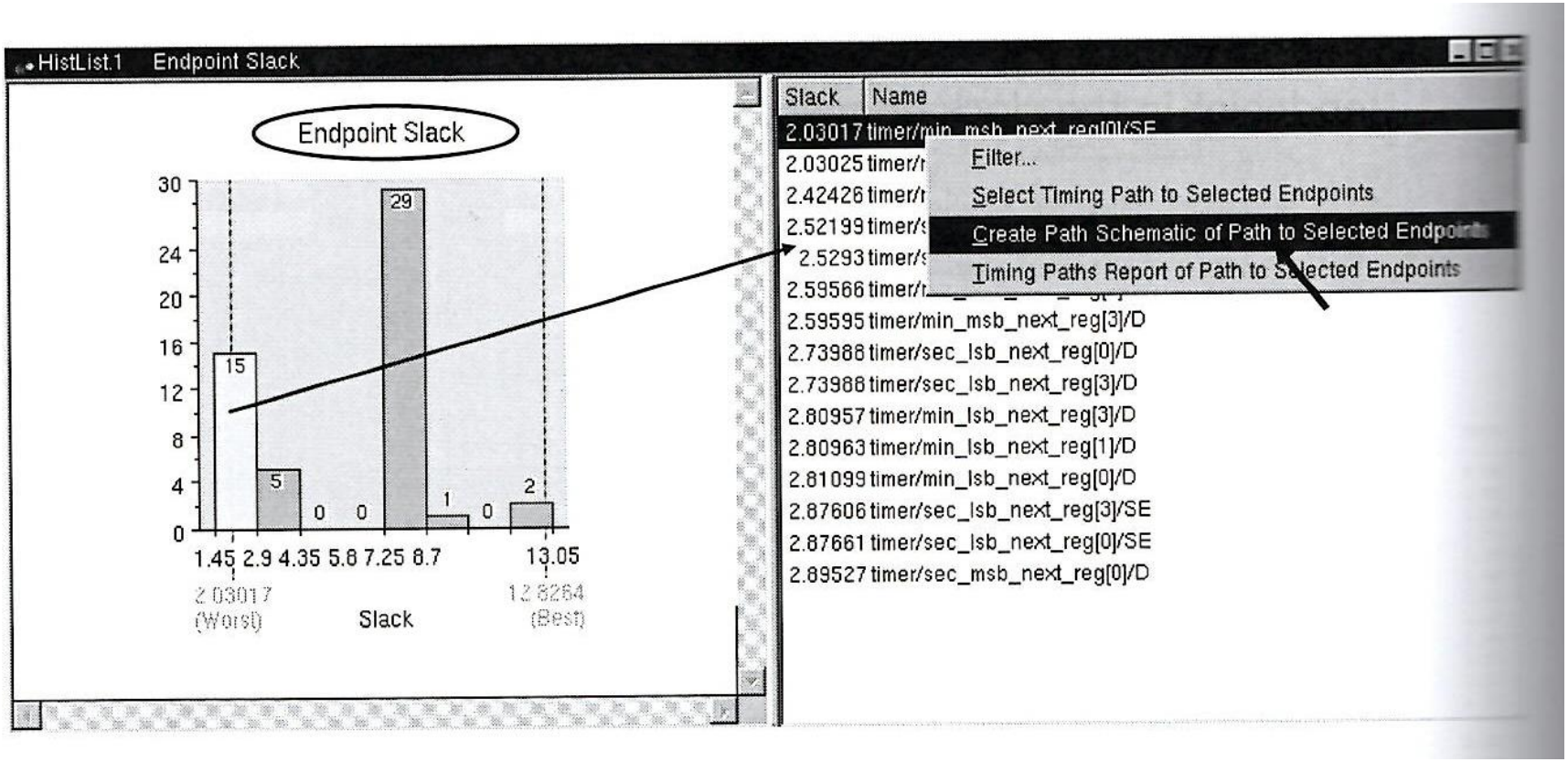
Timing Slack Histogram(1/3)

- Timing / Endpoint Slack



Timing Slack Histogram(2/3)



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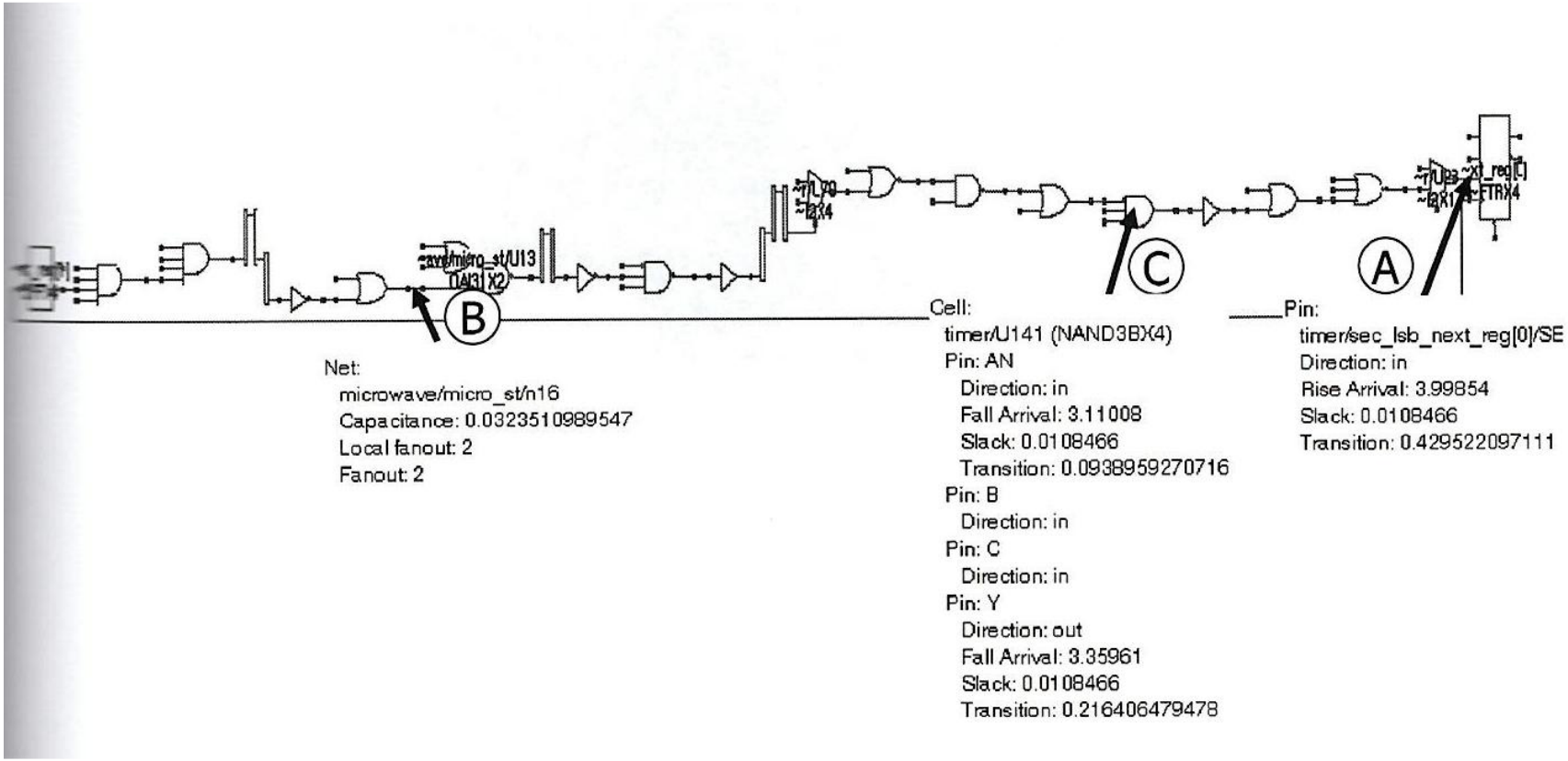
Timing Slack Histogram(3/3)



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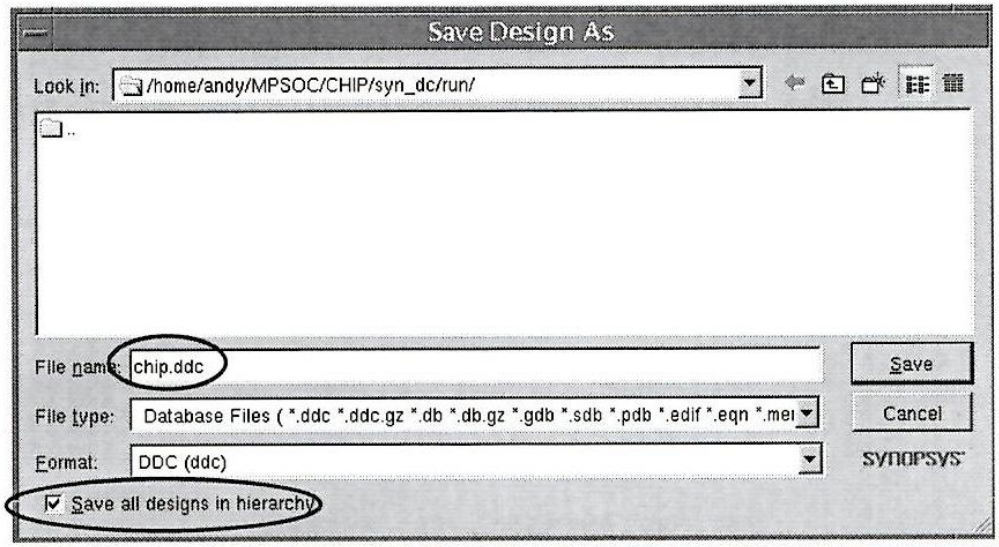
Save Design

- Save your design to file before you quit Design Compiler
- *File/Save* saves your design in the ddc (or db) format
- *File/Save As* can save your design in other Write formats:
  - PLA(Berkeley Espresso): .pla
  - Synopsys formats
    - equation: .eq
    - state table: .st
  - EDIF
  - Verilog: .v
  - VHDL: .vhd



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